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OIL AND NATURAL GAS CORPORATION LIMITED

'भारतीय पश्चिमी अपतट के कच्छ की खाड़ी क्षेत्र के "GK-28-10(PLATFORM)", प्लेटफॉर्म स्थान, के लिये मृदा अन्वेषण रिपोर्ट'

'SOIL INVESTIGATION REPORT FOR PLATFORM LOCATION "GK-28-10 (PLATFORM)", GULF OF KUTCH FIELD, WEST COAST OF INDIA'

प्रति (For):

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TITLE: REPORT ON "SOIL INVESTIGATION FOR PLATFORM LOCATION GK-28-10 (PLATFORM), GULF OF KUTCH"

EXECUTIVE SUMMARY

This is the geotechnical investigation report for the location GK-28-10 (PLATFORM) located in Gulf of Kutch, off the west coast of India. The purpose of the investigation was to ascertain the soil condition at the location for assessment of ultimate axial pile capacity, load deformation data for open-ended steel tubular piles for the piled platform, assessment of mud-mat bearing capacity and estimation of Jack-up leg penetration for Standard ONGC rig. The field investigation consisted of two boreholes - one shallow bore up to 29.96 m for CPTU (cone penetration test with pore pressure measurement) test and one deep borehole approximately 5.0 m apart for alternate CPTU, and sampling up to 125.03 m. Remote vane tests were conducted up to a depth of 20.8 m below seafloor in another borehole 5 m apart from original borehole. Approximate average water depth, not reduced to chart datum, at the location during the course of investigation was about 28.5 m.

Field investigation was carried out from 28th to 30th April and 2nd to 6th May, 2018 by M/s. Coastal Marine Construction and Engineering Company (COMACOE) from ONGC's Geotechnical vessel "Samudra Sarvekshak". Planning and performance of required additional laboratory testing, compilation of results, evaluation of soil design parameters, engineering analysis and report preparation was done by IEOT. Tests were specifically planned to provide soil parameters for analysis of mudmat bearing capacity, estimation of Jack-up leg penetration and pile design.

The soil profile is determined up to the investigated depth of 125.03 m below the mudline. The soil profile at the location GK-28-10 (PLATFORM) is clayey from seabed to 21.50 m and predominantly sandy till the investigated depth of 125.03 m below seafloor. Moderate to strong cementation is observed in some of the sand layers. Carbonate content in the soil layers varies from 3% to 97%.

Soil design parameters are presented for estimation of static ultimate bearing capacity of steel pipe piles. In addition, axial pile capacity curves for piles of 60", 66", 72" and 84" outer diameter, and, *p-y*, *q-z* and *t-z* data for a 60" OD, 2" thick open ended steel pipe pile have also been presented. The ultimate axial pile capacity of 35.0 MN in compression is achieved at a penetration of about 75.5 m for an open ended steel pipe pile of 60" OD and 2" wall thickness. The bearing capacity of seabed soil has been assessed for mud mat design. In case penetration of jacket leg extensions in to the seabed soil is envisaged during installation, it is recommended that rational analysis be performed to check that the desired penetration is achievable.

Jack up leg penetration analysis was carried out for standard ONGC spudcan of diameter 14.0 m and a maximum preload of 45.0 MN. The final penetration under full preload is estimated to be about 19.5 m below mudline. Rig-specific analysis should be carried out for assessment of suitability for individual rigs.

Month & Year	**August, 2018**

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REPORT TITLE: REPORT ON "SOIL INVESTIGATION FOR PLATFORM LOCATION 'GK-28-10 (PLATFORM)', GULF OF KUTCH FIELD, WEST COAST OF INDIA"

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CONTENTS

			Page
1.0	INTRODUCTION		1
2.0	SCOPE OF REPORT		1
3.0	FIELD OPERATIONS		2
	3.1	General	2
	3.2	Positioning	2
	3.3	Water Depth	3
	3.4	Drilling, Sampling and In-situ Testing	3
	3.4.1	Drilling and Sampling	3
	3.4.2	Peizocone Penetration Tests (CPTU)	4
	3.4.3	Remote (Field) Vane Tests	4
4.0	LABORATORY TESTING		5
	4.1	General	5
	4.2	Laboratory Testing Offshore	5
	4.3	Laboratory Testing Onshore	6
5.0	SOIL CONDITIONS		6
	5.1	General	6
	5.2	Stratigraphy	7
	5.3	Soil Properties and Design Parameters	8
	5.3.1	Clays	8
	5.3.2	Sands	9
	5.3.3	Silt	10
6.0	ENGINEERING ANALYSIS		10
	6.1	General	10
	6.2	Axial Pile Capacity	11
	6.3	Axial Load deformation data (t-z and q-z data)	12
	6.4	Lateral Load deformation data (p-y data)	12
	6.5	Mud-mat bearing Capacity	13
	6.6	Jack up leg penetration analysis	13
7.0	CONCLUSIONS AND RECOMMENDATIONS		14
8.0	LIST OF REFERENCES		15

ILLUSTRATIONS

APPENDIX A	**General Location Details**
APPENDIX B	**Laboratory Test Results and Photographs of Soil Samples**
APPENDIX C	**Summary of Soil Conditions**
APPENDIX D	**Boring Logs, Drilling Log and In-situ Test Results**
APPENDIX E	**General Information**

1.0 INTRODUCTION

Geotechnical field investigation for borehole designated GK-28-10 (PLATFORM) was performed by M/s. Coastal Marine Construction and Engineering Company from ONGC's Geotechnical vessel "Samudra Sarvekshak". The field investigation was carried out between 28th to 30th April and 2nd to 6th May, 2018. The borehole location is in Gulf of Kutch, off the West coast of India.

The purpose of the investigation was to ascertain the soil condition at the location for assessment of ultimate axial pile capacity, load deformation data for open-ended tubular piles for the piled platform, assessment of mudmat bearing capacity and estimation of Jack-up rig leg penetration for standard ONGC rig. CPTU (cone penetration test with pore pressure measurement) was carried out in a shallow bore continuously up to 29.96 m depth below seafloor. In a deeper borehole 5.0 m apart, alternate sampling and CPTU were later carried out up to a depth of 125.03 m. Remote vane tests were conducted up to a depth of 20.8 m below seafloor in another borehole 5 m apart from original borehole. Sampling interval was normally 1.0 m up to a depth of about 15.0 m which was increased subsequently. The in-situ test CPTU, FVT and on-board laboratory tests were performed by M/s. Coastal Marine Construction and Engineering Company (COMACOE, 2018).

Marine Survey Division referred the project to IEOT for additional laboratory testing, evaluation of design parameters, engineering analysis and preparation of final soil investigation report. Reference letter is placed at the end of Appendix E. Planning and performance of on-shore laboratory testing, compilation of tests results, evaluation of soil design parameters and engineering analysis were carried out by IEOT for a piled platform for the location.

The design parameters and the results of foundation engineering analysis are presented under "Illustrations" on Plates 1 to 15, which follow the text of this report. The SI system of units is used throughout this report. A conversion table from SI to other units is included at Plate E1.

The Appendices of this report include: -

Appendix A: General Location Details
Appendix B: Laboratory Test Results and Photographs of Soil Samples
Appendix C: Summary of Soil Conditions
Appendix D: Boring Logs, Drilling Log and In-situ Test Results
Appendix E: General Information

2.0 SCOPE OF REPORT

The scope of this report is to present:

- The results of the laboratory tests performed offshore and onshore

- Soil characteristics including boring logs showing the soil stratigraphy and the type and strength characteristics of various strata
- CPTU test results
- Axial pile capacities for open ended steel pipe piles of 60, 66, 72 and 84 inch OD and 2 inch wall thickness
- Axial load-deformation characteristics (*t-z* and *q-z* data) for open ended steel pipe piles of 60 inch OD and 2 inch wall thickness
- Lateral load-deformation characteristics (*p-y* data) for 60 inch OD open ended steel pipe piles
- Mud mat bearing capacity
- Jack-up leg penetration analysis for standard ONGC rig

3.0 FIELD OPERATIONS

3.1 General

The field work was performed by M/s Coastal Marine Construction and Engineering Company, with ONGC's Geotechnical Vessel "Samudra Sarvekshak" from 28th to 30th April and 2nd to 6th May, 2018.

The field investigation consisted of a borehole up to 29.96 m depth exclusively for CPTU test. In another borehole, 5.0 m apart, alternate sampling and CPTU up to 125.03 m depth below seafloor were carried out. Remote vane tests were conducted up to a depth of 20.8 m below seafloor in another borehole 5 m apart from original borehole.

3.2 Positioning

The GTV Samudra Sarvekshak was initially positioned at GK-28-10 (PLATFORM) location on 28th April, 2018 and CPTU operations commenced on the same day. As per on-board report, GTV was positioned later at the same location about 5.0 m apart on the same day for alternate sampling and CPTU. Remote vane tests in a borehole 5 m apart began on 5th May, 2018.

General location plan is presented on Plate A-1 and positioning data, including the released and actual co-ordinates of the location GK-28-10 (PLATFORM) are presented on Plate A-2. Plate A-3 presents location of GK-28-10 (PLATFORM) along with nearby borings performed earlier. GK-28-10 (PLATFORM) location falls in Zone 42, WGS 84 as per the on-board report.

3.3 Water Depth

The water depth at the location was measured at thirty minute interval throughout the field work with the vessel's Skipper Echo Sounder equipment. The water depths are presented in Table 1. **These depths have not been reduced to chart datum**:

Table 1: Water Depth measurements

Method	Water Depth (m)		
	05:00 hrs. and 11:25 hrs. on 28th April, 2018 (CPTU Borehole)	22:40 hrs. on 28th April and 08:40 hrs. on 2nd May, 2018 (Sampling borehole)	20:50 hrs. on 5th May, 2018 (Remote vane borehole)
Echo Sounder	26.8 and 28.9	29.50 and 27.40	27.80

The tidal variation, in graphical form, recorded during offshore investigation is presented in Plates A-4-I and A-4-II in Appendix A.

3.4 Drilling, Sampling and In Situ Testing

The sampling and in-situ testing programme consisted of the following:

- A borehole where CPTU was carried out continuously up to 29.96 m below the seafloor.
- Alternate CPTU and sampling were carried out in a separate borehole, about 5.0 m away from the CPTU borehole, up to a depth of 125.03 m below seafloor.
- Remote vane tests were conducted up to a depth of 20.8 m below seafloor in another borehole 5 m apart from original borehole.

3.4.1 Drlling and sampling

Drilling was performed using straight flush rotary drilling method with open center drag bit, using a motion compensated rig with pipes. Usually seawater was used to clean and stabilize the borehole. However, at times drilling mud made up of guar gum and seawater ballast was used to stabilize the hole, particularly in soft clays and sand strata. The wire line push sampler is lowered into the drill string where it latches into the BHA. The attached thin walled (76 mm OD & 71 mm ID) Shelby tube is hydraulically pushed (1m stroke) in to the ground ahead of the drill bit to

sample the soil. In Very stiff to hard clays, thick walled sampler was used. Before the push commences the piston head is located at the bottom of the sample tube to prevent and ingress of drill cuttings. This travels up the tube as it is pushed into the ground, minimizing sample disturbance and maximizing recovery.

3.4.2 Piezocone Penetration test (CPTU)

Cone penetration Testing with Pore Pressure Measurement (CPTU) was performed using standard 10 cm^2 A.P. van den Berg cone having an area ratio (α) of 0.75.

In situ Piezocone penetration tests (CPTU) were performed in the boreholes using a Wison down hole tool. This tool has a hydraulically controlled thrusting mechanism, which pushes an instrumented cone on a cone rod into the soil. A data cable connects the cone to a surface acquisition computer. To carry out a PCPT test, the hole is first drilled at the required test depth, ensuring that all the cuttings are flushed out of the hole either by seawater or drilling mud. The drill string is advanced to the required testing depth and clamped by the seabed frame to provide bottom weight reaction for the test. The Wison tool is then lowered into the drill string on its combined hydraulic/data/lift cable (umbilical) until it latches into the Bottom Hole Assembly of the drill string. When the tool is properly latched, the cone is just a few centimeters inside the drill bit at the bottom of the borehole. After taking the baseline (zero) reading, the test is started by activating the hydraulic flow from the power pack, which pushes the cone into the soil at a constant rate of 2 cm/sec for a stroke of 3 m or refusal, whichever occurs first. The data consisting of cone resistance, sleeve friction and pore pressure is continuously recorded and displayed in real time on the surface acquisition and computer control. At the end of the test the drill string is lifted to pull the cone rod free from the ground. At this point, the final baseline (zero) reading is recorded for the cone channels to monitor any drift in sensor offsets. The tool is then recovered from the drill string and returned to the drill deck.

Pore pressure measurements were taken at cone shoulder (u_2 position).

3.4.3 Remote (Field) Vane test

The remote (field) vane test is used to measures the in-situ undrained shear strength (and also remoulded strength, if required) of saturated clays and silts. The remote vane tests were performed using WIVATAP vane shear equipment of A.P. van den Berg make. Appropriate size of remote vane, which consists of 2 cross- blades at 90^o to each other with height to width ratio of 2, is selected based on the expected undrained shear strength of the soil. In this investigation, two vanes with sizes 130 mm x 65 mm x 2 mm, 130 mm x 50 mm x 2 mm thickness were used with a rotation rate of 0.2^o per second for intact undrained shear strength. For remoulded tests the rotation speed used is 1 degree per second (a faster speed) for a required number of rotations, then the test is run into slow speed till the torque line gets constant to measure remolded shear strength.

The 1-meter WIP tool accompanying the vane equipment is used to move the driving unit and the vanes down into the borehole. After taking zero reading, the vane is inserted into undisturbed soil and rotated at the specified rotation speed. The torque is measured to shear a cylindrical surface with the vane. The torque and angle are registered by the data acquisition system in real time through the connecting cables. After the soil has been sheared the equipment is switched to fast rotation speed turning for a required number of rotations, then the test is run into slow speed to measure remolded shear strength. After the test is completed, the drill string is lifted for more than one meter to be sure that the vane is out of the soil and zero readings after the test are taken. The tool is then recovered from the drill string and returned to drill deck.

The undrained shear strength, $s_{u,fv}$ is calculated from the following:

$$s_{u,fv} = 0.273 * (M_{max} / D^3)$$

where M_{max} represents maximum torque and D is the width of the cross-blades (diameter of the sheared cylindrical surface).

4.0 LABORATORY TESTING

4.1 General

The laboratory testing was performed in two stages –

- Laboratory testing offshore - performed by M/s. Coastal Marine Construction and Engineering Company (COMACOE).
- Laboratory testing onshore - performed by IEOT, ONGC.

Sections 4.2 and 4.3 describe the offshore laboratory testing and onshore laboratory testing respectively. The detailed description of the soil conditions based on the results of laboratory testing and CPTU are presented in Section 5.0 of this report.

4.2 Laboratory Testing Offshore

Samples recovered were extruded from the sample tubes, inspected and subjected to testing in the offshore laboratory onboard GTV "Samudra Sarvekshak". The tests performed were:

- Visual Classification
- Water Content
- Wet Unit Weight
- Undrained Shear Strength by Torvane, motorvane and Pocket Penetrometer
- Unconsolidated Undrained (UU) Triaxial Test

Parts of samples were sealed in plastic bags and sections of undisturbed samples of cohesive soils were waxed into cartons for further testing in onshore laboratory.

4.3 Laboratory Testing Onshore

An onshore laboratory testing programme was planned and performed on the samples available to confirm and supplement the findings of the offshore investigation. The tests are performed as per relevant Indian Standards. A summary of onshore laboratory testing programme is given in Table 2 below.

Table 2: Onshore Laboratory Tests

Tests	No. of Tests
Grain Size Distribution	15
Atterberg Limits	05
Carbonate Content	31
UU (Triaxial)	02

A summary of all offshore and onshore laboratory test results is given in Appendix B on Plates B1 to B2. The Field (Remote) Vane Test results are given in Plate B3. The Grain Size Distribution Curves are presented on Plates GS-01 to GS-15 for hydrometer tests and sieve analysis in Appendix B. The plots of Unconsolidated Undrained Triaxial Tests (UU) are presented in Appendix B on Plates LAB-UU-01 and LAB-UU-02 (02 onshore laboratory tests) and UU-1 to UU-2 (5 onboard tests). Photographs of selected samples have been included at the end of Appendix B.

5.0 SOIL CONDITIONS

5.1 General

The soil conditions at the location have been interpreted up to the investigated depth of 125.03 m from the CPTU test results, Remote Vane test results, on board test results and onshore laboratory test results.

A summary of laboratory tests is contained in Appendix B. Appendix C contains the summary of soil conditions. Detailed boring logs, including range of undrained shear strength (s_u) interpreted from CPTU results, are shown in Appendix D.

Soil classification and description is based on the laboratory classification test results, profiles of measured CPTU parameters and interpretation of CPTU data as per classification systems proposed by Robertson (1990) and Robertson and Wride (1998). Variation of Soil Behaviour Type (SBT) Index with depth is presented in

Plates C-1-1 and C-1-2 (Refer Appendix E for more details).

Section 5.2 gives layer-wise soil profile in a tabular form. The notations in parentheses following the colour description are according to the Standard soil colour chart. Section 5.3 contains a detailed description of the soil profile. The design soil parameters are tabulated in Plate 1 of Illustrations.

5.2 Stratigraphy

Soil profile of the boring GK-28-10 (PLATFORM) is described in Table 3 below:

Table 3: Description of soil layers

Sl. No:	Depth (m)	Soil Description
1	0.0 m – 4.5 m	Calcareous CLAY, silty, very soft, highly plastic, grayish olive (5Y-4/2) with occasional silt pockets at 4.0 m depth and occasional shell fragments.
2	4.5 m – 21.5 m	Calcareous CLAY, silty, very soft to firm, highly plastic, grayish olive (5Y-4/2) with occasional shell fragments and few sand pockets at 21.0 m depth.
3	21.5 m – 30.0 m	Carbonate SAND, medium to coarse grained, dense to very dense, pale yellow (5Y-8/4) moderately to well cemented with lots of shell fragments.
4	30.0 m – 45.3 m	Calcareous silica SAND, fine to medium grained, dense to very dense, olive (5Y-6/6) moderately to well cemented with lots of shell fragments. Traces of clay pockets at 34.0 m depth.
5	45.3 m – 52.0 m	Calcareous overconsolidated CLAY, hard, low to medium plastic, gray (5Y-5/1). Layer is very silty, sandy towards bottom.
6	52.0 m – 69.0 m	Calcareous silica SAND, fine to medium grained, dense, gray (7.5Y-4/1) with occasional shell fragments. Layer is silty, clayey around 66.0 m depth.
7	69.0 m – 71.0 m	Overconsolidated CLAY, silty, hard, low to medium plastic, gray (7.5Y-5/1).
8	71.0 m – 125.03 m	Calcareous silica SAND, fine to medium grained, dense, grayish olive (5Y-4/2) with few pockets and partings of clay and occasional shell fragments.

5.3 Soil properties and design parameters

The soil profile at the location GK-28-10(PLATFORM) is clayey from seabed to 21.5 m and predominantly sandy till the investigated depth of 125.03 m below seafloor. Sections 5.3.1, 5.3.2 and 5.3.3 present the soil conditions in more details for clays, sands and silts respectively.

Plates C-2, C-3 and C-4 of Appendix C present the variation of water content, unit weight and overburden pressure respectively.

Design parameters have been assigned keeping in view the requirement of design and installation of foundations related to the fixed jacket platforms with tubular steel piles. More importance is normally given to results of triaxial tests and CPTU for estimating the strength parameters of clays. In case of sands, beside the interpreted parameters from CPTU, the carbonate content, inclusion of fine grained soil and cementation are taken into consideration in assigning the design parameters. The Design Parameters are presented in Plate 1 of Illustrations.

Visual observation of colour and reaction to HCl are also considered while classifying the soils. It is to be mentioned that the bag samples of clay used for the carbonate content determinations generally contain sand/silt pockets, partings etc. When samples are not recovered at certain depths, properties of layers have been assigned at such depths based on other available data such as CPTU and drilling log. If variation is observed in the depth of strata in the exclusive CPTU log and samples from alternate borehole, precedence is generally given to CPTU log while finalizing the depth of layers.

Detailed soil boring logs are presented on Plates D–1 to D–13 in Appendix D. Drilling log is presented on Plate D–14 followed by CPTU plots.

5.3.1 Clays

A very soft silty clay layer of 4.5 m thick is observed in the soil profile starting at the seafloor followed by a thick very soft to firm silty clay layer extending up to 21.5 m below seabed. A very hard over-consolidated clay layer is encountered from depths of 45.3 m and 52.0 m depth. Another thin, very hard over-consolidated clay layer exists from 69.0 m to 71.0 m depth.

The clay layers show increase in strength or constant strength with increase in depth.

Based on the results of carbonate content tests conducted on clay samples the clays have been classified as calcareous clays as per modified Clark and Walker (1977) classification system.

Plasticity chart (Plate C-5) shows the clays to be mostly highly plastic. The Activity Chart (Plate C-6) shows most of the clays to be inactive. The variation of

liquidity index, normalized undrained shear strength and sensitivity with depth is also presented graphically in Appendix C in Plates C-7, C-8 and C-9 respectively. The liquidity Index shows a trend of decrease with depth. OCR interpreted from CPTU results (refer Appendix E for details) is presented on Plates C-10-1 and C-10-2.

The clays have been examined on Nagaraj's intrinsic properties framework (refer Appendix E for details). Plate C-11 presents the natural state of clays in Nagaraj space and shows that the clays generally are normally consolidated or slightly overconsolidated.

A lot of scatter is observed in the measured sensitivity values (Plate C-9). Sensitivity value of 3.0 is recommended for the top clay layers.

The cone penetrometer (CPTU) results were used for interpretation of undrained shear strengths of clay layers. The raw CPTU data received from Samudra Sarvekshak were first corrected to bring all down-the-hole test results to the same reference i.e. seabed. In addition, the data were corrected for pore pressure effect i.e. for unequal end areas of the cone (Lunne et al, 1997). After the correction, undrained shear strength, s_u has been interpreted using the corrected cone resistance, q_t from the expression:

$$s_u = (q_t - \sigma_{v0})/N_{kt}$$

where σ_{v0} is total overburden pressure and N_{kt} is cone factor (Lunne et al, 1997). N_{kt} range used is 15 to 20 for upper and lower bound interpretation respectively.

The design shear strength profile has been derived based on the results of various laboratory tests with more weightage on UU tests, Remote Vane tests and interpretation from CPTU.

5.3.2 Sands

First sand layer is observed between depths of 21.5 m and 30.0 m, which is dense to very dense. This sand layer is observed to be moderately to well cemented. Subsequently another moderately to well cemented dense to very dense sand layer is encountered from 30.0 m to 45.3 m depth below seabed. The next sand layer is dense between depths of 52.0 m and 69.0 m. The last sand layer is very thick and dense starting from 71.0 m till the final investigated depth of 125.03 m.

Carbonate content, measured on the available samples of sand layers, indicates sand layers to be silica sands, calcareous silica sands or carbonate sand as per modified Clark and Walker (1977) classification system. Shell fragments are normally present in the sand layers in varying quantities. The presence of carbonate, cementation and shell fragments influence their engineering behaviour. Layers, where samples were not recovered, CPTU and drilling data were used for interpretation.

The design friction angles of the sands were evaluated from the estimated densities and soil type. The relative densities of sand layers were estimated based on the correlation proposed by Jamiolkowski et al., 2001 (refer Appendix E for details) based on cone resistance and are presented on Plates C-10-1 and C-10-2. However this correlation is based on silica sands and the nature of calcareous/carbonate sands - lower crushing strength, higher compressibility and arching effect causing low residual radial stresses after pile driving - has been given due consideration while selecting design friction angles.

While prescribing end bearing values, approaches proposed in API (2007) for direct pile design methods based on CPT (IEOT, 2012b) have also been examined with due consideration given to cementation and calcareous nature of sands.

The design friction angles vary from 30 to 35 degrees for the sand layers. Limiting skin friction of 20 to 80 kPa is recommended for the sand layers. The limiting unit end bearing values were selected on the basis of relative density, degree of cementation and cone resistance. The recommended limiting end bearing values for sand layers vary from 5.0 to 8.0 MPa.

End bearing resistance of sands may not be uniform within a layer especially in layers which are relatively thick and/or have cementation. Considering lateral soil variability and variability in extent and strength of cementation, it is recommended that for drivability analysis unit end bearing value of 7.0 MPa be considered for sand layer nos. 3, 4 and 6.

5.3.3 Silts

No silt layers are observed in the soil profile up to the investigated depth of 125.03 m.

6.0 ENGINEERING ANALYSIS

6.1 General

This section reports the results of the following engineering analyses:

- Axial pile capacities for open ended steel pipe piles of 60, 66, 72 and 84 inch OD and 2 inch wall thickness
- Axial load-deformation data for open ended steel pipe piles of 60 inch OD and 2 inch wall thickness (*t-z* and *q-z* data)
- Lateral load-deformation data for open ended steel pipe piles of 60 inch OD and 2 inch wall thickness (*p-y* data)
- Mudmat bearing capacity

- Jack-up leg penetration analysis

Plates 1 to 15, which summarize the soil design parameters and the results of engineering analysis, are presented immediately after the text.

6.2 Axial Pile Capacity

Axial pile capacities were calculated after an evaluation of the suitability of various design methods for the type of soils that occur at the location. The API criteria were selected as the general design method with the following modifications and restrictions:

In sands: Ultimate axial bearing capacity in sands is calculated based on API RP2A - 1993 using k=0.7 in compression and k=0.5 in tension. Also, limiting unit skin friction and unit end bearing values as given in the design parameters on Plate nos. 1 and 2 are used in engineering analysis.

In clays: Ultimate axial bearing capacity in clays is calculated based on API RP2A - 1986 method 2 with the following modifications:

For normally consolidated clays with $Su/p_o' < 0.45$, α method (Tomlinson, 1977) has been used with $\alpha=0.7$. For over consolidate clays with $Su/p_o' > 0.45$, API RP 2A 1986 Method 2 has been used.

Unit skin friction on the inside of the pile is considered same as the unit skin friction on the outside of the pile. Plug length to pile penetration ratio has been assumed to be 0.9 for calculation of inner skin friction.

For evaluating axial pile capacity in tension (pull out capacity), only the outer skin friction has been considered.

Self weight of pile and the weight of soil plug inside the pile have not been considered in the calculation of ultimate axial capacity both in tension and compression which need to be taken into consideration during design.

In calculation of axial pile capacity in compression, transition from weak to strong and strong to weak layers has been accounted for. The transition from weak clay layer to a strong clay layer and vice versa is accounted for by linearly increasing N_c value from 6 at the interface of the layers to 9 at a depth of 3 pile diameter. Similarly, in case of a strong sand layer, transition is accounted for by linearly increasing the limiting end bearing value from q_{lim} (weak layer) at the interface of the layers to q_{lim} (strong layer) at a depth of 3 pile diameters.

Local scour has been considered in the analysis for a depth of 1.5 times the external diameter of pile in the absence of the local experience in the area.

Unit values for skin friction and end bearing as derived from the design parameters are presented in graphical form on Plate nos. 2 and 3 respectively. The ultimate

capacity in tension and compression of a 60" dia. and 2" thick open-ended pipe pile is presented on Plate no.4. Ultimate axial capacity curves for 66, 72 and 84 inch piles are presented in Plate nos. 5, 6 and 7 respectively. Typical ultimate axial capacity in compression of 35.0 MN is achieved at a penetration depth of about 75.5 m for an open ended steel tubular pile with outer diameter of 60 inches and wall thickness 2 inches. The capacity is governed by plugged behaviour.

Soil condition below the depth of 125.03 m is not known. Therefore, depth of maximum pile penetration should be restricted to keep the pile tip at least 3 times the pile diameter above 125.03 m to avoid any chance of reduction of axial capacity due to presence of any weaker soil layer below.

6.3 Axial Load-Deformation Data (*t-z* and *q-z* data)

In this report, skin friction *vs* deflection (*t-z*) data has been schematized as curves of five points. It is assumed that the residual ultimate skin friction is reached at 1% of the pile diameter (Vijayvergia, 1977).The ultimate values are consistent with the unit skin friction values given on Plate No.2. The *t-z* data are presented on Plate No. 8.

The pile tip load vs. deflection (*q-z*) data is schematized as curves of five points. The *q-z* data has been derived based on load test data (Beringen, 1979 and Vijayvergia, 1977) and correlation with laboratory tests on calcareous/carbonate soils. The end bearing is developed at much larger deflection than outer skin friction. The ultimate values are consistent with the unit end bearing values given on Plate no. 3. Transition from weak to strong and strong to weak layers has been accounted for in the analysis. The *q-z* data have been given for potential bearing layers below 50.0 m to the end of the borehole. The *q-z* data are presented on Plate Nos. 9 to 11.

6.4 Lateral Load-Deformation Data (*p-y* data)

Bending moments and deflections induced in laterally loaded piles are usually evaluated numerically with techniques similar to those for axial pile analyses. Input information for such spring supported beam-column models includes curves of lateral soil pressure per unit pile length versus lateral deflection (*p-y*) data.

The design parameters used in the analysis are initial subgrade reaction moduli *k*, for sands and strain at 50% of stress at failure,ε_{50}, for clays, which is presented on Plate No. 1. Design ε_{50} values of 0.5% to 1.5 % were selected for clays from an evaluation of the soil properties and design recommendations for evaluation of *p-y* data. The design *k* values range from 16.6 MN/m^3 to 34.6 MN/m^3 for the sand layers.

The *p-y* data is generated based on Matlock (1970) approach for soft clays and Reese et al (1975) approach for stiff clays. The transition undrained shear strength level between soft and stiff clays is considered as 96.0 kPa as per API recommendation.

For sands, *p-y* data is generated using O'Neil and Murchison (1983) approach. This approach has also been applied for cemented soil layers. The initial slope of the *p-y*

curve is very important and is a function of *k*, initial modulus of subgrade reaction. A check is applied on the first point (*p1, y1*) to ensure that the computed *k* value is within 5% of the design *k* value. Also, *p1*, *p2*, *p3* and *p4* values are limited to a value of limiting pile end bearing of the concerned layer times the pile outer diameter.

The *p-y* data up to 50.3 m depth, generated for 60-inch OD piles, are presented on Plate Nos. 12 and 13.

6.5 Mud-mat bearing capacity

Mudmat ultimate bearing capacities for purely vertical and concentric loading have been computed for square, rectangular and triangular mudmats without considering any factor of safety and assuming that penetration of mudmat into the sea floor does not occur. Mudmat bearing capacity varies with configuration of mudmat as well as design forces and moments applied on the mudmat. As such, it is recommended that final bearing capacity for design should be based on the design loads and actual geometry of the mudmat. Required factors of safety should be applied on soil resistance for calculation of stability of the jacket for installation.

In the location GK-28-10 (PLATFORM), soil condition near the seafloor is very soft clay gradually increasing in strength as the depth below seafloor increases. The ultimate bearing capacity for purely vertical and concentric load is plotted versus side B for square mud-mats, side B for triangular mudmats of isosceles right-angular shapes and with smaller side W for rectangular mudmats, (L/W=2). Ultimate bearing capacity for mudmat has been calculated following current API RP 2GEO. The recommended ultimate bearing capacity is shown on Plate No.14 in graphical form.

For installation of jacket with jacket leg extension below the level of mudmat, it is recommended that rational analysis be performed to check that the desired penetration is achievable.

6.6 Jack up leg penetration analysis

Analysis for penetration of the spudcan of a jack up rig was carried out for the location GK-28-10 (PLATFORM). Standard ONGC spudcan with a diameter of 14.0 m and maximum preload of 45 MN was considered for the analysis. The bearing capacity curve for this spudcan is presented in Plate No. 15. The bearing capacity at any depth along the bearing capacity curve corresponds to the depth of the bottom of the spudcan below the seafloor. The analysis does not account for the lateral variability of the soil strata.

Soil condition near the seafloor is very soft clay gradually increasing in strength as the depth below seafloor increases. Clayey soil exists below the seafloor up to the depth of 21.5 m. Below 21.5 m, sandy soil exists up to 45.3 m. The penetration of the analysed spudcan is estimated to be about 19.5 m.

Rig-specific analysis should be carried out for assessment of spudcan penetration

and suitability for individual rigs.

7.0 CONCLUSIONS AND RECOMMENDATIONS

The soil profile at the location GK-28-10 (PLATFORM) is clayey from seabed to 21.50 m and predominantly sandy till the investigated depth of 125.03 m below seafloor. Moderate to strong cementation is observed in some of the sand layers.

The typical ultimate compressive capacity of 35.0 MN is achieved at a penetration depth of about 75.5 m below seafloor for an open ended steel pipe pile of 60 inch outer diameter and 2 inches wall thickness. The ultimate axial pile capacity is governed by plugged behavior. Ultimate axial capacity curves are also presented for piles of outer diameter of 66, 72 and 84 inches. Soil condition below the depth of 125.03 m is not known. Therefore, depth of maximum pile penetration should be restricted to keep the pile tip at least 3 times the pile diameter above 125.03 m to avoid any chance of reduction of axial capacity due to presence of any weaker soil layer below.

Axial and lateral load-deformation data are also presented in the report for an open ended steel pipe pile of 60 inch outer diameter and 2 inch wall thickness.

It is recommended to carry out detailed drivability studies giving due consideration to likely set up effect in clays.

In the location GK-28-10 (PLATFORM), soil condition near the seafloor is very soft clay gradually increasing in strength as the depth below seafloor increases. The ultimate bearing capacity for purely vertical and concentric load is plotted versus side B for square mud-mats, side B for triangular mudmats of isosceles right-angular shapes and with smaller side W for rectangular mudmats, (L/W=2). Ultimate bearing capacity for mudmat has been calculated following current API RP 2GEO. In case penetration of jacket leg extensions in to the seabed soil is envisaged, it is recommended that rational analysis be performed to check that the desired penetration is achievable.

The analysis for the standard ONGC spudcan (flat bottom) indicates that the spudcan bottom with maximum preload (45 MN) is expected to stop at about 19.5 m below the seafloor.

Rig specific analysis should be carried out for assessment of suitability and leg penetration depths for individual rigs.

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ILLUSTRATIONS

ILLUSTRATIONS

	Plate
Design Soil Parameters	1
Ultimate Unit Skin Friction and End Bearing	2 - 3
Ultimate Bearing Capacity Curves for 60" OD Pile	4
Ultimate Bearing Capacity Curves for 66" OD Pile	5
Ultimate Bearing Capacity Curves for 72" OD Pile	6
Ultimate Bearing Capacity Curves for 84"OD Pile	7
t-z Data for 60" inch OD Pile	8
q-z Data for 60" inch OD Pile	9 - 11
p-y Data for 60" inch OD Pile	12 - 13
Ultimate Mudmat Bearing Capacity	14
Jack -up Footing Penetration Curve	15

DESIGN PARAMETERS

BORING **GK-28-10 (PLATFORM)** FIELD **GULF OF KUTCH**

Layer Number	Depth m	Soil Type	Plast. OR Density	PHI (Φ) [deg]	s_u kPa	Eff. Unit weight kN/m³	f_{lim} kPa	q_{lim} MPa	ε_{50} %	k MN/m³
1	0.00 4.50	CLAY	hp		2 7	6.0			1.5	
2	4.50 21.50	CLAY	hp		7 35	7.0			1.5	
3	21.50 30.00	SAND	d	35		10.5	20	5		34.6
4	30.00 45.30	SAND	d	35		10.5	20	6		34.6
5	45.30 52.00	CLAY	lp-mp		475 600	10.5			0.5	
6	52.00 69.00	SAND	d	35		10.8	60	6		34.6
7	69.00 71.00	CLAY	lp-mp		450	10.8			0.5	
8	71.00 125.03	SAND	d	30		10.5	80	8		16.6

NOTE: For Pile Driveability Analysis, unit end bearing of 7 MPa should be considered for sands layer no. 3, 4, and 6.

k = rate of increase with depth of initial modulus of subgrade reaction for lateral soil resistance

CALC./SANDST. = Calcarenite/Sandstone

PLAST. = lp/mp/hp : low /medium/ high plasticity

DENSITY = l / m / d : loose / medium / dense sand

ε_{50} = strain at 50% failure stress

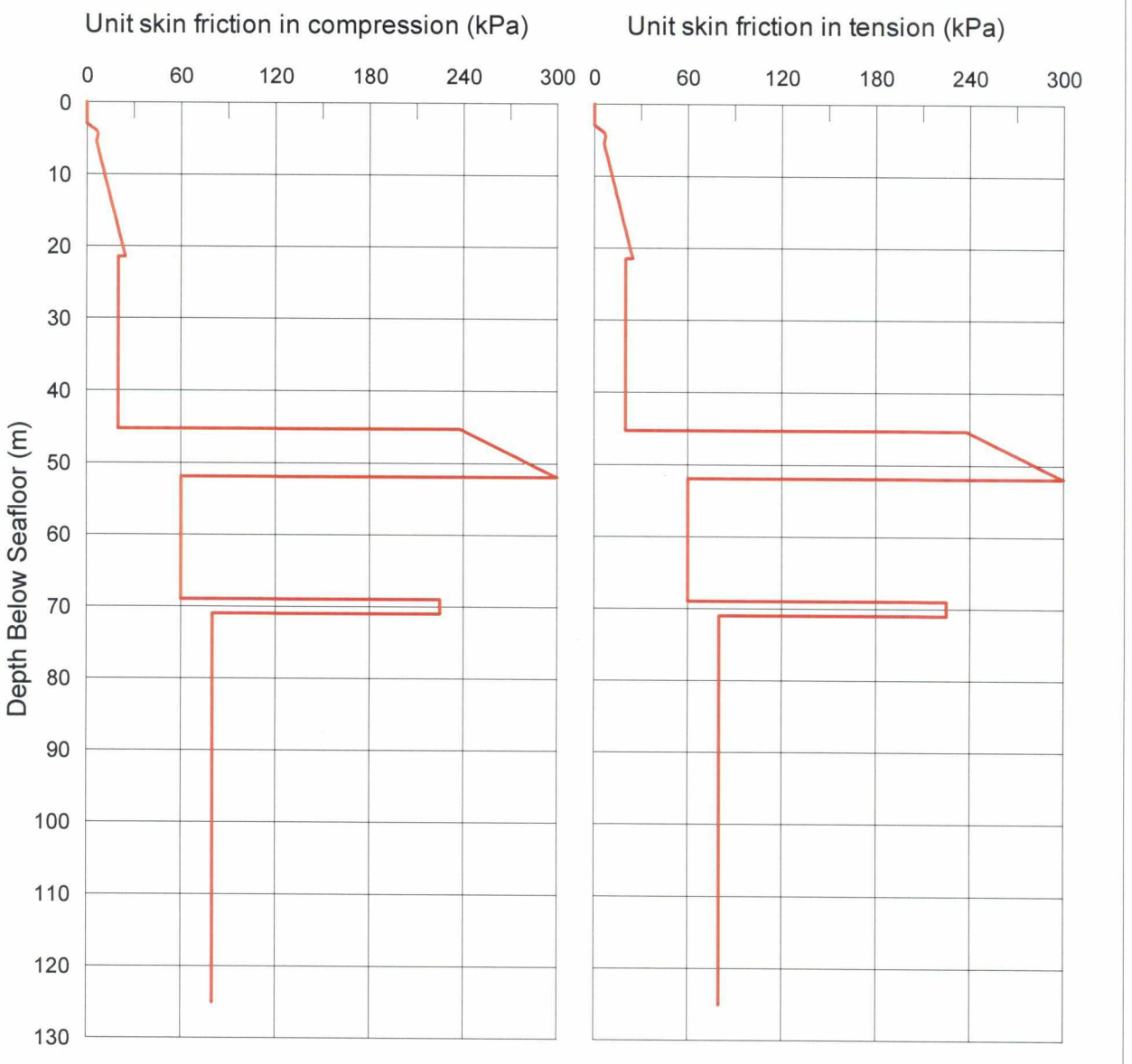
q_{lim} = limit unit end bearing pressure

f_{lim} = limit unit skin friction

s_U = undrained shear strength

ϕ = effective angle of internal friction

Report No.	IEOT/GEO/07/2018-19		
Boring	GK-28-10 (PLATFORM)	IEOT, ONGC	
Plate No.	1	Date	13.08.2018
Prep. By	Balram Nayak	Chkd. By	T.K.Roy



UNIT SKIN FRICTION VALUES		PLATE	2
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ NO :IEOT/GEO/07/2018-19 REP NO :IEOT/GEO/07/2018-19 LOCATION :**GK-28-10 (PLATFORM)** FIELD :**Gulf of Kutch**	DATE	17/08/2018
		PREPARED BY	(T.K.Roy)
		CHECKED BY:	(R.Mahanta)

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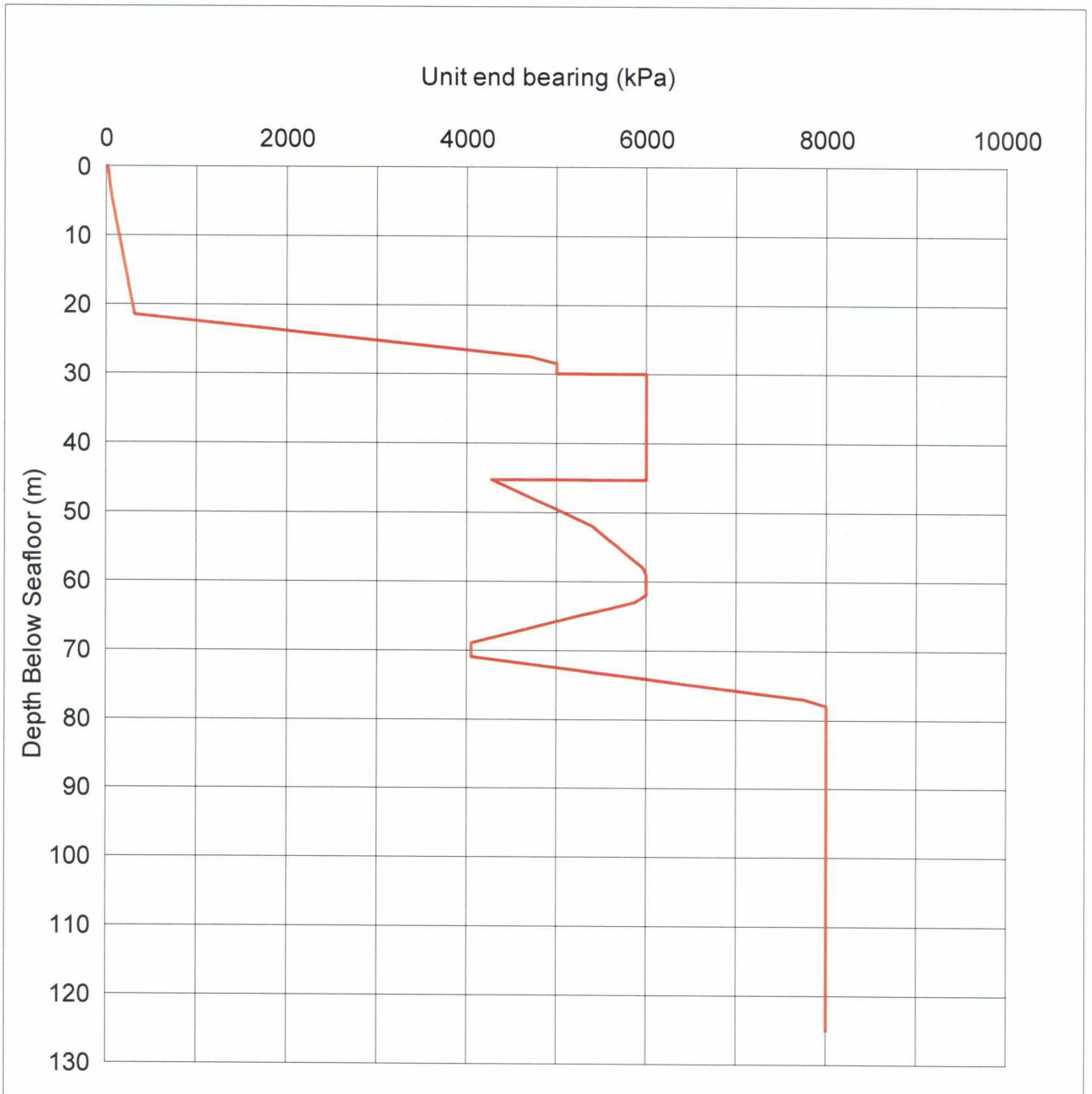

UNIT END BEARING VALUES		PLATE	3
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		PREPARED BY	(T.K.Roy)
		CHECKED BY:	(R.Mahanta)

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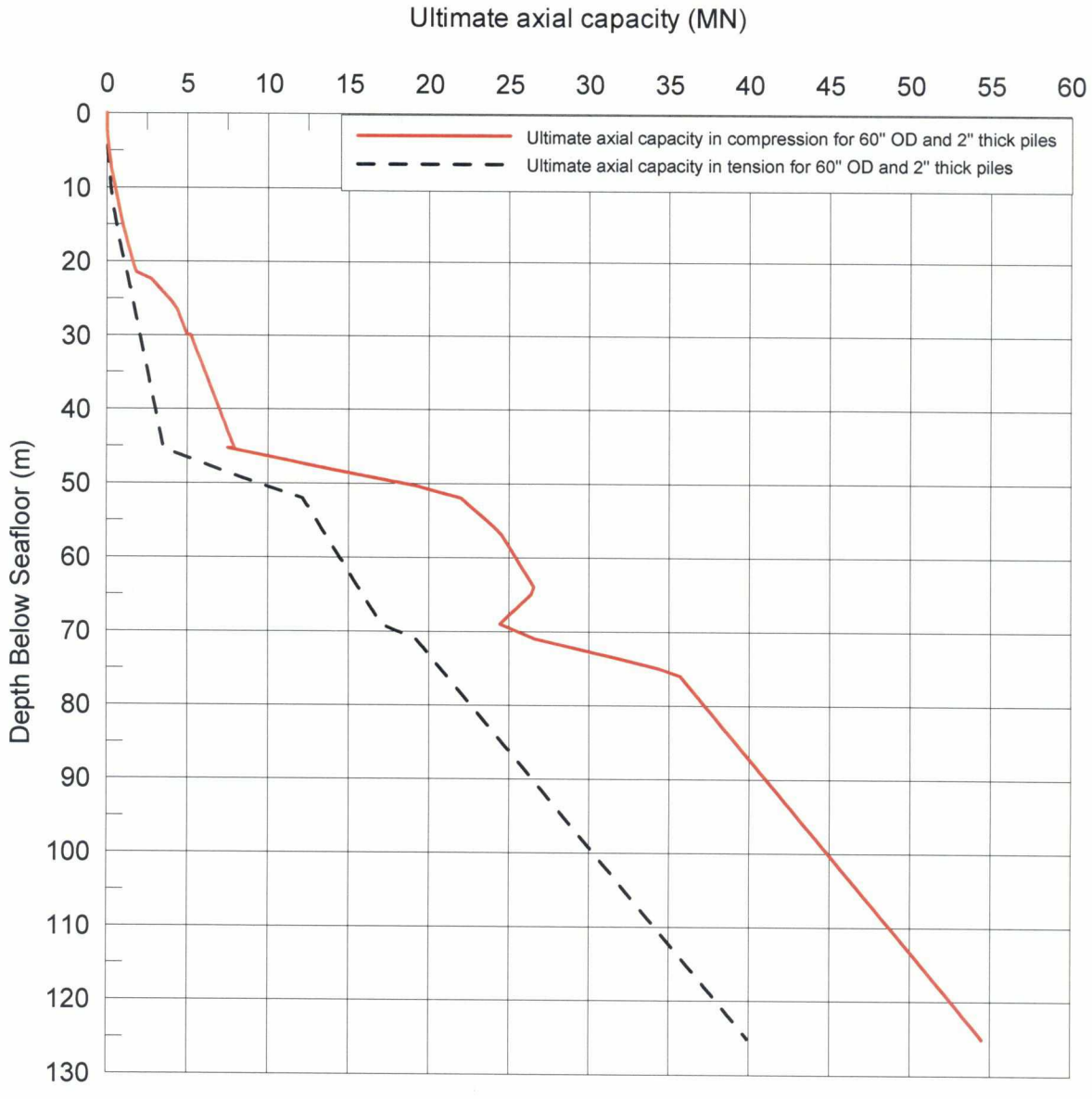
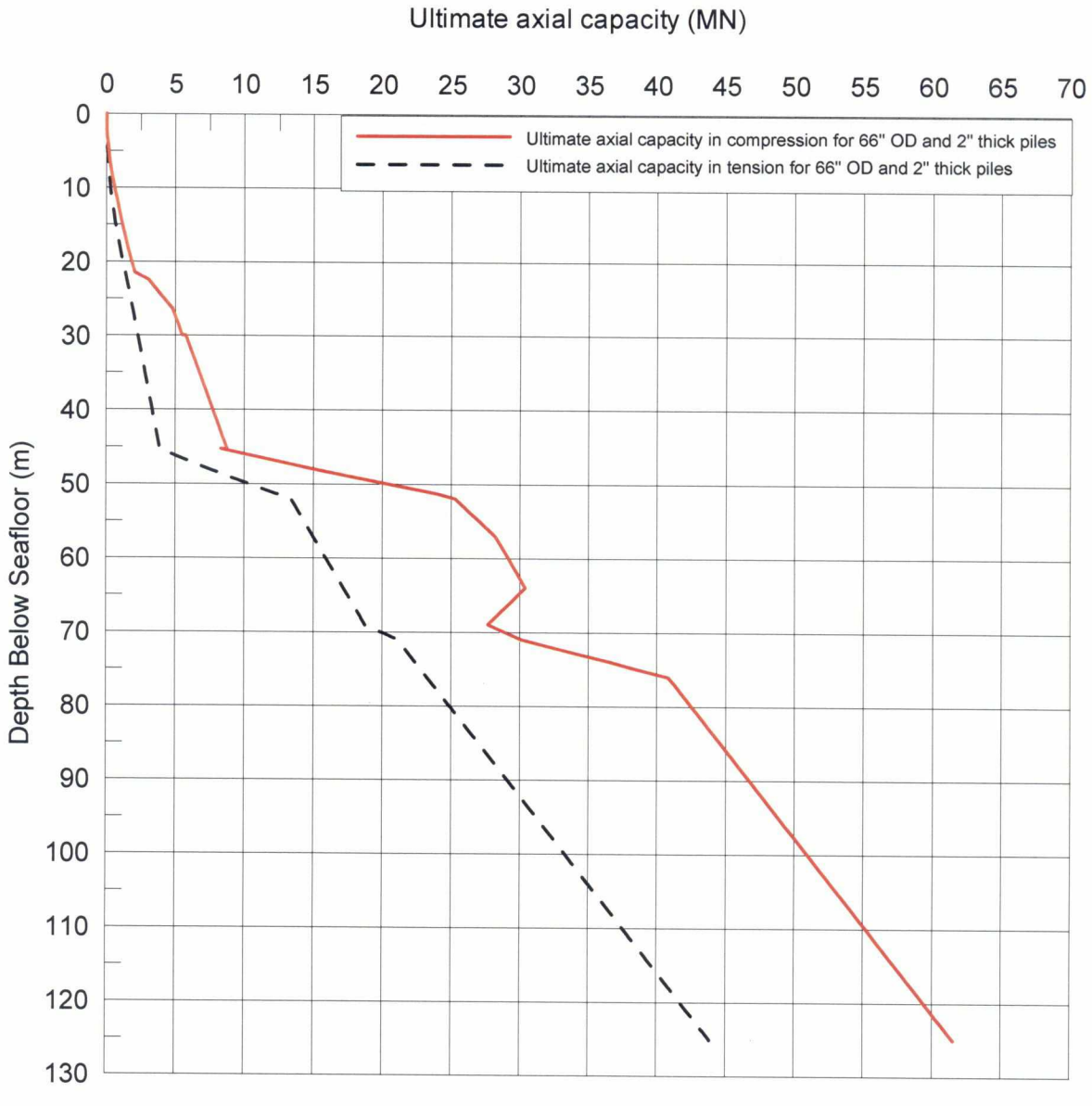
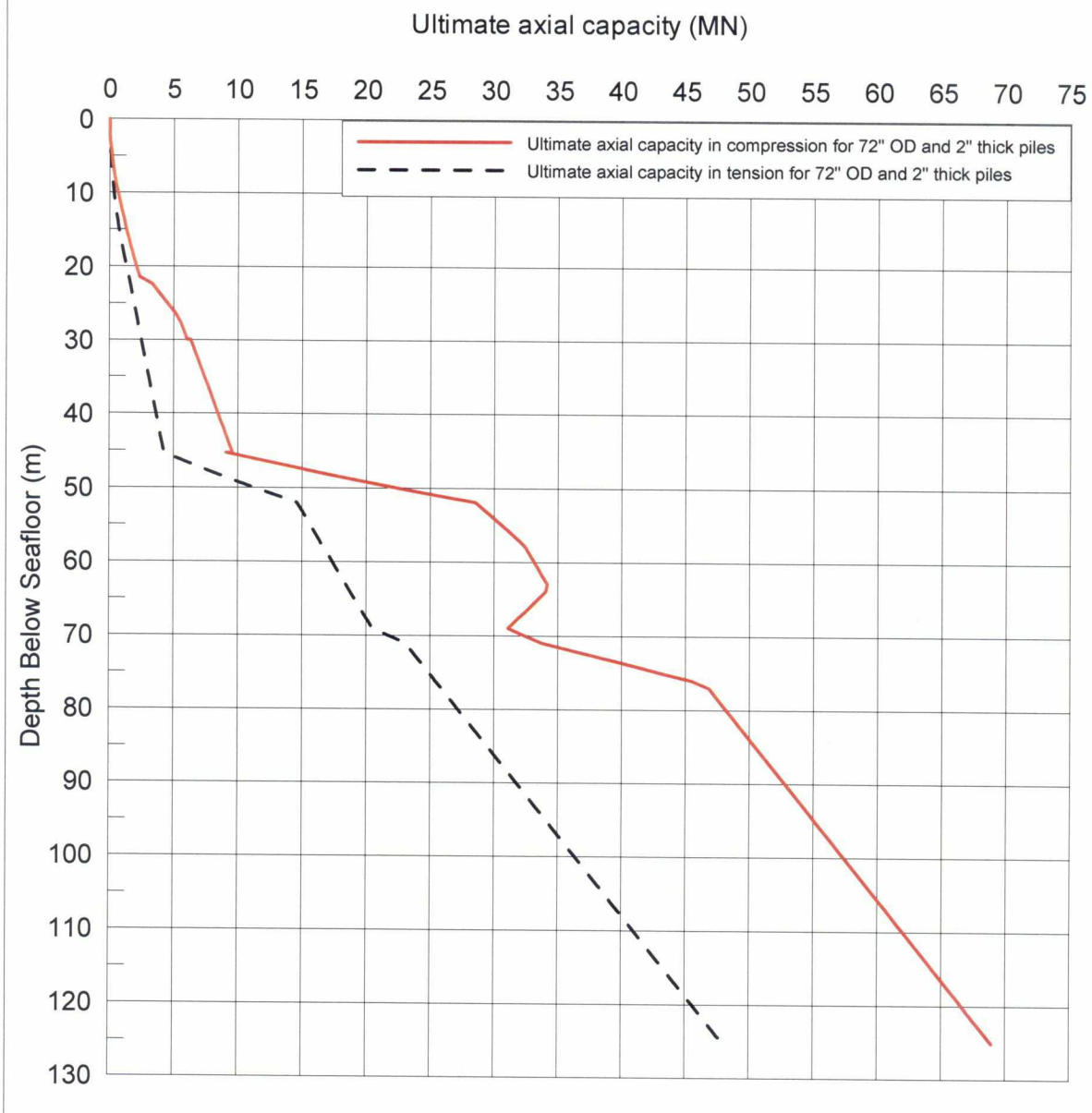


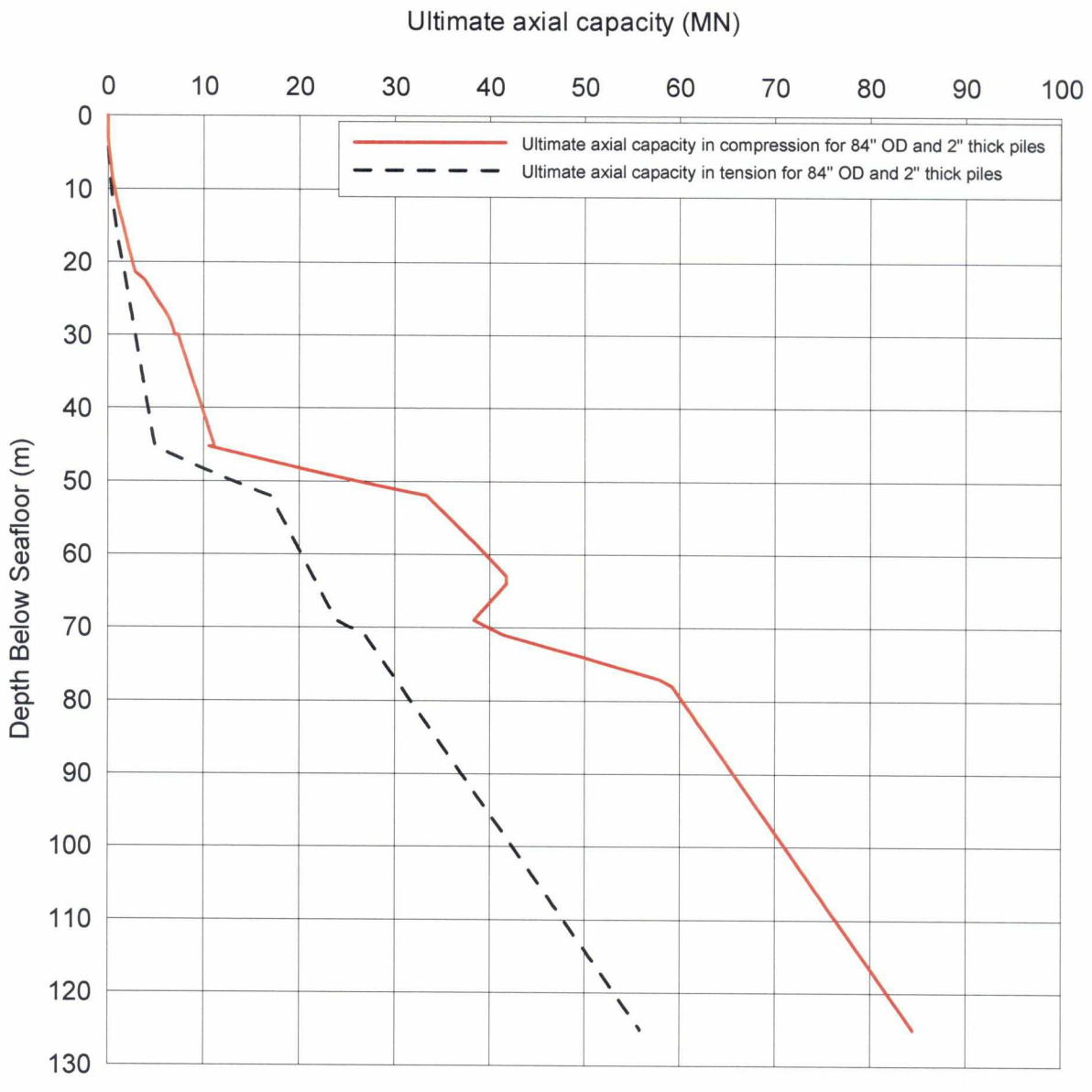
ULTIMATE AXIAL CAPACITY		PLATE	4
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ NO :IEOT/GEO/07/2018-19 REP NO :IEOT/GEO/07/2018-19 LOCATION :**GK-28-10 (PLATFORM)** FIELD :**Gulf of Kutch**	DATE	17/08/2018
		PREPARED BY	(T.K.Roy)
		CHECKED BY:	(R.Mahanta)



ULTIMATE AXIAL CAPACITY		PLATE	5
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ NO :IEOT/GEO/07/2018-19 REP NO :IEOT/GEO/07/2018-19 LOCATION :**GK-28-10 (PLATFORM)** FIELD :**Gulf of Kutch**	DATE	17/08/2018
		PREPARED BY	(T.K.Roy)
		CHECKED BY:	(R.Mahanta)



ULTIMATE AXIAL CAPACITY		PLATE	6
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ NO :IEOT/GEO/07/2018-19 REP NO :IEOT/GEO/07/2018-19 LOCATION :**GK-28-10 (PLATFORM)** FIELD :**Gulf of Kutch**	DATE	17/08/2018
		PREPARED BY	(T.K.Roy)
		CHECKED BY:	(R.Mahanta)



ULTIMATE AXIAL CAPACITY		PLATE	7
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ NO :IEOT/GEO/07/2018-19 REP NO :IEOT/GEO/07/2018-19 LOCATION :**GK-28-10 (PLATFORM)** FIELD :**Gulf of Kutch**	DATE	17/08/2018
		PREPARED BY	(T.K.Roy)
		CHECKED BY:	(R.Mahanta)

T-Z DATA **LOCATION: GK-28-10 (PLATFORM), Gulf of Kutch**

PILE DIAMETER = 1.524 m
WALL THICKNESS = 0.051 m
PLUG LENGTH : PILE PENETRATION RATIO = 0.90
CLAYS : API 1986 METHOD 2 / TOMLINSON (1977) METHOD
SANDS : API 1993 METHOD (K = 0.7 compression, 0.5 tension)
t-z DATA : VIJAYVERGIYA (1977) METHOD

LOCAL SCOUR DEPTH = 2.286 m
OVERBURDEN REDUCTION ZONE = 9.144 m
t/c = Tension/compression
t1/t5 = Mobilized skin friction (MN/m)
z1/z5 = Pile-soil displacement (mm)

Depth m	Soil type	t/c	t1	z1	t2	z2	t3	z3	t4	z4	t5	z5
0.00	CLAY	c	0.000	1.9	0.000	3.8	0.000	7.6	0.000	11.4	0.000	15.2
		t	0.000	1.9	0.000	3.8	0.000	7.6	0.000	11.4	0.000	15.2
4.50	CLAY	c	0.014	1.9	0.018	3.8	0.021	7.6	0.023	11.4	0.023	15.2
		t	0.014	1.9	0.018	3.8	0.021	7.6	0.023	11.4	0.023	15.2
4.50	CLAY	c	0.014	1.9	0.018	3.8	0.021	7.6	0.023	11.4	0.023	15.2
		t	0.014	1.9	0.018	3.8	0.021	7.6	0.023	11.4	0.023	15.2
21.50	CLAY	c	0.068	1.9	0.088	3.8	0.107	7.6	0.115	11.4	0.117	15.2
		t	0.068	1.9	0.088	3.8	0.107	7.6	0.115	11.4	0.117	15.2
21.50	SAND	c	0.056	1.9	0.072	3.8	0.088	7.6	0.094	11.4	0.096	15.2
		t	0.056	1.9	0.072	3.8	0.088	7.6	0.094	11.4	0.096	15.2
30.00	SAND	c	0.056	1.9	0.072	3.8	0.088	7.6	0.094	11.4	0.096	15.2
		t	0.056	1.9	0.072	3.8	0.088	7.6	0.094	11.4	0.096	15.2
30.00	SAND	c	0.056	1.9	0.072	3.8	0.088	7.6	0.094	11.4	0.096	15.2
		t	0.056	1.9	0.072	3.8	0.088	7.6	0.094	11.4	0.096	15.2
45.30	SAND	c	0.056	1.9	0.072	3.8	0.088	7.6	0.094	11.4	0.096	15.2
		t	0.056	1.9	0.072	3.8	0.088	7.6	0.094	11.4	0.096	15.2
45.30	CLAY	c	0.662	1.9	0.853	3.8	1.040	7.6	1.117	11.4	1.137	15.2
		t	0.662	1.9	0.853	3.8	1.040	7.6	1.117	11.4	1.137	15.2
52.00	CLAY	c	0.836	1.9	1.077	3.8	1.313	7.6	1.411	11.4	1.436	15.2
		t	0.836	1.9	1.077	3.8	1.313	7.6	1.411	11.4	1.436	15.2
52.00	SAND	c	0.167	1.9	0.215	3.8	0.263	7.6	0.282	11.4	0.287	15.2
		t	0.167	1.9	0.215	3.8	0.263	7.6	0.282	11.4	0.287	15.2
69.00	SAND	c	0.167	1.9	0.215	3.8	0.263	7.6	0.282	11.4	0.287	15.2
		t	0.167	1.9	0.215	3.8	0.263	7.6	0.282	11.4	0.287	15.2
69.00	CLAY	c	0.627	1.9	0.808	3.8	0.985	7.6	1.058	11.4	1.077	15.2
		t	0.627	1.9	0.808	3.8	0.985	7.6	1.058	11.4	1.077	15.2
71.00	CLAY	c	0.627	1.9	0.808	3.8	0.985	7.6	1.058	11.4	1.077	15.2
		t	0.627	1.9	0.808	3.8	0.985	7.6	1.058	11.4	1.077	15.2
71.00	SAND	c	0.223	1.9	0.287	3.8	0.350	7.6	0.376	11.4	0.383	15.2
		t	0.223	1.9	0.287	3.8	0.350	7.6	0.376	11.4	0.383	15.2
125.03	SAND	c	0.223	1.9	0.287	3.8	0.350	7.6	0.376	11.4	0.383	15.2
		t	0.223	1.9	0.287	3.8	0.350	7.6	0.376	11.4	0.383	15.2

INST. OF ENGG. & OCEAN TECH. ONGC	PLATE NO. 8
PREP. BY (T.K.Roy) CHKD. BY (R.Mahanta)	DATE 15/08/2018

Q-Z DATA

LOCATION: GK-28-10 **(PLATFORM)** , Gulf of Kutch

PILE DIAMETER = **1.524 m**
WALL THICKNESS = 0.051 m
PLUG LENGTH : PILE PENETRATION RATIO = 0.90

CLAYS : API 1986 METHOD 2
SANDS : API 1993 METHOD (K = 0.7 compression, 0.5 tension)
q-z DATA : VIJAYVERGIYA (1977) METHOD

LOCAL SCOUR DEPTH = 2.286 m
OVERBURDEN REDUCTION ZONE = 9.144 m
Tip 0 = unplugged, 1 = plugged
q1/q5 = Mobilized end bearing (MN)
z1/z5 = Pile-soil displacement (mm)

Depth m	Soil type	tip	q1	z1	q2	z2	q3	z3	q4	z4	q5	z5
50.30	CLAY	1	5.78	15.2	7.00	30.5	8.12	61.0	8.68	91.4	9.33	152.4
51.30	CLAY	1	5.97	15.2	7.23	30.5	8.38	61.0	8.96	91.4	9.64	152.4
52.00	CLAY	1	6.11	15.2	7.39	30.5	8.57	61.0	9.16	91.4	9.85	152.4
52.00	SAND	1	6.11	15.2	7.39	30.5	8.57	61.0	9.16	91.4	9.85	152.4
53.00	SAND	1	6.26	15.2	7.57	30.5	8.78	61.0	9.38	91.4	10.09	152.4
54.00	SAND	1	6.40	15.2	7.75	30.5	8.99	61.0	9.61	91.4	10.33	152.4
55.00	SAND	1	6.55	15.2	7.93	30.5	9.19	61.0	9.83	91.4	10.57	152.4
56.00	SAND	1	6.70	15.2	8.11	30.5	9.40	61.0	10.05	91.4	10.81	152.4
57.00	SAND	1	6.79	15.2	8.21	30.5	9.52	61.0	10.18	91.4	10.94	152.4
58.00	SAND	1	6.79	15.2	8.21	30.5	9.52	61.0	10.18	91.4	10.94	152.4
59.00	SAND	1	6.79	15.2	8.21	30.5	9.52	61.0	10.18	91.4	10.94	152.4
60.00	SAND	1	6.79	15.2	8.21	30.5	9.52	61.0	10.18	91.4	10.94	152.4
61.00	SAND	1	6.79	15.2	8.21	30.5	9.52	61.0	10.18	91.4	10.94	152.4
62.00	SAND	1	6.79	15.2	8.21	30.5	9.52	61.0	10.18	91.4	10.94	152.4
63.00	SAND	1	6.79	15.2	8.21	30.5	9.52	61.0	10.18	91.4	10.94	152.4
64.00	SAND	1	6.79	15.2	8.21	30.5	9.52	61.0	10.18	91.4	10.94	152.4
65.00	SAND	1	6.51	15.2	7.87	30.5	9.13	61.0	9.76	91.4	10.50	152.4
66.00	SAND	1	6.03	15.2	7.29	30.5	8.46	61.0	9.04	91.4	9.72	152.4
67.00	SAND	1	5.55	15.2	6.71	30.5	7.78	61.0	8.32	91.4	8.94	152.4
68.00	SAND	1	5.06	15.2	6.12	30.5	7.10	61.0	7.59	91.4	8.17	152.4
69.00	SAND	1	4.58	15.2	5.54	30.5	6.43	61.0	6.87	91.4	7.39	152.4
69.00	CLAY	1	4.58	15.2	5.54	30.5	6.43	61.0	6.87	91.4	7.39	152.4
70.00	CLAY	1	4.58	15.2	5.54	30.5	6.43	61.0	6.87	91.4	7.39	152.4
71.00	CLAY	1	4.58	15.2	5.54	30.5	6.43	61.0	6.87	91.4	7.39	152.4
71.00	SAND	1	4.58	15.2	5.54	30.5	6.43	61.0	6.87	91.4	7.39	152.4
72.00	SAND	1	5.56	15.2	6.72	30.5	7.80	61.0	8.34	91.4	8.96	152.4
73.00	SAND	1	6.53	15.2	7.90	30.5	9.17	61.0	9.80	91.4	10.54	152.4
74.00	SAND	1	7.51	15.2	9.09	30.5	10.54	61.0	11.27	91.4	12.12	152.4
75.00	SAND	1	8.49	15.2	10.27	30.5	11.91	61.0	12.73	91.4	13.69	152.4

...continued

INST. OF ENGG. & OCEAN TECH. ONGC		PLATE NO. 9
PREP. BY (T.K.Roy)	CHKD. BY (R.Mahanta)	DATE 15/08/2018

Q-Z DATA

LOCATION: GK-28-10 (**PLATFORM**), Gulf of Kutch

...continued from last page

Tip 0 = unplugged, 1 = plugged
q1/q5 = Mobilized end bearing (MN)
z1/z5 = Pile-soil displacement (mm)

Depth m	Soil type	tip	q1	z1	q2	z2	q3	z3	q4	z4	q5	z5
76.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
77.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
78.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
79.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
80.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
81.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
82.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
83.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
84.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
85.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
86.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
87.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
88.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
89.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
90.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
91.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
92.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
93.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
94.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
95.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
96.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
97.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
98.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
99.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
100.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
101.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
102.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
103.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
104.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
105.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
106.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
107.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
108.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
109.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
110.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4

...continued

INST. OF ENGG. & OCEAN TECH. ONGC		PLATE NO. 10
PREP. BY (T.K.Roy)	CHKD. BY (R.Mahanta)	DATE 15/08/2018

Q-Z DATA

LOCATION: GK-28-10 (**PLATFORM**), Gulf of Kutch

...continued from last page

Tip 0 = unplugged, 1 = plugged
q1/q5 = Mobilized end bearing (MN)
z1/z5 = Pile-soil displacement (mm)

Depth m	Soil type	tip	q1	z1	q2	z2	q3	z3	q4	z4	q5	z5
111.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
112.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
113.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
114.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
115.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
116.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
117.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
118.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
119.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
120.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
121.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
122.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
123.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
124.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
125.00	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4
125.03	SAND	1	9.05	15.2	10.94	30.5	12.70	61.0	13.57	91.4	14.59	152.4

INST. OF ENGG. & OCEAN TECH. ONGC		PLATE NO. 11
PREP. BY (T.K.Roy)	CHKD. BY (R.Mahanta)	DATE 15/08/2018

P-Y DATA **LOCATION: GK-28-10 (PLATFORM), Gulf of Kutch**

PILE DIAMETER = 1.524 m

TRANSITION Su BETWEEN SOFT & STIFF CLAY = 96.00 kPa

LOCAL SCOUR DEPTH = 2.286 m
OVERBURDEN REDUCTION ZONE = 9.144 m

CLAYS : Soft - Matlock (1970)
Stiff - Reese et al (1975)
SANDS : O`Neil and Murchison (1983)

CYCLIC P-Y DATA
p1/p4 = Mobilized soil resistance (MN/m)
y1/y4 = Lateral Pile displacement (mm)

Depth (m)	Soil type	p1	y1	p2	y2	p3	y3	p4	y4
3.000	CLAY	0.011	12.573	0.019	57.150	0.027	171.450	0.018	857.250
4.000	CLAY	0.018	12.573	0.029	57.150	0.042	171.450	0.031	857.250
4.500	CLAY	0.021	12.573	0.035	57.150	0.050	171.450	0.038	857.250
4.500	CLAY	0.021	12.573	0.035	57.150	0.050	171.450	0.043	857.250
5.500	CLAY	0.028	12.573	0.047	57.150	0.068	171.450	0.060	857.250
6.500	CLAY	0.036	12.573	0.060	57.150	0.086	171.450	0.079	857.250
7.500	CLAY	0.044	12.573	0.073	57.150	0.105	171.450	0.099	857.250
8.500	CLAY	0.052	12.573	0.087	57.150	0.125	171.450	0.120	857.250
9.500	CLAY	0.060	12.573	0.100	57.150	0.144	171.450	0.142	857.250
10.500	CLAY	0.068	12.573	0.113	57.150	0.163	171.450	0.163	857.250
11.500	CLAY	0.076	12.573	0.127	57.150	0.183	171.450	0.183	857.250
12.500	CLAY	0.083	12.573	0.138	57.150	0.199	171.450	0.199	857.250
13.500	CLAY	0.090	12.573	0.150	57.150	0.216	171.450	0.216	857.250
14.500	CLAY	0.097	12.573	0.161	57.150	0.232	171.450	0.232	857.250
15.500	CLAY	0.103	12.573	0.172	57.150	0.248	171.450	0.248	857.250
16.500	CLAY	0.110	12.573	0.184	57.150	0.264	171.450	0.264	857.250
17.500	CLAY	0.117	12.573	0.195	57.150	0.281	171.450	0.281	857.250
18.500	CLAY	0.124	12.573	0.206	57.150	0.297	171.450	0.297	857.250
19.500	CLAY	0.130	12.573	0.217	57.150	0.313	171.450	0.313	857.250
20.500	CLAY	0.137	12.573	0.229	57.150	0.329	171.450	0.329	857.250
21.500	CLAY	0.144	12.573	0.240	57.150	0.346	171.450	0.346	857.250
21.500	SAND	3.419	4.835	7.620	15.117	7.620	25.400	7.620	57.150
22.500	SAND	3.847	5.201	7.620	15.300	7.620	25.400	7.620	57.150
23.500	SAND	4.242	5.486	7.620	15.443	7.620	25.400	7.620	57.150
24.500	SAND	4.707	5.842	7.620	15.621	7.620	25.400	7.620	57.150
25.500	SAND	5.166	6.160	7.620	15.780	7.620	25.400	7.620	57.150
26.500	SAND	5.534	6.350	7.620	12.700	7.620	25.400	7.620	57.150
27.500	SAND	5.751	6.350	7.620	12.700	7.620	25.400	7.620	57.150
28.500	SAND	5.967	6.350	7.620	12.700	7.620	25.400	7.620	57.150
29.500	SAND	6.184	6.350	7.620	12.700	7.620	25.400	7.620	57.150

...continued

INST. OF ENGG. & OCEAN TECH. ONGC		PLATE NO. 12
PREP. BY (T.K.Roy)	CHKD. BY (R.Mahanta)	DATE 15/08/2018

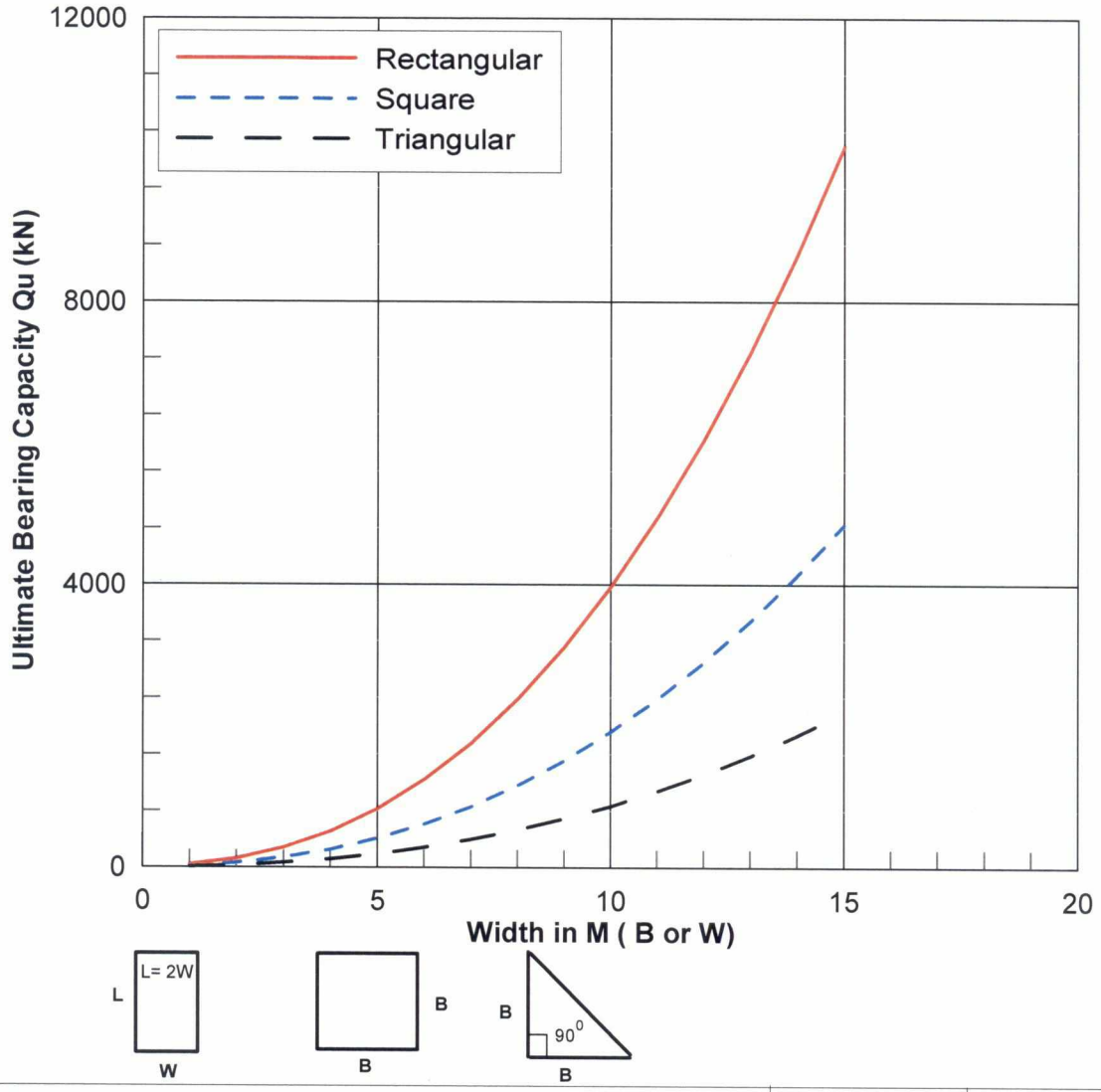
P-Y DATA **LOCATION: GK-28-10 (PLATFORM), Gulf of Kutch**

...continued from last page

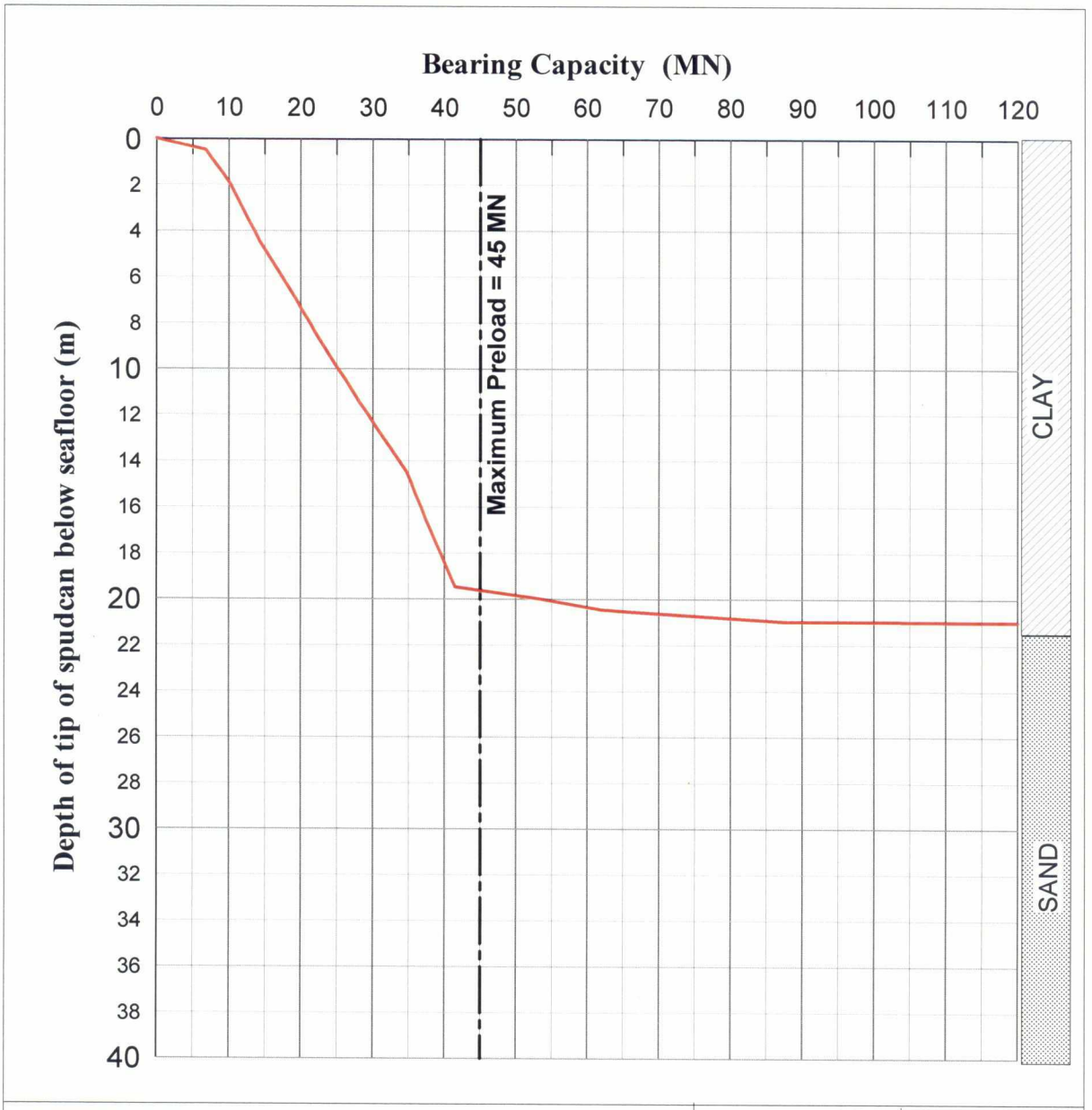
p1/p4 = Mobilized soil resistance (MN/m)
y1/y4 = Lateral Pile displacement (mm)

Depth (m)	Soil type	p1	y1	p2	y2	p3	y3	p4	y4
30.000	SAND	6.292	6.350	7.620	12.700	7.620	25.400	7.620	57.150
30.000	SAND	6.292	6.350	9.144	12.700	9.144	25.400	9.144	57.150
31.000	SAND	6.508	6.350	9.144	12.700	9.144	25.400	9.144	57.150
32.000	SAND	6.724	6.350	9.144	12.700	9.144	25.400	9.144	57.150
33.000	SAND	6.939	6.350	9.144	12.700	9.144	25.400	9.144	57.150
34.000	SAND	7.155	6.350	9.144	12.700	9.144	25.400	9.144	57.150
35.000	SAND	7.370	6.350	9.144	12.700	9.144	25.400	9.144	57.150
36.000	SAND	7.586	6.350	9.144	12.700	9.144	25.400	9.144	57.150
37.000	SAND	7.801	6.350	9.144	12.700	9.144	25.400	9.144	57.150
38.000	SAND	8.016	6.350	9.144	12.700	9.144	25.400	9.144	57.150
39.000	SAND	8.231	6.350	9.144	12.700	9.144	25.400	9.144	57.150
40.000	SAND	8.446	6.350	9.144	12.700	9.144	25.400	9.144	57.150
41.000	SAND	8.661	6.350	9.144	12.700	9.144	25.400	9.144	57.150
42.000	SAND	8.876	6.350	9.144	12.700	9.144	25.400	9.144	57.150
43.000	SAND	9.091	6.350	9.144	12.700	9.144	25.400	9.144	57.150
44.000	SAND	9.144	6.287	9.144	15.843	9.144	25.400	9.144	57.150
45.000	SAND	9.144	6.160	9.144	15.780	9.144	25.400	9.144	57.150
45.300	SAND	9.144	6.096	9.144	15.748	9.144	25.400	9.144	57.150
45.300	CLAY	1.218	2.103	1.959	4.206	2.389	8.458	0.239	33.741
46.300	CLAY	1.266	2.103	2.036	4.206	2.483	8.458	0.248	33.741
47.300	CLAY	1.314	2.103	2.113	4.206	2.577	8.458	0.258	33.741
48.300	CLAY	1.362	2.103	2.190	4.206	2.670	8.458	0.267	33.741
49.300	CLAY	1.410	2.103	2.267	4.206	2.764	8.458	0.276	33.741
50.300	CLAY	1.458	2.103	2.344	4.206	2.858	8.458	0.286	33.741

INST. OF ENGG. & OCEAN TECH. ONGC		PLATE NO. 13
PREP. BY (T.K.Roy)	CHKD. BY (R.Mahanta)	DATE 15/08/2018



MUDMAT ULTIMATE BEARING CAPACITY		PLATE	14
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ NO: IEOT/GEO/07/2018-19 REP NO : IEOT/GEO/07/2018-19 **LOCATION: GK-28-10 (Platform)** **FIELD: Gulf of Kutch**	DATE	13/08/2018
		PREPARED BY	(R. Mahanta)
		CHECKED BY:	(T. K. Roy)



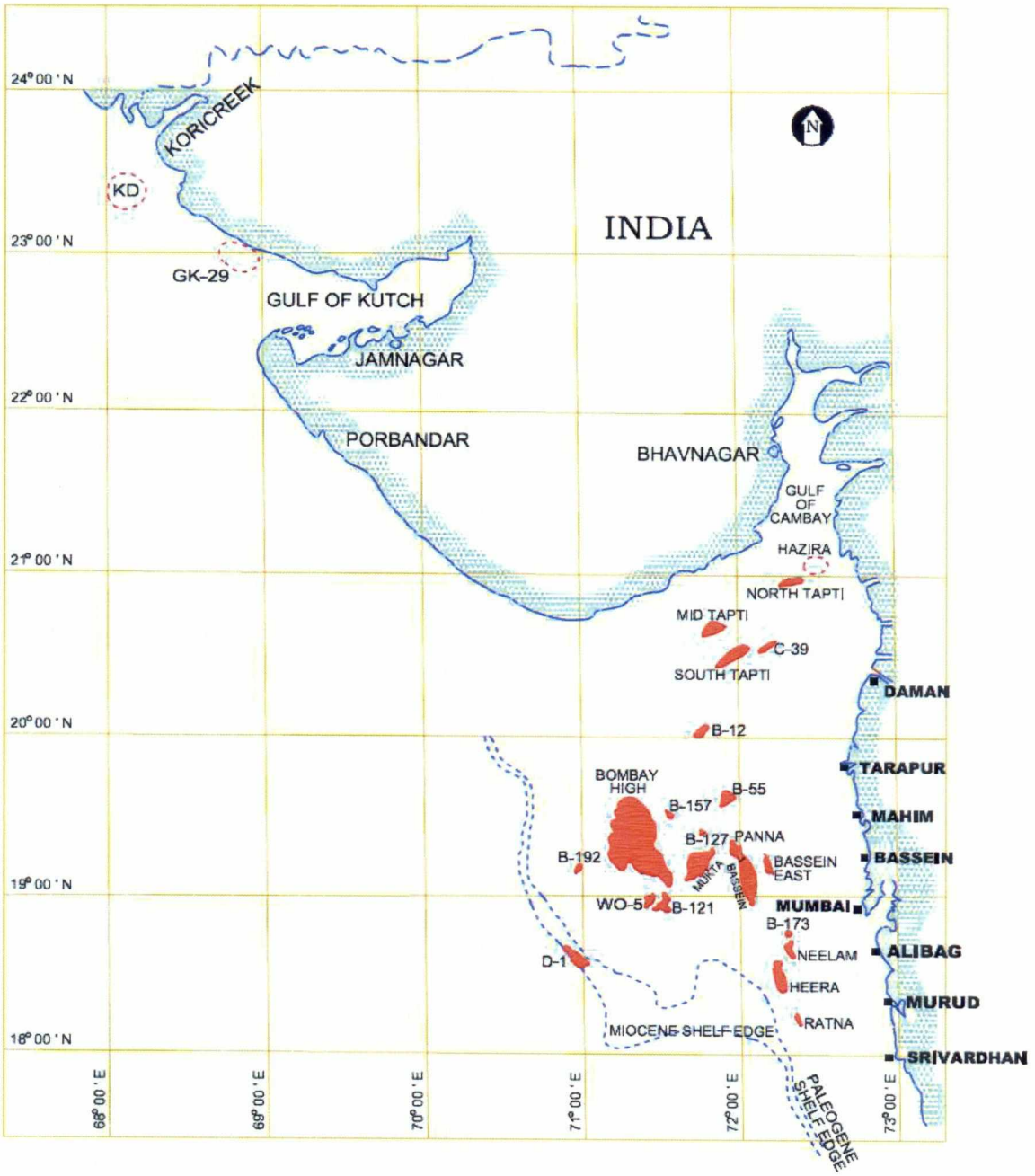
JACKUP FOOTING PENETRATION CURVE		PLATE	15
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ NO: IEOT/GEO/07/2018-19 REP NO : IEOT/GEO/07/2018-19 LOCATION: GK-28-10 (PLATFORM) FIELD: GULF OF KUTCH RIG NAME: STANDARD ONGC Eq.Diameter of spudcan = 14 m Volume of spudcan = 300 cum Max. preload = 45 MN Tip extension = NIL	DATE	13/08/2018
		PREPARED BY	(Rupam Mahanta)
		CHECKED BY	(T K Roy)

APPENDIX-A

APPENDIX - A

GENERAL LOCATION DETAILS

General Location Map	**A-1**
Positioning Data	**A-2**
New Platform Location GK-28-10 and Nearby Locations	**A-3**
Tidal Variation Chart	**A-4-I & A-4-II**



GENERAL LOCATION PLAN WESTERN OFFSHORE, INDIA	PLATE	A-1

OIL AND NATURAL GAS CORPORATION LIMITED
INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY

POSITIONING DATA
GK-28-10 (PLATFORM)

WGS-84

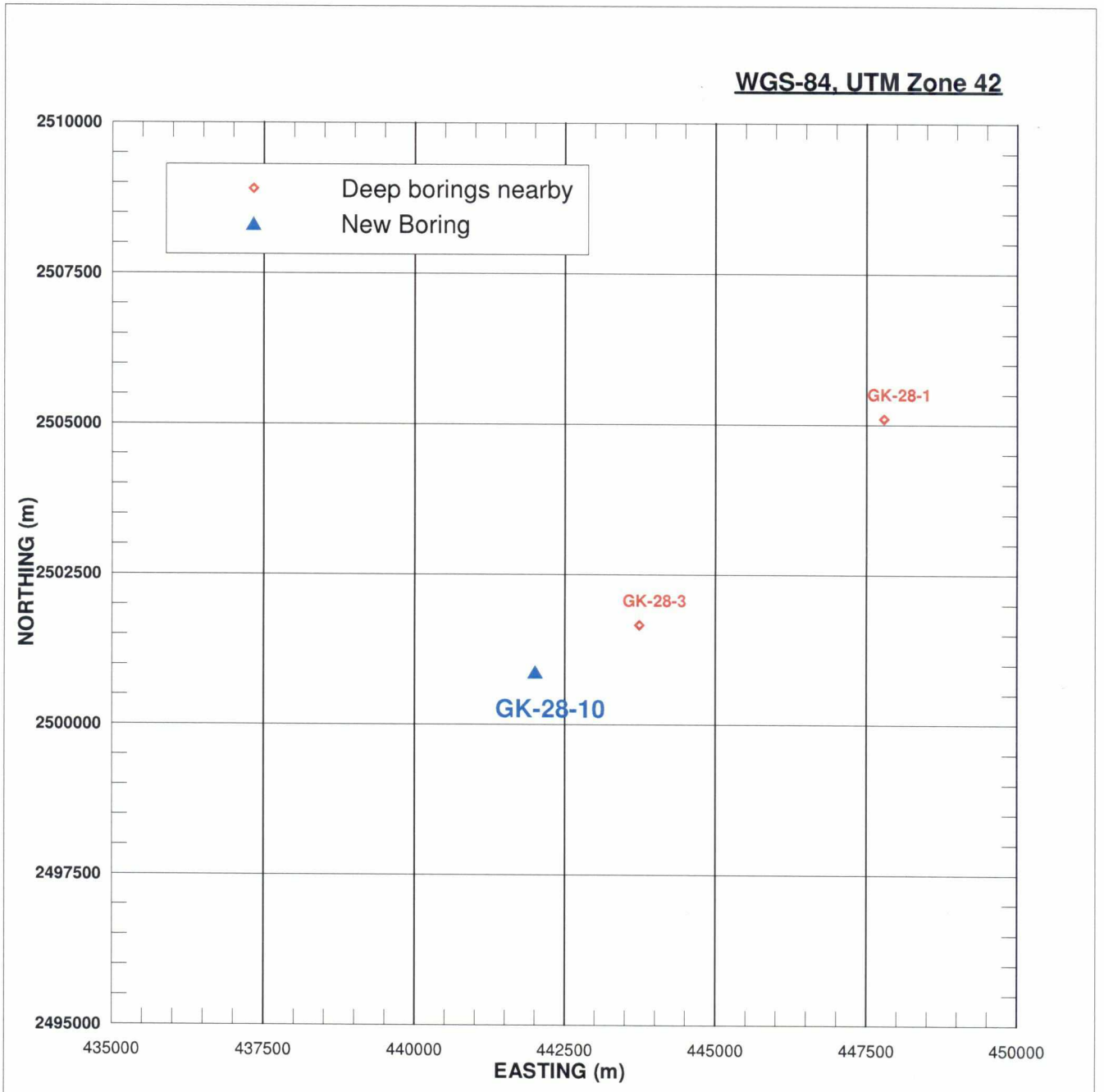
BORING	GEOGRAPHICAL CO-ORDINATES		UTM CO-ORDINATES (CM:69) ZONE - 42	
	LATITUDE - N	LONGITUDE - E	EASTING (in Meter)	NORTHING (in Meter)
RELEASED	22° 36' 49.602"	068° 26' 08.317"	442 000.0	2 500 876.0
POSITIONED	22° 36' 49.614"	068° 26' 08.305"	441 999.64	2 500 876.37

EVEREST-1830

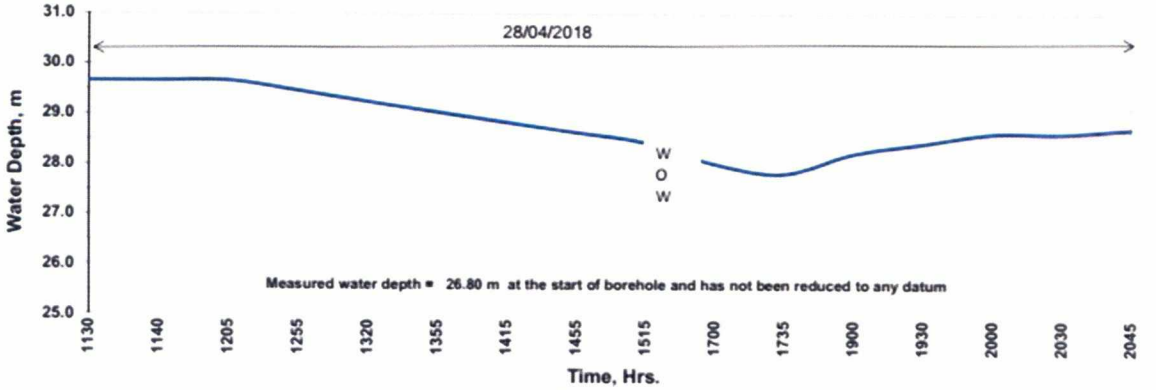
BORING	GEOGRAPHICAL CO-ORDINATES		UTM CO-ORDINATES (CM:69) ZONE - 42	
	LATITUDE - N	LONGITUDE - E	EASTING (in Meter)	NORTHING (in Meter)
RELEASED	22° 36' 48.068"	068° 26' 07.355"	442155.410	2500628.120
POSITIONED	22° 36' 48.080"	068° 26' 07.343"	442155.070	2500628.490

Offset from Released location	
Distance drill head to Released location (m)	0.52
Bearing (Degrees)	315.8° T
Heading(Degrees)	210.0°

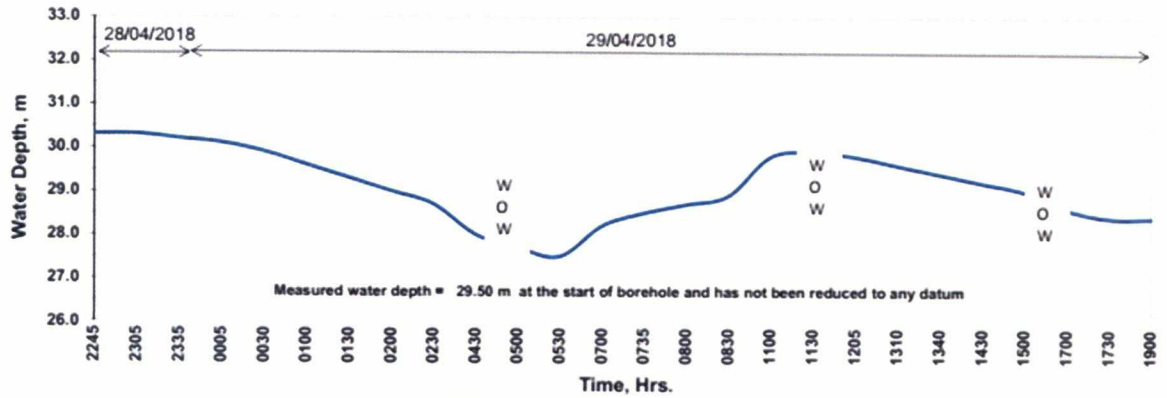
REPORT NO : IEOT/GEO/7/2018-19	
LOCATION : GK-28-10 (PLATFORM)	IEOT,ONGC
PLATE: A-2	DATE : 20/8/2018
PREP. BY : Sri Harsha Gamidi	CHKD. BY : Alankar Srivastava



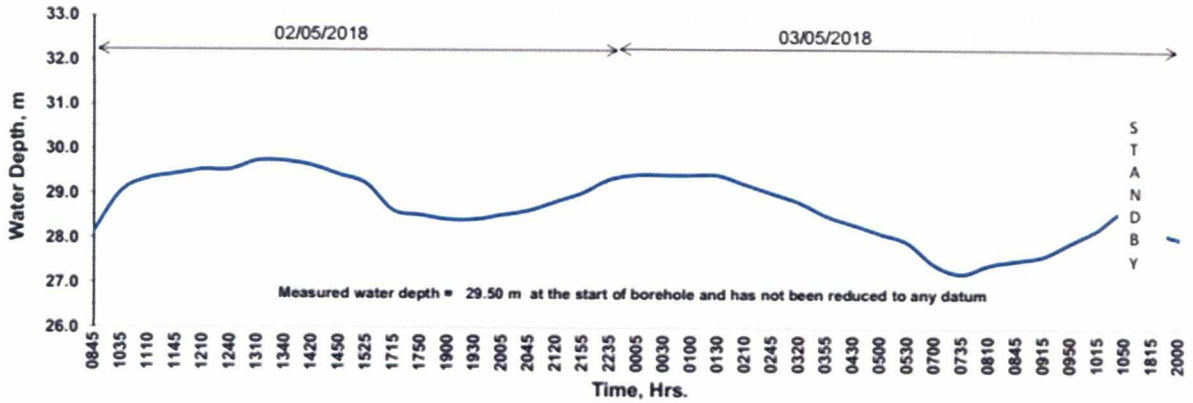
New and Nearby Locations		PLATE	A-3
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 Platform	DATE	25-07-2018
		PREPARED BY	
		CHECKED BY	



TIDAL VARIATION CHART
GK-28-10 (PLATFORM) (CPT)
GULF OF KUTCH, WEST COAST, OFFSHORE INDIA



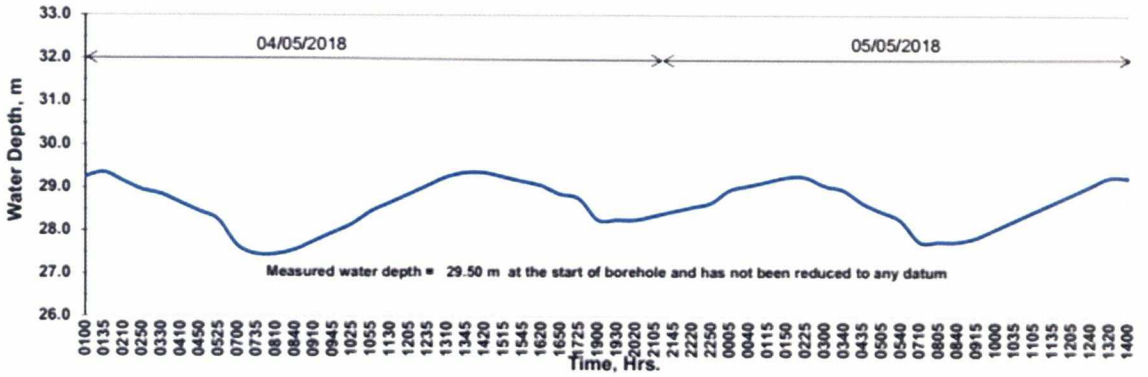
TIDAL VARIATION CHART
GK-28-10 (PLATFORM) (SAMPLING-I)
GULF OF KUTCH, WEST COAST, OFFSHORE INDIA



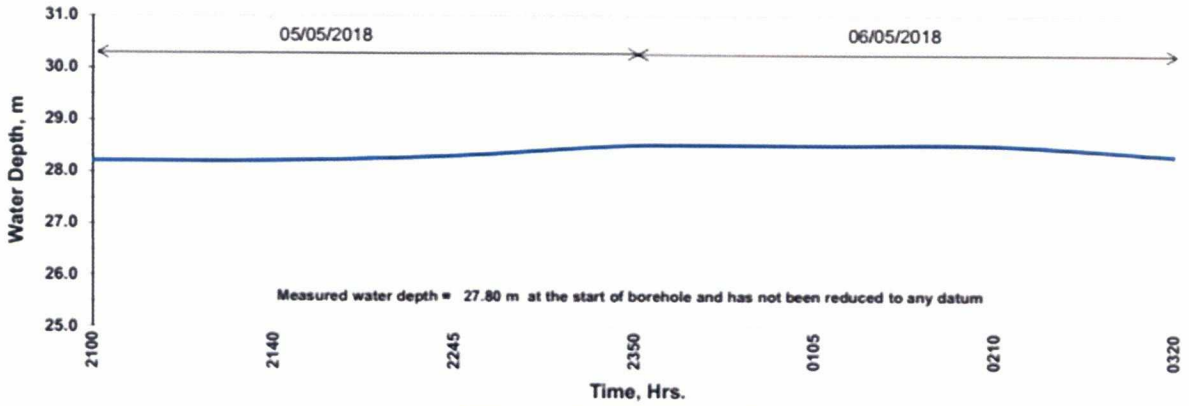
TIDAL VARIATION CHART
GK-28-10 (PLATFORM) (SAMPLING-II)
GULF OF KUTCH, WEST COAST, OFFSHORE INDIA

TIDAL VARIATION CHART	PLATE	A-4-I

OIL AND NATURAL GAS CORPORATION LIMITED
INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOG



TIDAL VARIATION CHART
GK-28-10 (PLATFORM) (SAMPLING-III)
GULF OF KUTCH, WEST COAST, OFFSHORE INDIA



TIDAL VARIATION CHART
GK-28-10 (PLATFORM) (FVT)
GULF OF KUTCH, WEST COAST, OFFSHORE INDIA

TIDAL VARIATION CHART	PLATE	A-4-II

OIL AND NATURAL GAS CORPORATION LIMITED
INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY

APPENDIX-B

APPENDIX - B

LABORATORY TEST RESULTS AND PHOTOGRAPHS OF SOIL SAMPLES

Summary of Laboratory Tests	**B 1 – B2**
Field Vane Shear Test Results	**B3**
Grain Size Distribution Curves	**GS 1 - GS 15**
Unconsolidated Undrained Triaxial Tests (On Board)	**UU 01 - UU 02**
Unconsolidated Undrained Triaxial Tests (On Shore)	**LABUU 01 – LABUU 02**
Selected Photos of Soil Samples	**PH-1 & PH-2**

SUMMARY OF TEST RESULTS

BORING: GK-28-10 (PLATFORM) — **FIELD : GULF OF KUTCH**

Samp. No.	Depth (m)	Wet unit wt. (KN/m3)	Dry unit wt. (KN/m3)	Back calculated unit wt (KN/m3)	Wat. cont. (%)	Carb. cont. (%)	Sp. grav	Clay frac (%)	INDEX TESTS: Atterberg Limits LL (%)	PL (%)	PI (%)	Shear strength: MV KPa	MVr KPa	FC KPa	FCr KPa	FV KPa	FVr KPa	TV KPa	PP KPa	TRIAXIAL TESTS: Type	S_u KPa	ε_{50} (%)	Remarks
W-01	0.50	15.6	8.9	15.8	75																		
	0.60	15.7	9.0																				
FV-1	0.80															2.8	1.2						
W-02	1.50	16.0	9.5			16		60	70	28	42							7					
	1.60	15.9	9.5	16.19	68													7					
	1.70																	7					
	1.80																	7					
W-03	2.50	16.2	9.5															8					
	2.60	16.2	9.4															8					
	2.70			15.99	72													8					
W-04	3.10																	8	3				
	3.20	16.5	10.9	17.38	51													8	5				
	3.40																	8	5				
	3.50																	8	5				
	3.55											7	2										
W-05	4.10																	10	5				
	4.20																	10	5				
	4.30	15.5	10.1															10	5				
	4.40																	10	5				
	4.60																	10	5				
	4.65			17.27	53							8	2										
FV-2	4.80															7.2	2.3						
W-06	5.10																	11	6				
	5.20																	11	6				
	5.30	16.6	11.1	17.53	50													11	6				
	5.50																	11	6				
	5.60											8	3					11	6				
W-07	6.20	16.2	10.5	17.21	54													11	6				
	6.30																	11	6				
	6.40																	11	6				
	6.50																	11	6				
	6.65											9	3					11	6				
W-08	7.10																	13	6				
	7.20	16.7	11.3	17.68	48													13	6				
	7.30																	13	6				
	7.40																	13	6				
	7.50																	13	6				
	7.55											10	5										
W-09	8.20																	11	6				
	8.30	16.7	10.8															12	6				
	8.40			17.21	54													12	6				
	8.50																	12	6				
	8.60											11	4					13	6				
FV-3	8.80															17.9	5.7						
W-10	9.20																	12	6				
	9.30					18		53	50	19	31							12	8				
	9.40																	12	8				
	9.50	16.4	11.2	17.77	47													12	8				
	9.60											12	5					12	8				
FV-4	9.80															22.9	1.7						
W-11	10.20																	13	9				
	10.30																	13	9				
	10.40	16.8	11.2	17.44	51													14	9				
	10.60																	14	9				
	10.70											12	5					14	9				
W-12	11.20																	13	9				
	11.30	16.7	10.8	17.12	55													14	9				
	11.50																	14	9				
	11.60																	14	9				
	11.70											12	5					14	9				
W-13	12.20																	15	10				
	12.40	16.4	10.9	17.5	50													15	11				
	12.50																	16	11				
	12.60																	16	11				
FV-5	12.80											16	5			25.7	7.3	16	11				
W-14	13.30																	20	10				
	13.40																	20	11				
	13.60	17.2	11.5	17.6	49													20	11				
	13.70																	20	11				
	13.80											15	5					20	11				
W-15	14.30																	22	13				

LL LIQUID LIMIT
MV MOTOR VANE
PP POCKET PENTROMETER
FV FIELD VANE

PL PLASTIC LIMIT
FC FALL CONE
FCr FALL CONE REMOULDED
FVr FIELD VANE REMOULDED

PI PLASTICITY INDEX
TV TOR VANE
UU UNCONSOLIDATED UNDRAINED TRIAXIAL

REPORT NO: IEOT/GEO/07/2018-19		
BORING	GK-28-10 (PLATFORM)	IEOT,ONGC
PLATE	B1	DATE: 08.08.2018
PREPARED BY		CHECKED BY
	B.Nayak	T.K.Roy

SUMMARY OF TEST RESULTS

BORING: GK-28-10 (PLATFORM) — **FIELD : GULF OF KUTCH**

Samp. No.	Depth (m)	Wet unit wt. (KN/m3)	Dry unit wt. (KN/m3)	Back calculated unit wt (KN/m3)	Wat. cont. (%)	Carb. cont. (%)	Sp. grav	Clay frac (%)	INDEX TESTS: Atterberg Limits LL (%)	PL (%)	PI (%)	Shear strength: MV KPa	MVr KPa	FC KPa	FCr KPa	FV KPa	FVr KPa	TV KPa	PP KPa	TRIAXIAL TESTS: Type	S_u KPa	ε_{50} (%)	Remarks
	14.60																	20	12				
	14.70																	20	12				
	14.80											20	5					20	12				
W-16	15.30																	20	12				
	15.40	16.8	10.9	17.17	54													20	12				
	15.60																	20	12				
	15.70																	20	12				
	15.80											21	6					20	12				
W-17	16.70	16.3	10.7															24	12				
FV-6	16.80															50.4	12.1						
	16.87	16.2	10.7															23	12				
	17.04	16.8	11.1	17.35	52													23	12				
	17.30											24	7										
W-18	18.20																	16	20				
	18.37	16.6	11.2															20	20	UU	26	1.4	OB
	18.54	16.8	11.3															25	20				
	18.71			17.65	48													26	20				
	18.85											20	7										
W-19	19.64	16.9	11.1	17.59	49									39	12			30	20	UU	30	1.9	LAB
	19.81	17.0	11.1			21		67	56	25	31							31	25	UU	28	1.4	OB
	19.98	17.0	11.1															32	25				
	20.15			17.28	53													31	25				
	20.30											30											
FV-7	20.80															73.3	18.1						
W-20	21.12	17.1	11.2															34.3	27	UU	37.5	0.7	OB
	21.29			17.29	53													34.3	27				
	21.50	19.4	15.1			84																	
	21.70			20.04	28																		
W-21	22.50			22.53	16	91																	
W-22	24.00	21.5	18.4	22.22	17																		
W-23	25.55	19.8	16.2	21.12	22																		
	25.70	19.7	16.1			95																	
W-24	27.00			20.4	26	97																	
W-25	28.50			21.66	19	96																	
W-26	30.00			22.7	15	94																	
W-27	34.00	21.6	18.1	21.62	20																		
	34.15	19.4	15.1			18																	
	34.20			20.03	28																		
W-28	38.00	20.2	15.9	20.23	27	23																	
W-29	42.00	20.6	17.7	22.29	17	57																	
W-30	46.12	21.4	18.0			17		60	38	23	15									UU	482	3.6	OB
	46.29	21.0	17.7	21.14	22															UU	477	2.46	LAB
	46.46			21.81	19																		
W-31	50.00	22.4	19.5	22.73	15	27		21															
W-32	54.10	20.6	16.0	20.07	28																		
W-33	58.00	18.7	15.6	21.46	20																		
	58.10	20.3	16.2	20.55	25	12																	
W-34	62.00	21.3	17.5																				
	62.10			21.3	21																		
W-35	66.00	20.9	17.3			27		50															
	66.10			21.36	21																		
W-36	70.20	21.0	17.5			12		61	40	22	18									UU	435	2.5	OB
	70.37																						
	70.40			21.46	20																		
W-37	74.10	20.7	16.8			8																	
	74.20			20.93	23																		
W-38	78.00	19.6	15.2			8																	
	78.10			20.01	29																		
W-39	82.00	20.0	16.0	20.52	25	6																	
	82.10	21.6	18.1	21.66	19	15																	
W-40	86.00	20.3	16.2	20.57	25	3																	
W-41	90.00	20.7	17.0	21.22	22	8																	
W-42	94.00	21.0	16.7	20.57	25	6																	
W-43	98.00	18.3	14.5	20.37	26	4																	
W-44	102.00	19.9	16.0	20.8	24	7																	
W-45	106.00	19.4	15.5	20.54	25	11																	
W-46	110.00	19.5	15.6	20.5	26	6																	
W-47	N/R																						
W-48	114.00	21.4	16.8	20.15	28	6																	
W-49	122.00	21.2	17.5	21.34	21	12																	
W-50	125.00			22.63	15	24																	

LL LIQUID LIMIT
MV MOTOR VANE
PP POCKET PENTROMETER
FV FIELD VANE

PL PLASTIC LIMIT
FC FALL CONE
FCr FALL CONE REMOULDED
FVr FIELD VANE REMOULDED

PI PLASTICITY INDEX
TV TOR VANE
UU UNCONSOLIDATED UNDRAINED TRIAXIAL

REPORT NO: IEOT/GEO/07/2018-19		
BORING	GK-28-10 (PLATFORI	IEOT,ONGC
PLATE	B2	DATE: 08.08.2018
PREPARED BY		CHECKED BY
	B.Nayak	T.K.Roy

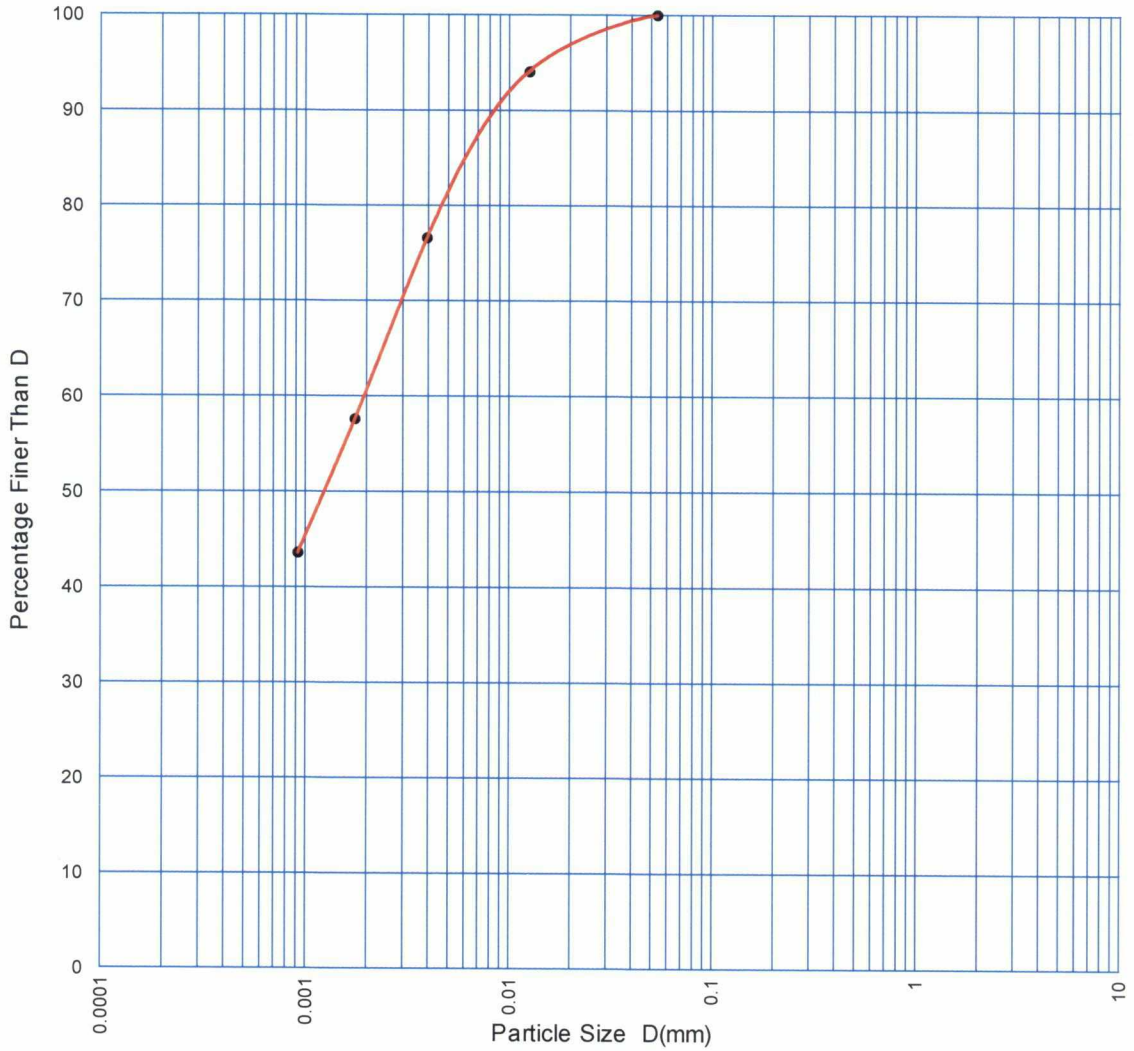
SUMMARY OF FIELD VANE TEST RESULTS

BORING: GK-28-10 (PLATFORM) **FIELD : GULF OF KUTCH**

Test No.	Depth (m)	s_u (kPa)	s_{ur} (kPa)	s_t	Remarks
1	0.80	2.80	1.20	2.33	Vane Size 130x65x2 mm
2	4.80	7.20	2.30	3.13	Vane Size 130x65x2 mm
3	8.80	17.90	5.70	3.14	Vane Size 130x65x2 mm
4	9.80	22.90	1.70	13.47	Vane Size 100x50x2 mm
5	12.80	25.70	7.30	3.52	Vane Size 100x50x2 mm (Carried out during high swell)
6	16.80	50.40	12.10	4.17	Vane Size 100x50x2 mm (Carried out during high swell)
7	20.80	73.30	18.10	4.05	Vane Size 100x50x2 mm (Carried out during high swell)

S_u: UNDRAINED SHEAR STRENGTH S_{ur}: REMOULDED SHEAR STRENGTH S_t: SENSITIVITY

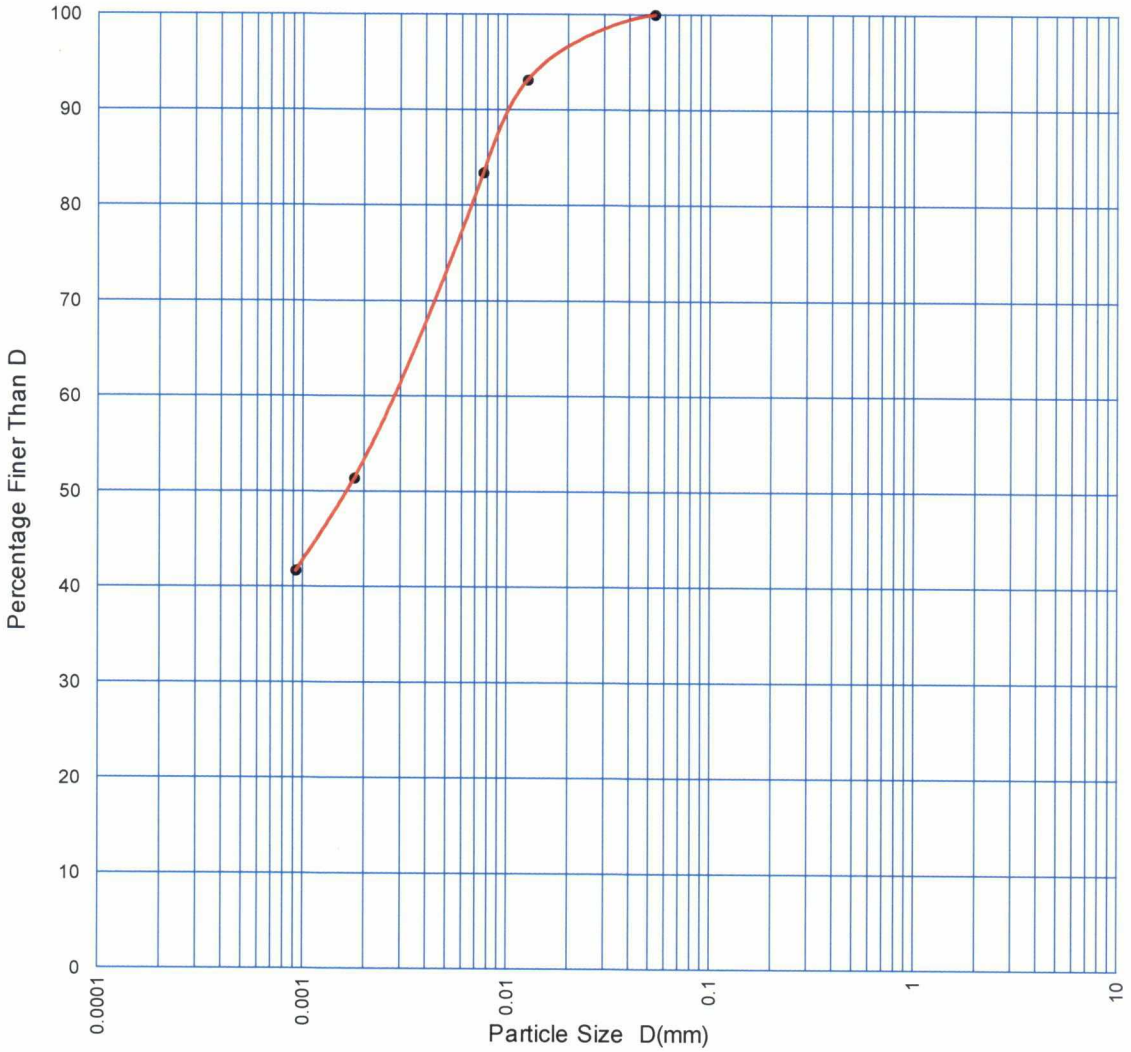
REPORT NO: IEOT/GEO/07/2018-19	
BORING : GK-28-10 (PLATFORM)	IEOT,ONGC
PLATE : B3	DATE: 08.08.2018
PREPARED BY:	CHECKED BY:
B.Nayak	T.K.Roy



CLAY	SILT	FINE SAND	MEDIUM SAND	COARSE SAND	GRAVEL

GRAIN SIZE DISTRIBUTION

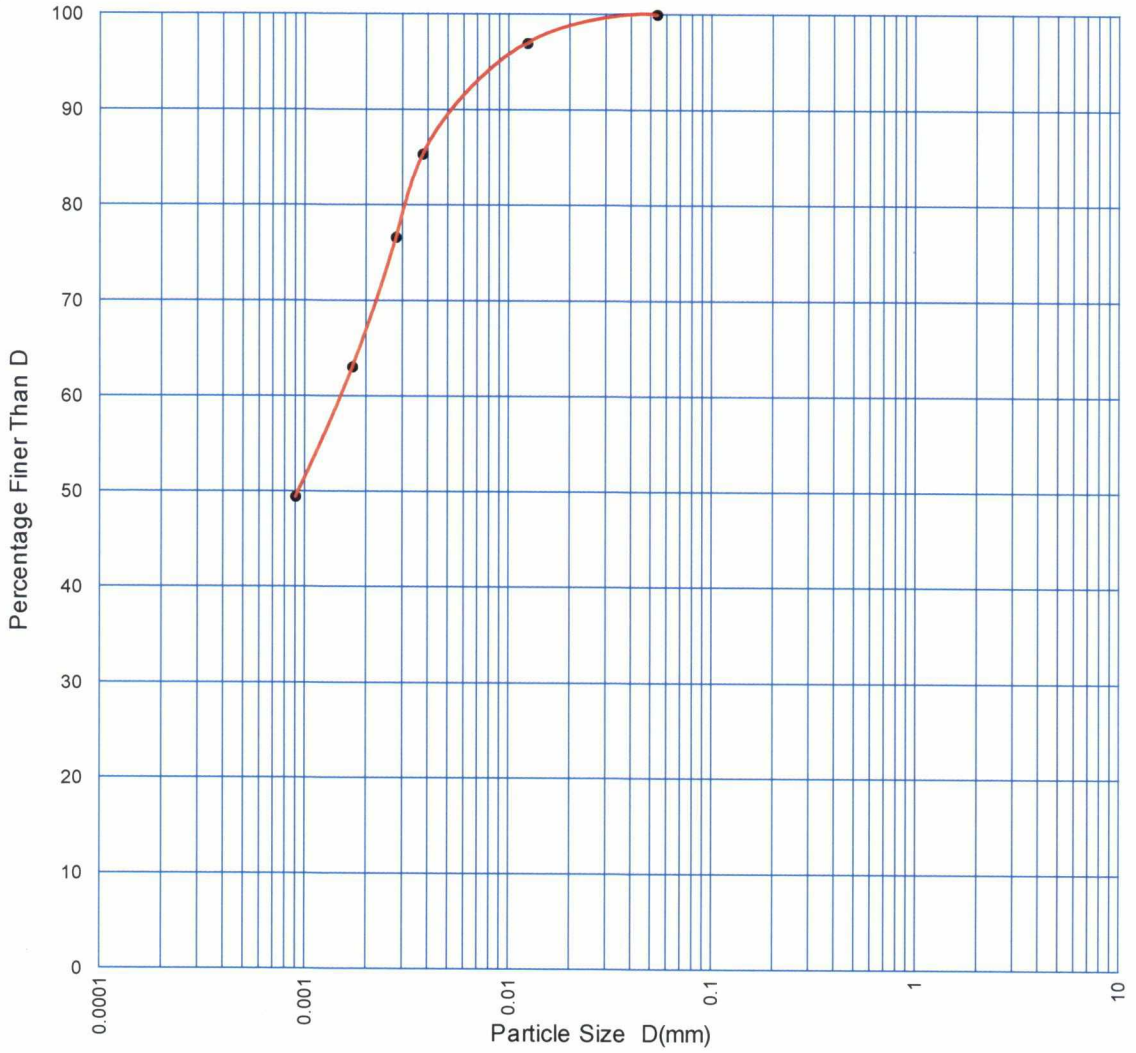
		PLATE	GS-01
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ. No.: IEOT/GEO/07/2018-19 REP. No. : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (Platform) FIELD : Gulf of Kutch SAMPLE: W-02 DEPTH : 1.50m	DATE	01/08/18
		PREPARED BY	D Lakshminarasimhan
		CHECKED BY	Alankar Srivastava



CLAY	SILT	FINE SAND	MEDIUM SAND	COARSE SAND	GRAVEL

GRAIN SIZE DISTRIBUTION

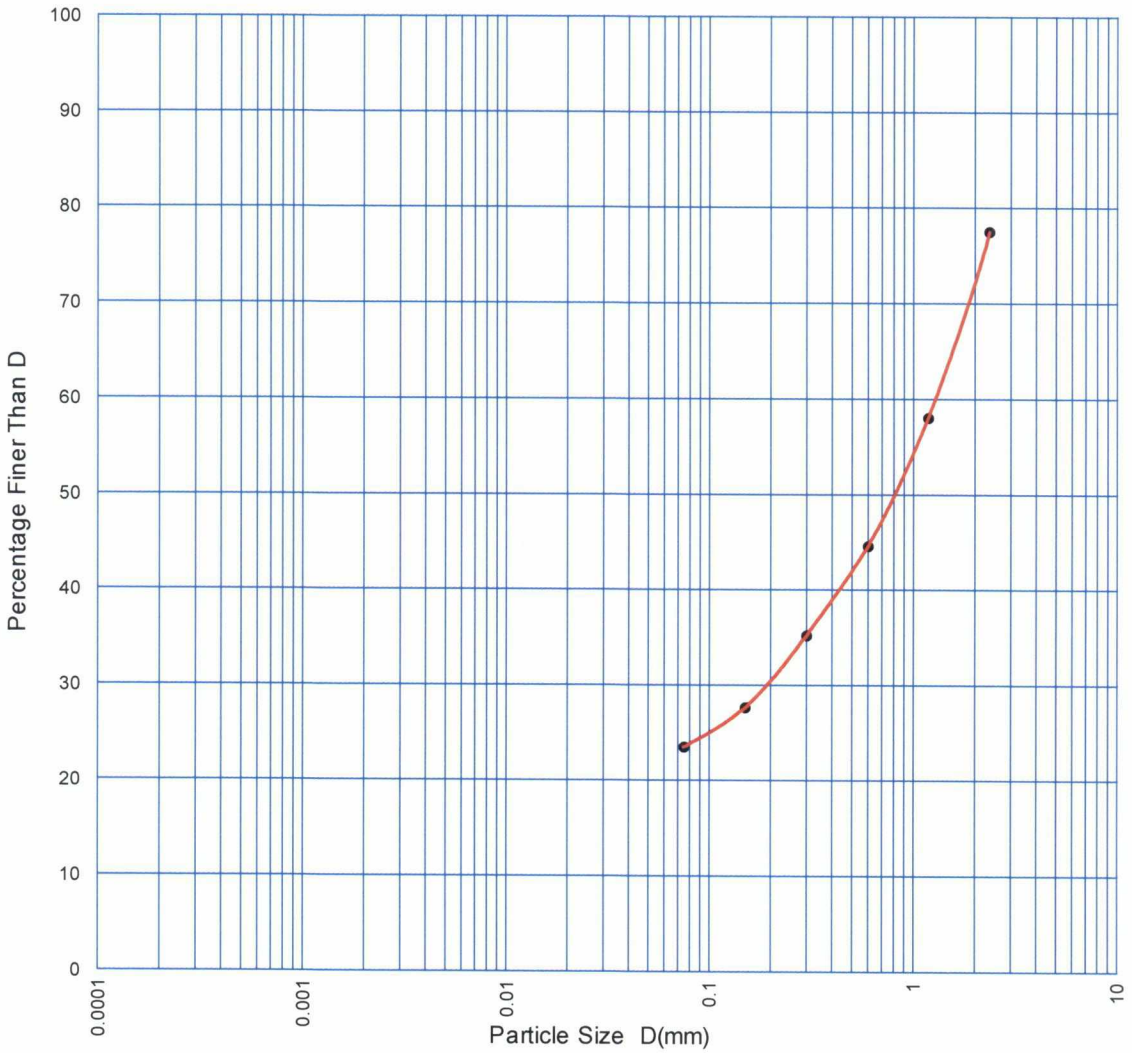
		PLATE	GS-02
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ. No.: IEOT/GEO/07/2018-19 REP. No. : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (Platform) FIELD : Gulf of Kutch SAMPLE: W-10 DEPTH : 9.30m	DATE	01/08/18
		PREPARED BY	D Lakshminarasimhan
		CHECKED BY	Alankar Srivastava



CLAY	SILT	FINE SAND	MEDIUM SAND	COARSE SAND	GRAVEL

GRAIN SIZE DISTRIBUTION

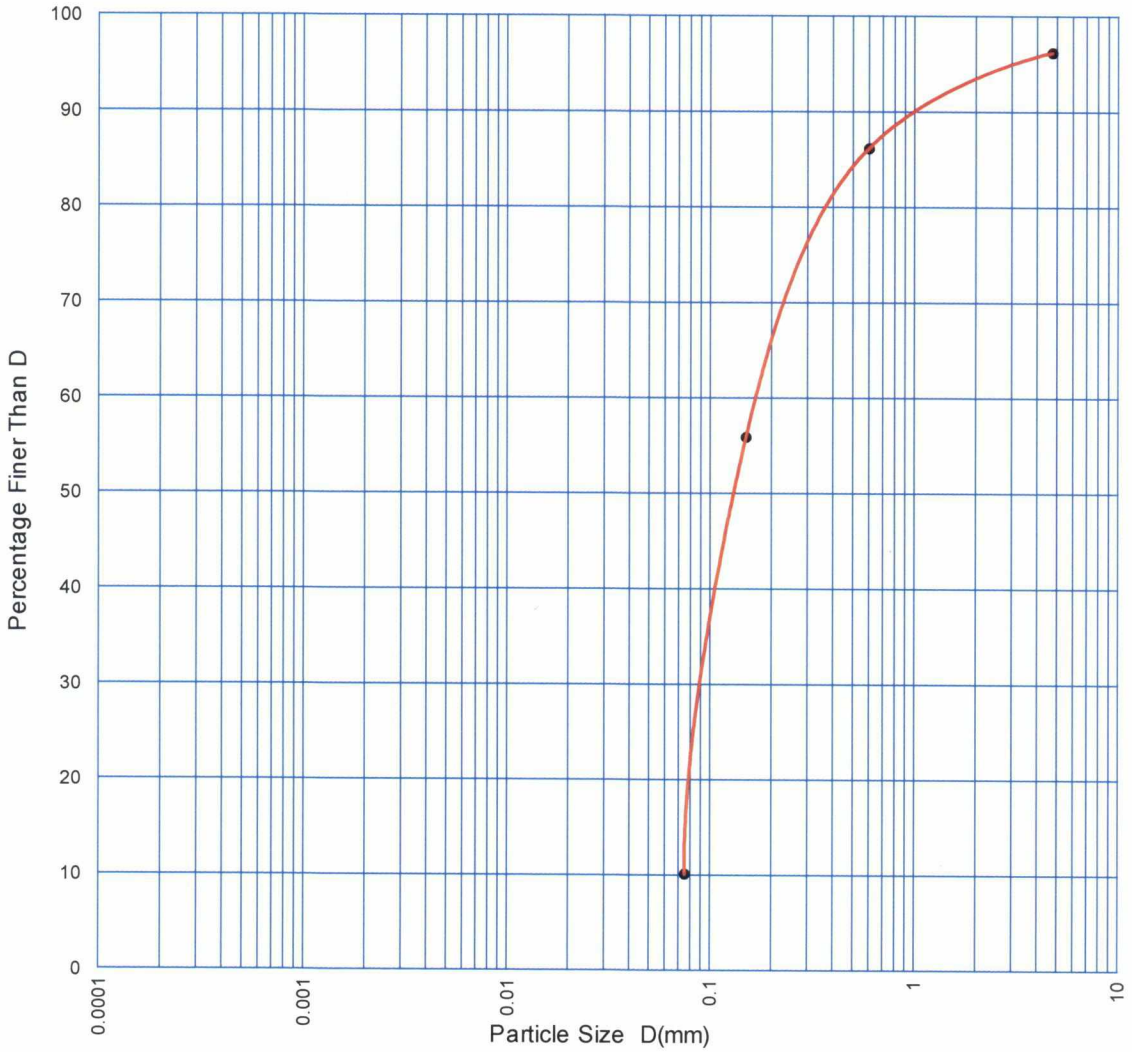
		PLATE	GS-03
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ. No.: IEOT/GEO/07/2018-19 REP. No. : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (Platform) FIELD : Gulf of Kutch SAMPLE: W-19 DEPTH : 19.81m	DATE	01/08/18
		PREPARED BY	D Lakshminarasimhan
		CHECKED BY	Alankar Srivastava



CLAY	SILT	FINE SAND	MEDIUM SAND	COARSE SAND	GRAVEL

GRAIN SIZE DISTRIBUTION

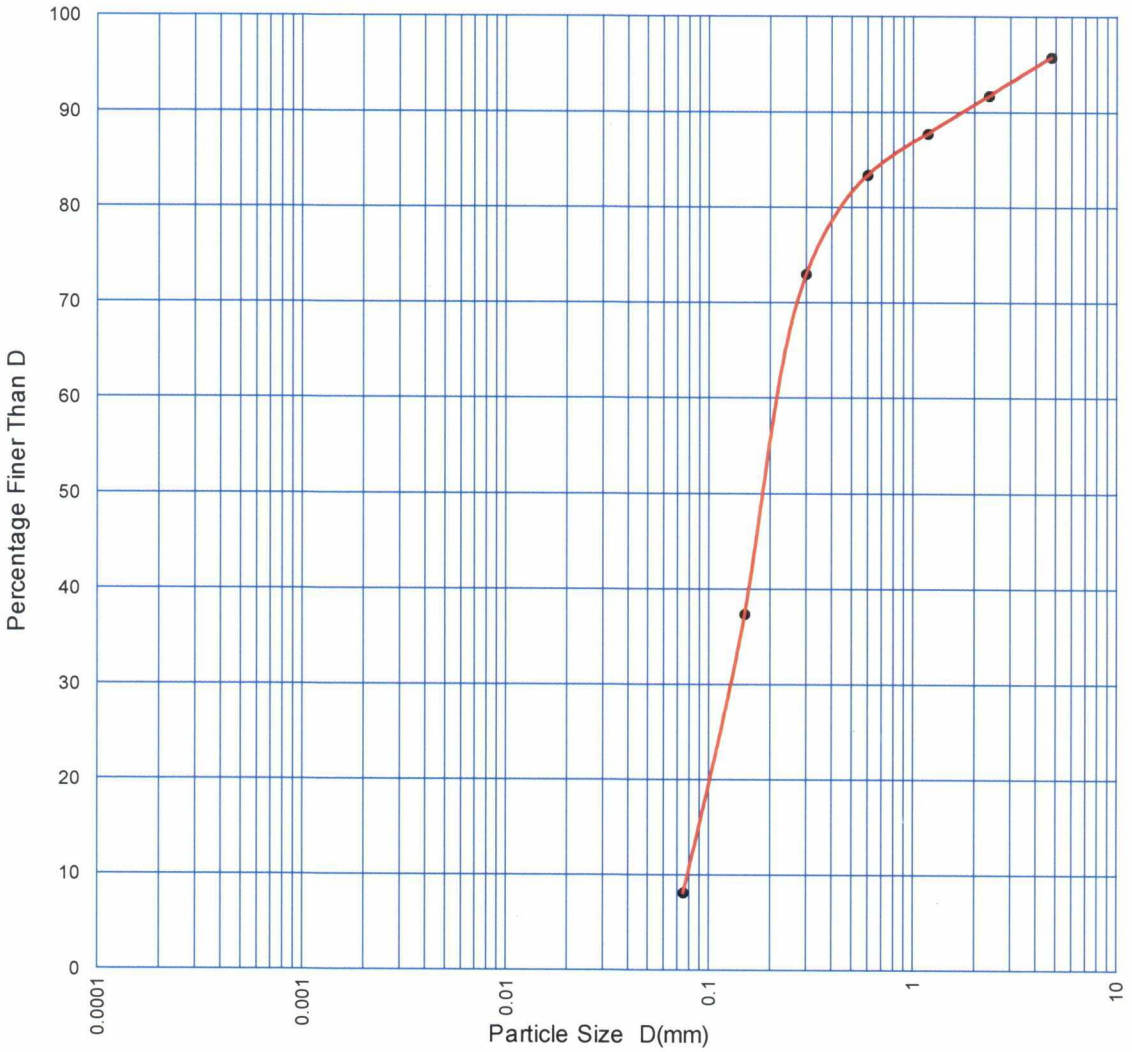
		PLATE	GS-04
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ. No.: IEOT/GEO/07/2018-19 REP. No. : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (Platform) FIELD : Gulf of Kutch SAMPLE: W-20-II DEPTH : 21.50m	DATE	01/08/18
		PREPARED BY	D Lakshminarasimhan
		CHECKED BY	Alankar Srivastava



CLAY	SILT	FINE SAND	MEDIUM SAND	COARSE SAND	GRAVEL

GRAIN SIZE DISTRIBUTION

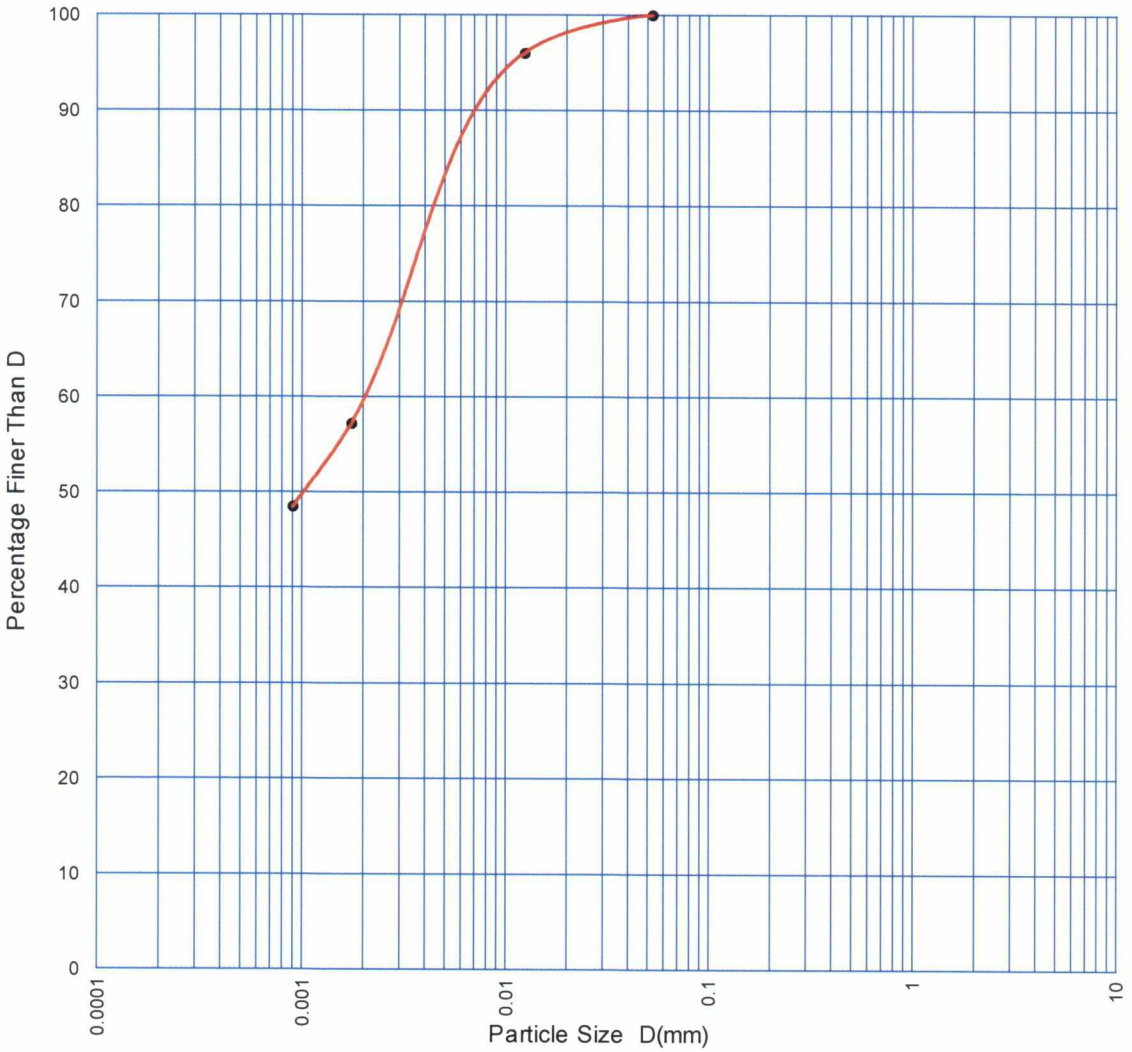
		PLATE	GS-05
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ. No.: IEOT/GEO/07/2018-19 REP. No. : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (Platform) FIELD : Gulf of Kutch SAMPLE: W-27-II DEPTH : 34.15m	DATE	01/08/18
		PREPARED BY	D Lakshminarasimhan
		CHECKED BY	Alankar Srivastava



CLAY	SILT	FINE SAND	MEDIUM SAND	COARSE SAND	GRAVEL

GRAIN SIZE DISTRIBUTION

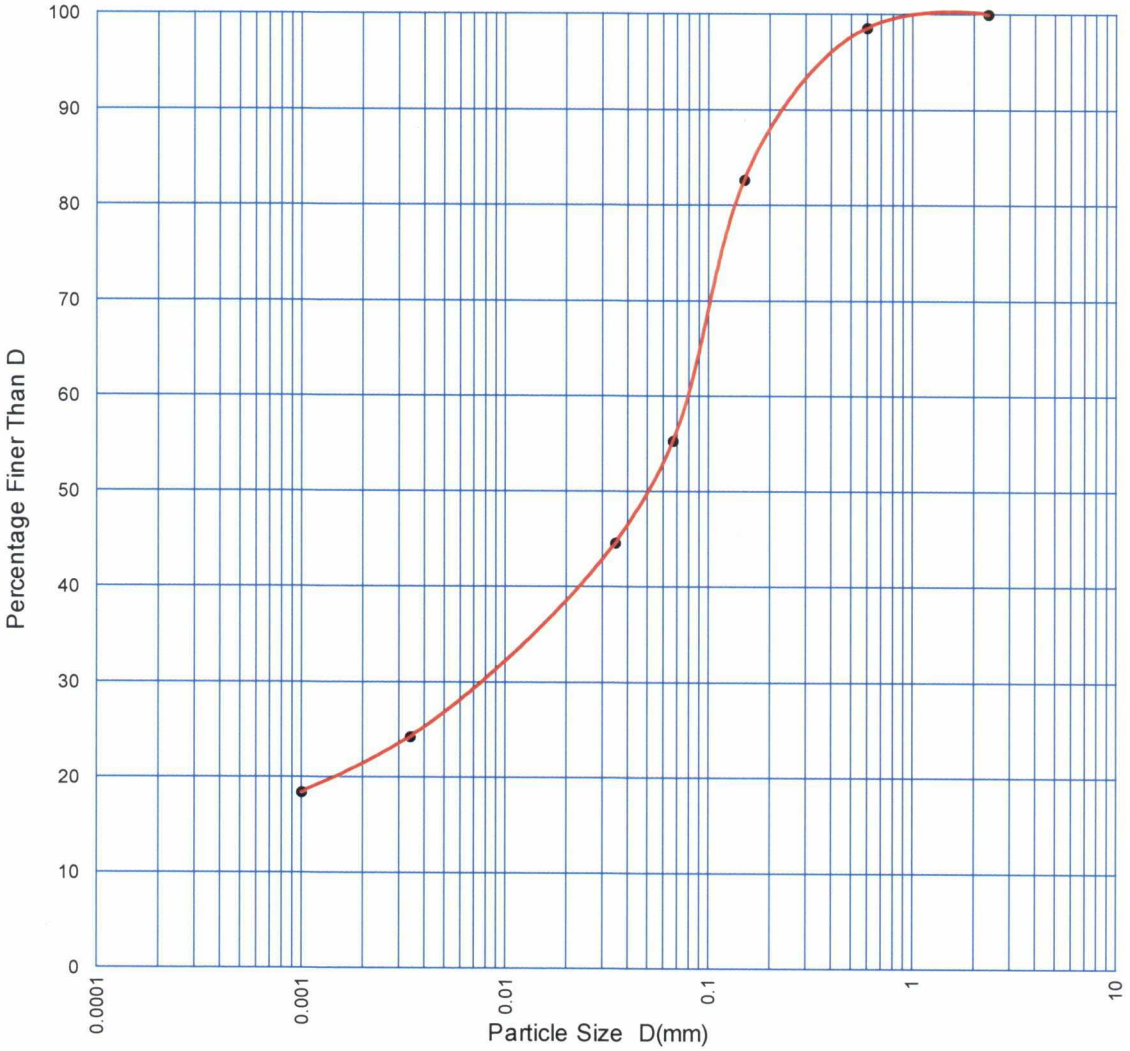
		PLATE	GS-06
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ. No.: IEOT/GEO/07/2018-19 REP. No. : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (Platform) FIELD : Gulf of Kutch SAMPLE: W-28 DEPTH : 38.00m	DATE	01/08/18
		PREPARED BY	D Lakshminarasimhan
		CHECKED BY	Alankar Srivastava



CLAY	SILT	FINE SAND	MEDIUM SAND	COARSE SAND	GRAVEL

GRAIN SIZE DISTRIBUTION

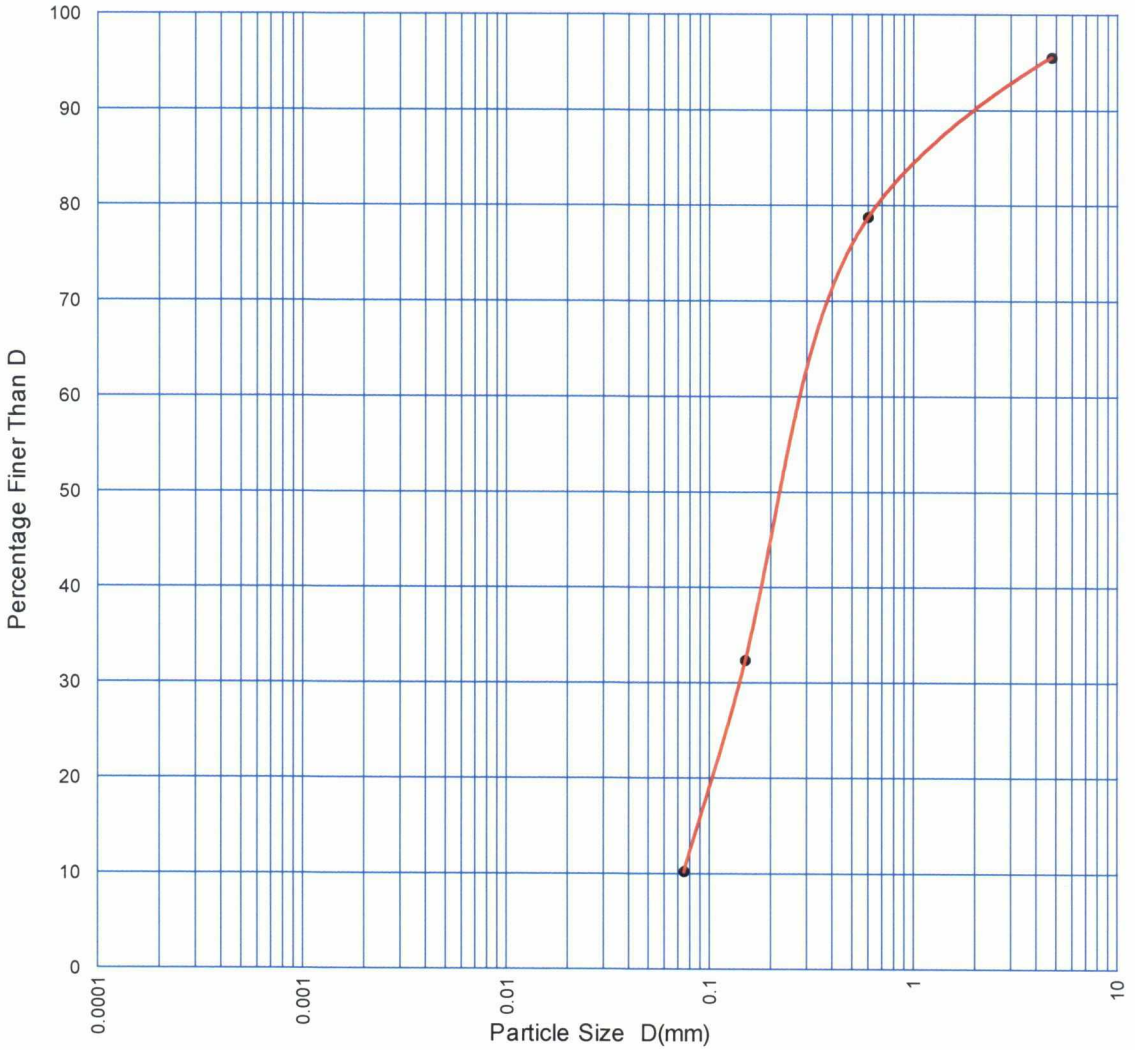
		PLATE	GS-07
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ. No.: IEOT/GEO/07/2018-19 REP. No. : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (Platform) FIELD : Gulf of Kutch SAMPLE: W-30 DEPTH : 46.12m	DATE	01/08/18
		PREPARED BY	D Lakshminarasimhan
		CHECKED BY	Alankar Srivastava



CLAY	SILT	FINE SAND	MEDIUM SAND	COARSE SAND	GRAVEL

GRAIN SIZE DISTRIBUTION

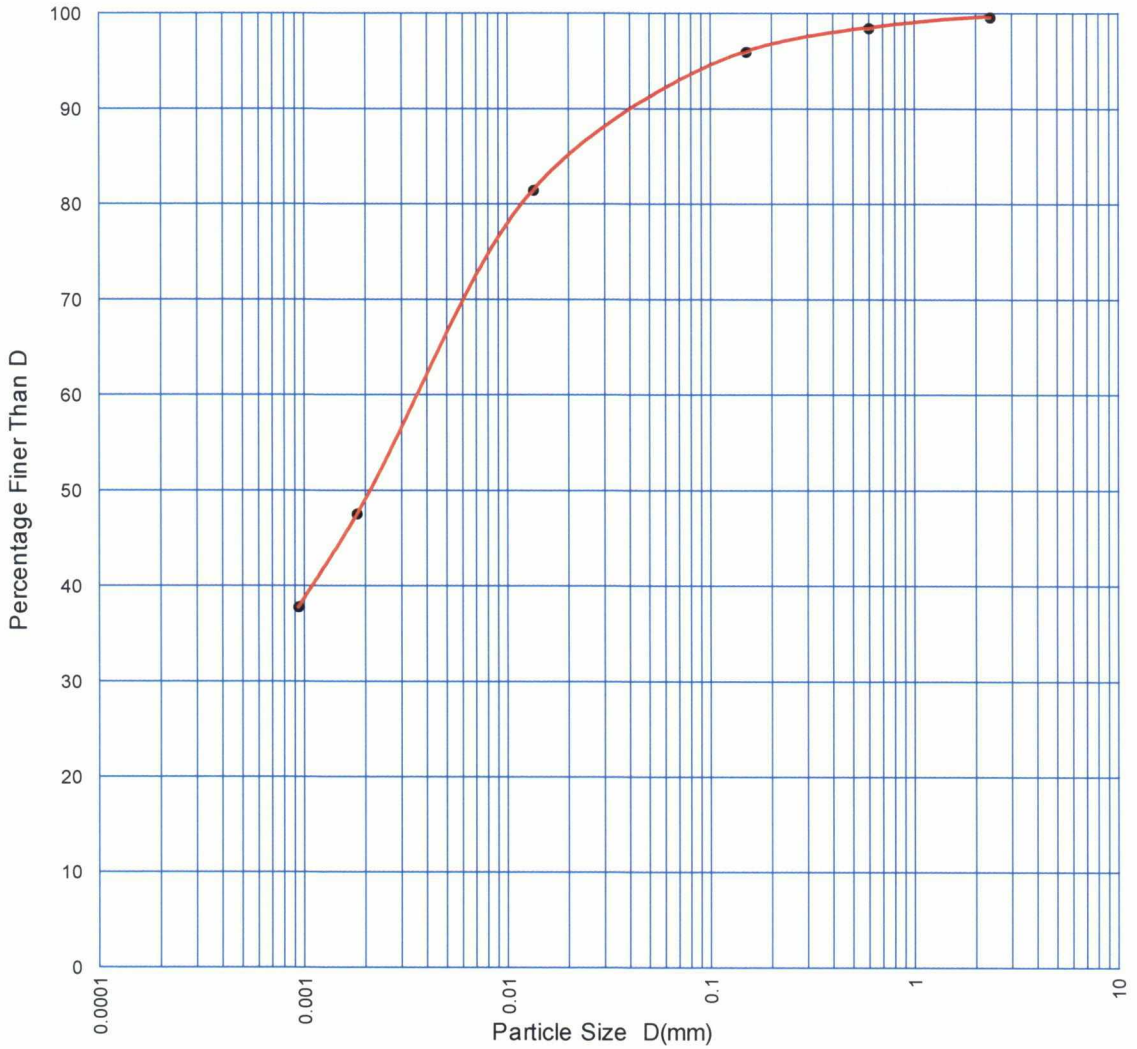
		PLATE	GS-08
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ. No.: IEOT/GEO/07/2018-19 REP. No. : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (Platform) FIELD : Gulf of Kutch SAMPLE: W-31 DEPTH : 50.00m	DATE	01/08/18
		PREPARED BY	D Lakshminarasimhan
		CHECKED BY	Alankar Srivastava



CLAY	SILT	FINE SAND	MEDIUM SAND	COARSE SAND	GRAVEL

GRAIN SIZE DISTRIBUTION

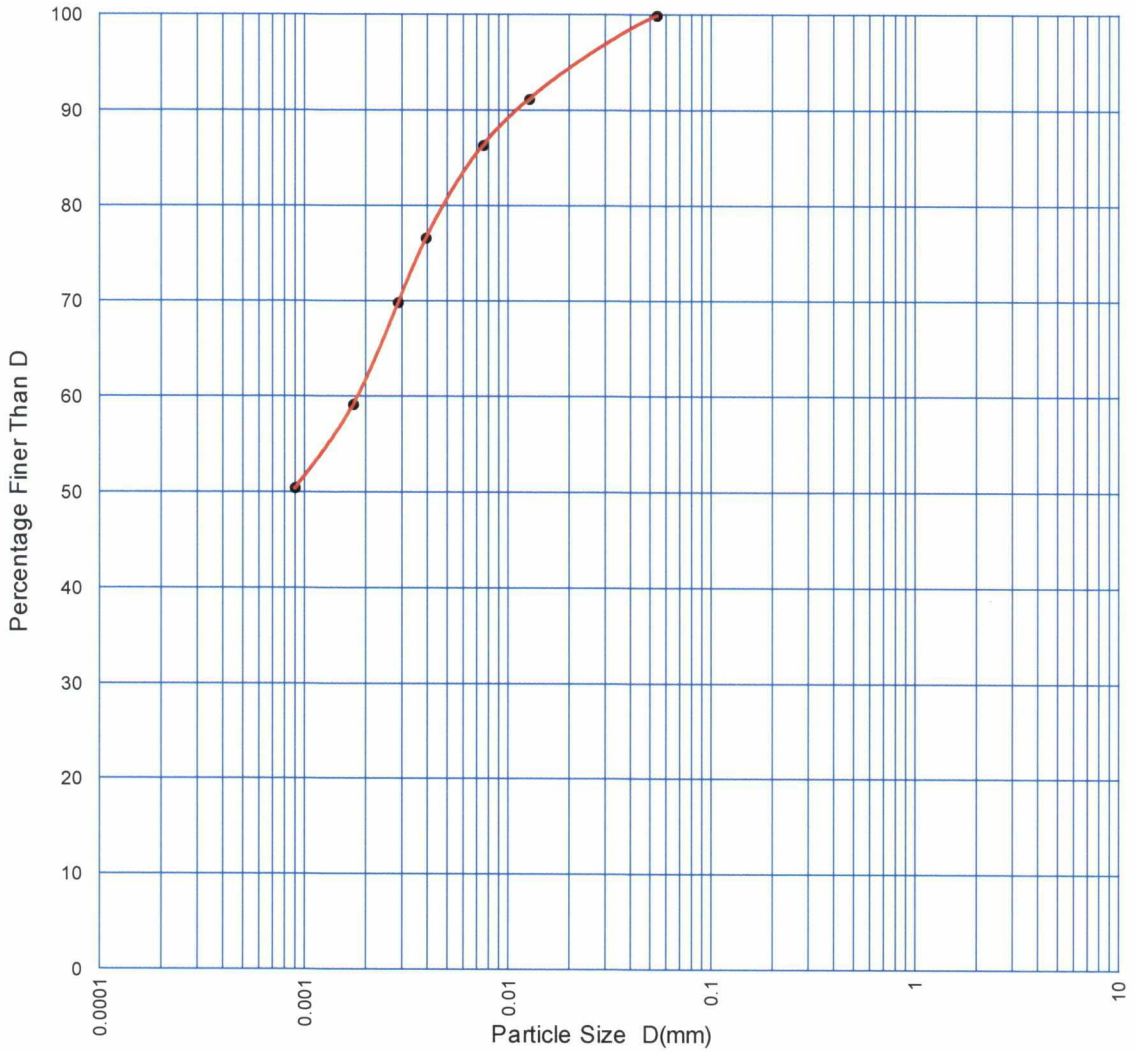
		PLATE	GS-09
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ. No.: IEOT/GEO/07/2018-19 REP. No. : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (Platform) FIELD : Gulf of Kutch SAMPLE: W-33-II DEPTH : 58.10m	DATE	01/08/18
		PREPARED BY	D Lakshminarasimhan
		CHECKED BY	Alankar Srivastava



CLAY	SILT	FINE SAND	MEDIUM SAND	COARSE SAND	GRAVEL

GRAIN SIZE DISTRIBUTION

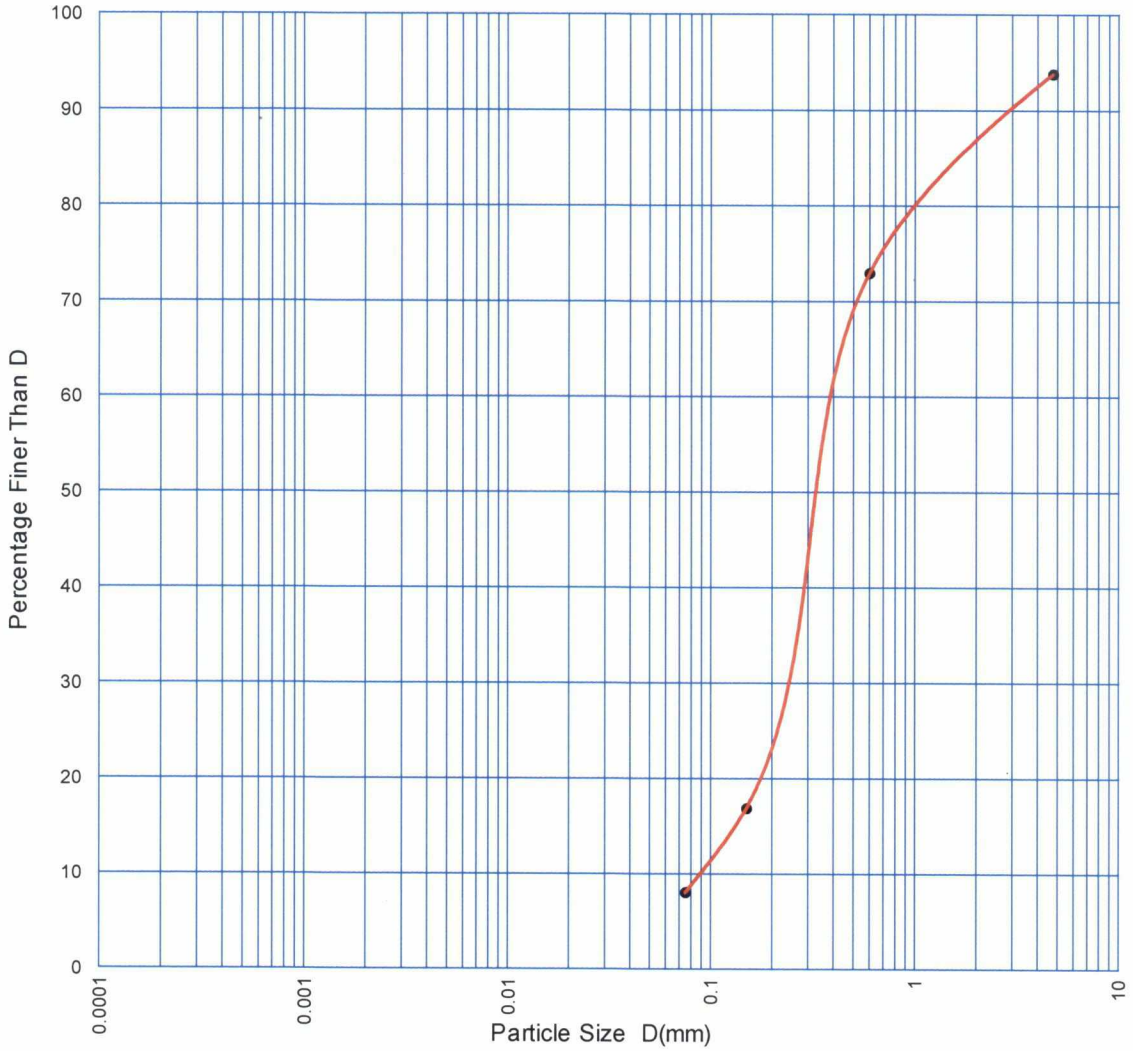
		PLATE	GS-10
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ. No.: IEOT/GEO/07/2018-19 REP. No. : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (Platform) FIELD : Gulf of Kutch SAMPLE: W-35 DEPTH : 66.00m	DATE	01/08/18
		PREPARED BY	D Lakshminarasimhan
		CHECKED BY	Alankar Srivastava



CLAY	SILT	FINE SAND	MEDIUM SAND	COARSE SAND	GRAVEL

GRAIN SIZE DISTRIBUTION

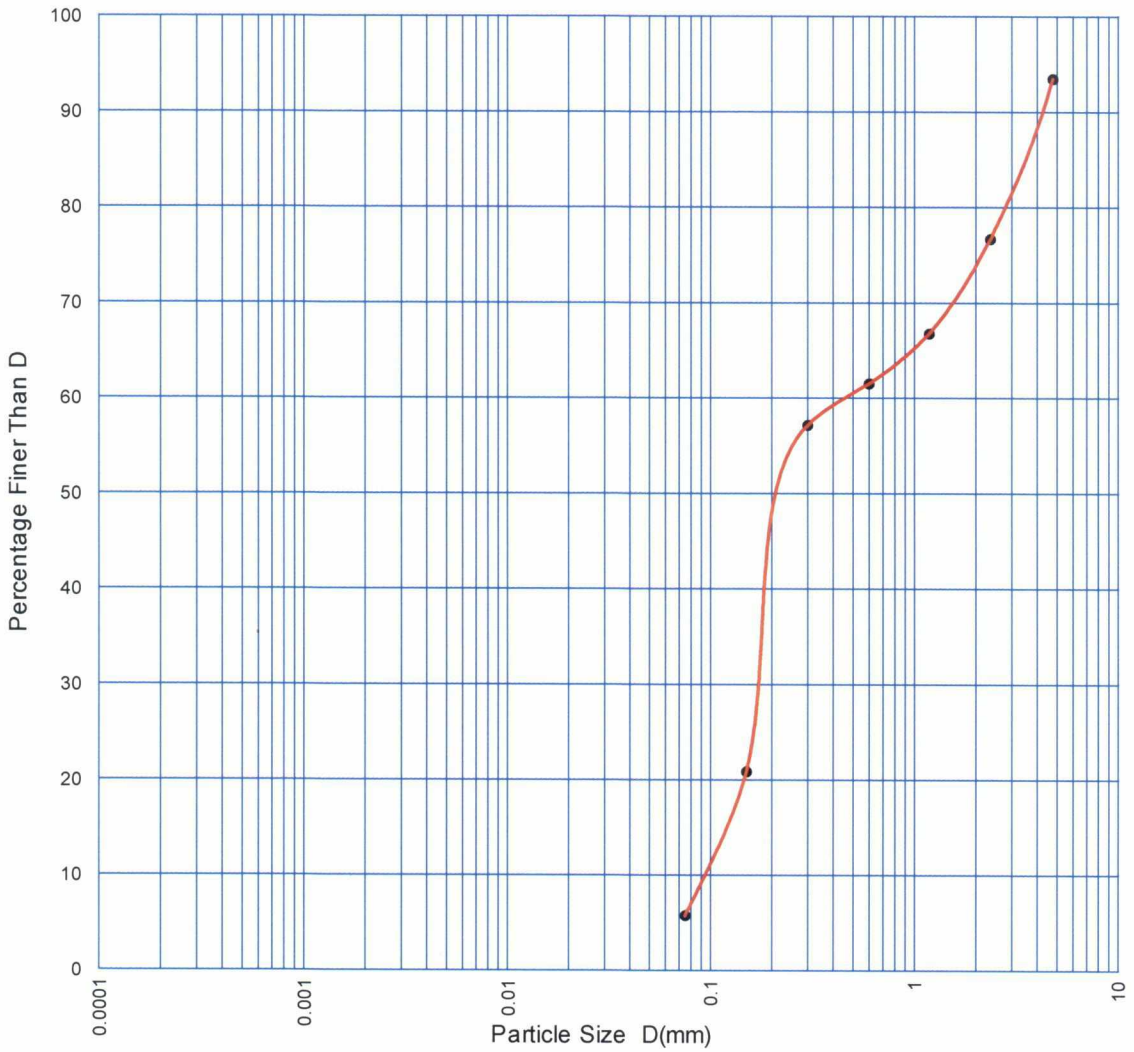
		PLATE	GS-11
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ. No.: IEOT/GEO/07/2018-19 REP. No. : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (Platform) FIELD : Gulf of Kutch SAMPLE: W-36 DEPTH : 70.20m	DATE	01/08/18
		PREPARED BY	D Lakshminarasimhan
		CHECKED BY	Alankar Srivastava



CLAY	SILT	FINE SAND	MEDIUM SAND	COARSE SAND	GRAVEL

GRAIN SIZE DISTRIBUTION

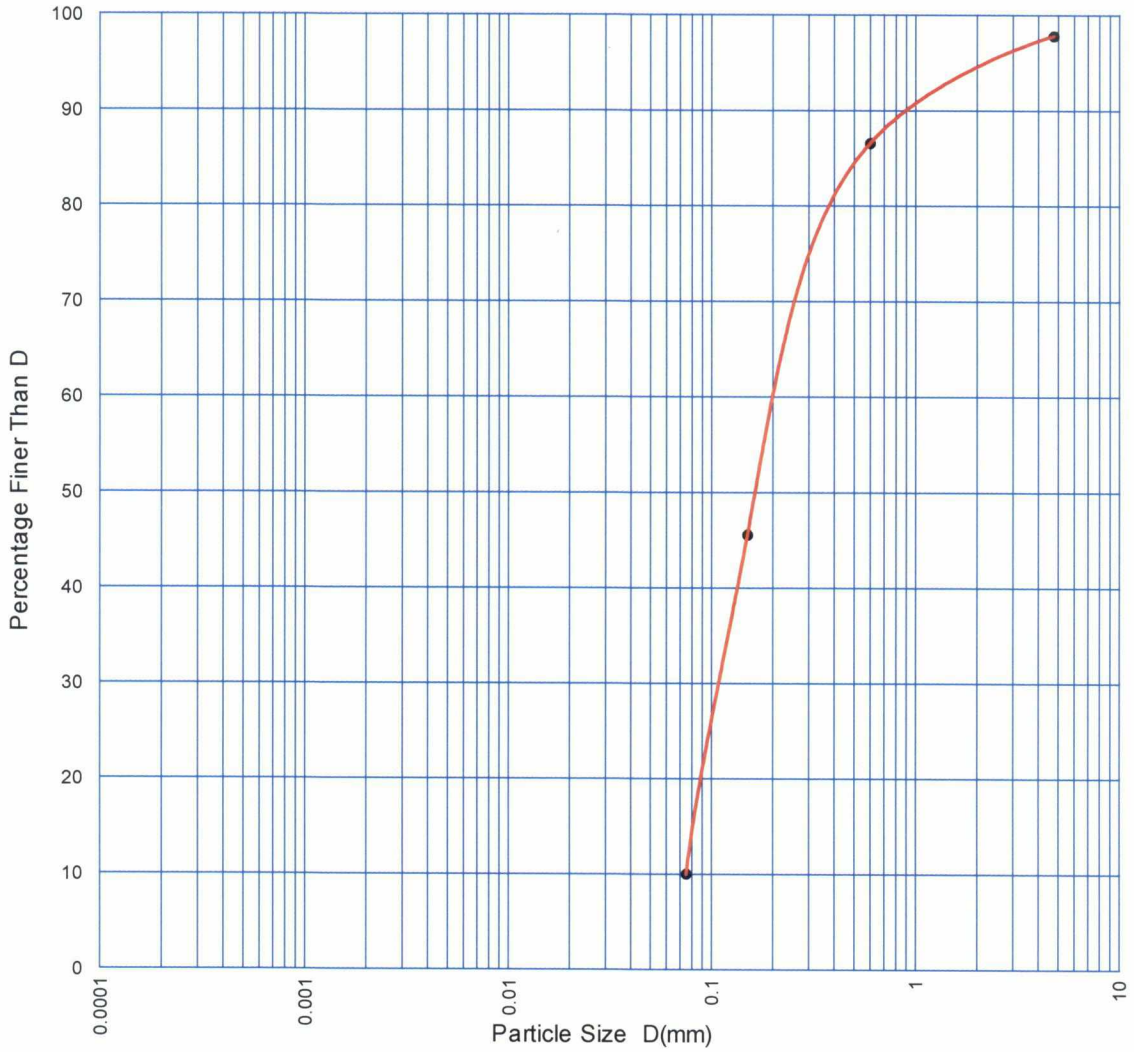
		PLATE	GS-12
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ. No.: IEOT/GEO/07/2018-19 REP. No. : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (Platform) FIELD : Gulf of Kutch SAMPLE: W-37 DEPTH : 74.10m	DATE	01/08/18
		PREPARED BY	D Lakshminarasimhan
		CHECKED BY	Alankar Srivastava



CLAY	SILT	FINE SAND	MEDIUM SAND	COARSE SAND	GRAVEL

GRAIN SIZE DISTRIBUTION

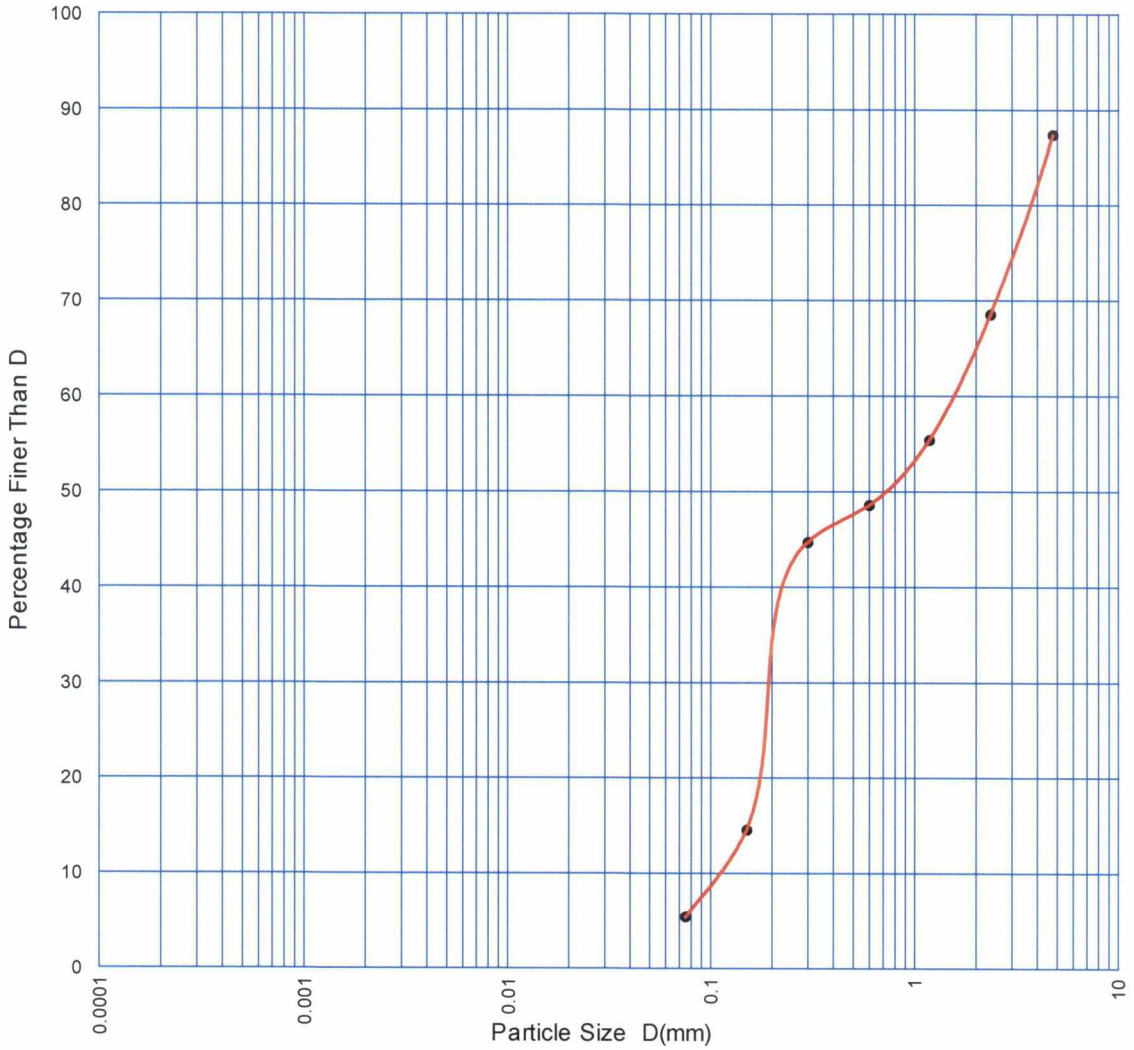
		PLATE	GS-13
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ. No.: IEOT/GEO/07/2018-19 REP. No. : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (Platform) FIELD : Gulf of Kutch SAMPLE: W-41 DEPTH : 90.00m	DATE	01/08/18
		PREPARED BY	D Lakshminarasimhan
		CHECKED BY	Alankar Srivastava



CLAY	SILT	FINE SAND	MEDIUM SAND	COARSE SAND	GRAVEL

GRAIN SIZE DISTRIBUTION

		PLATE	GS-14
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ. No.: IEOT/GEO/07/2018-19 REP. No. : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (Platform) FIELD : Gulf of Kutch SAMPLE: W-46 DEPTH : 110.00m	DATE	01/08/18
		PREPARED BY	D Lakshminarasimhan
		CHECKED BY	Alankar Srivastava

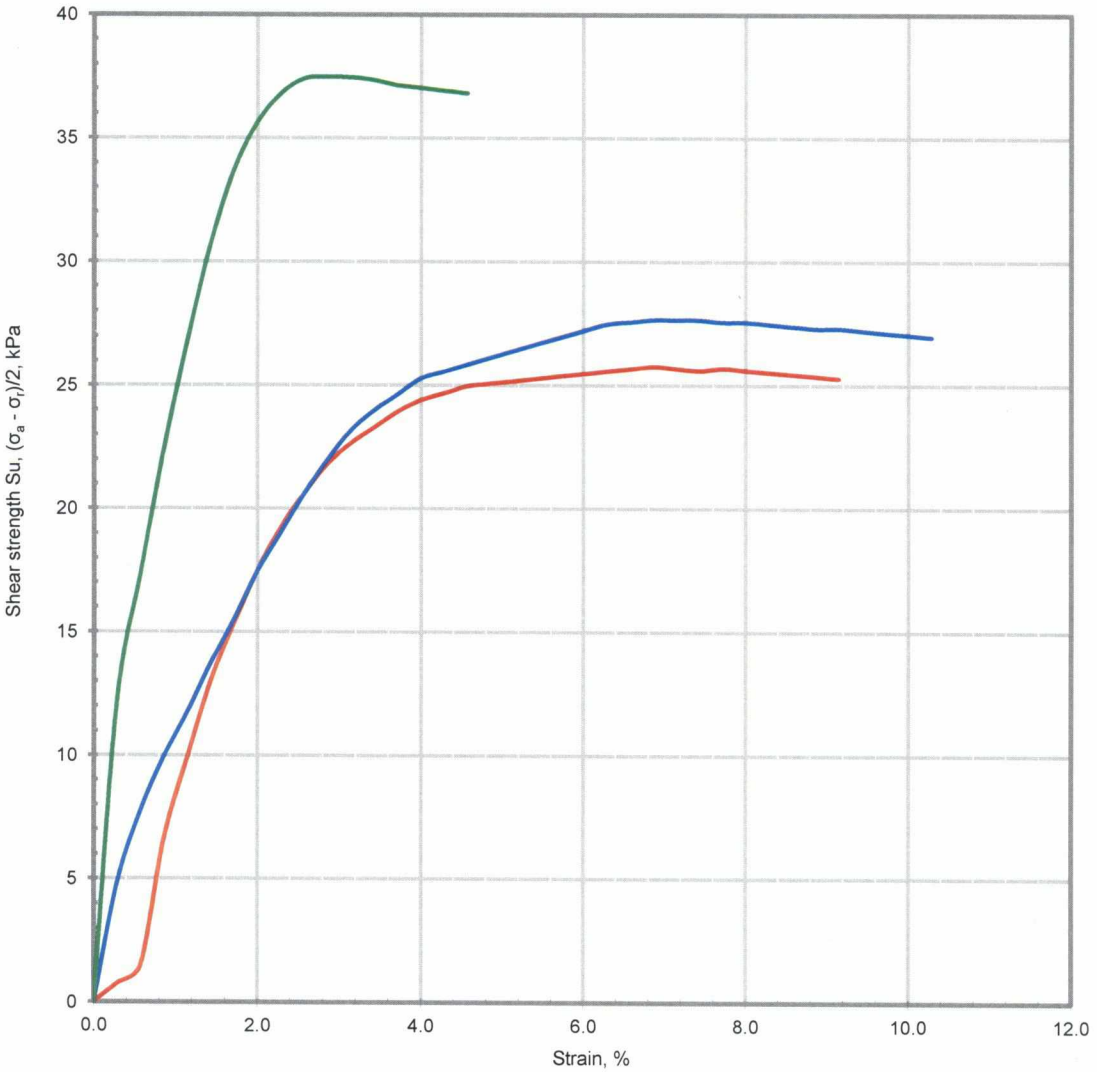


CLAY	SILT	FINE SAND	MEDIUM SAND	COARSE SAND	GRAVEL

GRAIN SIZE DISTRIBUTION

		PLATE	GS-15
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJ. No.: IEOT/GEO/07/2018-19 REP. No. : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (Platform) FIELD : Gulf of Kutch SAMPLE: W-49 DEPTH : 122.00m	DATE	01/08/18
		PREPARED BY	D Lakshminarasimhan
		CHECKED BY	Alankar Srivastava

Stress - Strain Curves
Unconsolidated Undrained Triaxial Compression Test



Failure Type	
A	Bulge
B	Inclined Shear
C	Multiple Shear
D	Vertical Shear

Symbol	Sample No	Depth m	ϕ mm	Ht mm	W %	γ' kN/m^3	σ_r kN/m^2	S_u kN/m^2	ε_f %	ε_{50} %	Type of Failure
	UU - 18	18.37	72	140	48.3	5.8	600	25.7	6.86	1.40	A, B
	UU - 19	19.81	72	140	52.6	6.9	640	27.6	6.86	1.44	A, B
	UU - 20	21.12	72	140	52.5	7.1	660	37.5	2.86	0.65	A, B

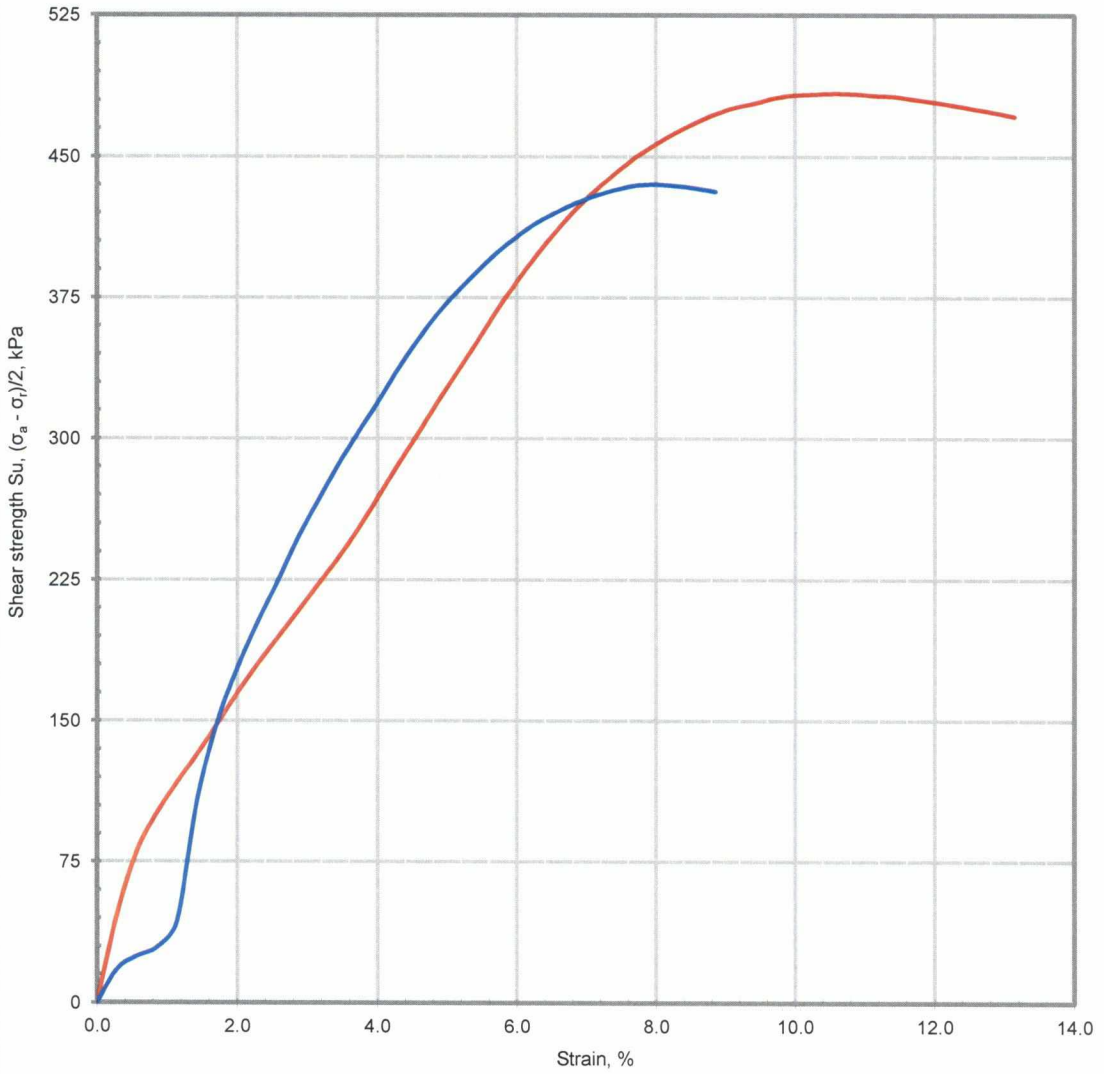
ONGC
INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY

PROJECT NO:IEOT/GEO/07/2018-19
PREPARED BY: COMACOE
CHECKED BY: IEOT
LOCATION: GK-28-10 (PLATFORM), GULF OF KUTCH

(Alankar Srivastava)

PLATE: UU-1
(ONBOARD)

Stress - Strain Curves
Unconsolidated Undrained Triaxial Compression Test



Failure Type	
A	Bulge
B	Inclined Shear
C	Multiple Shear
D	Vertical Shear

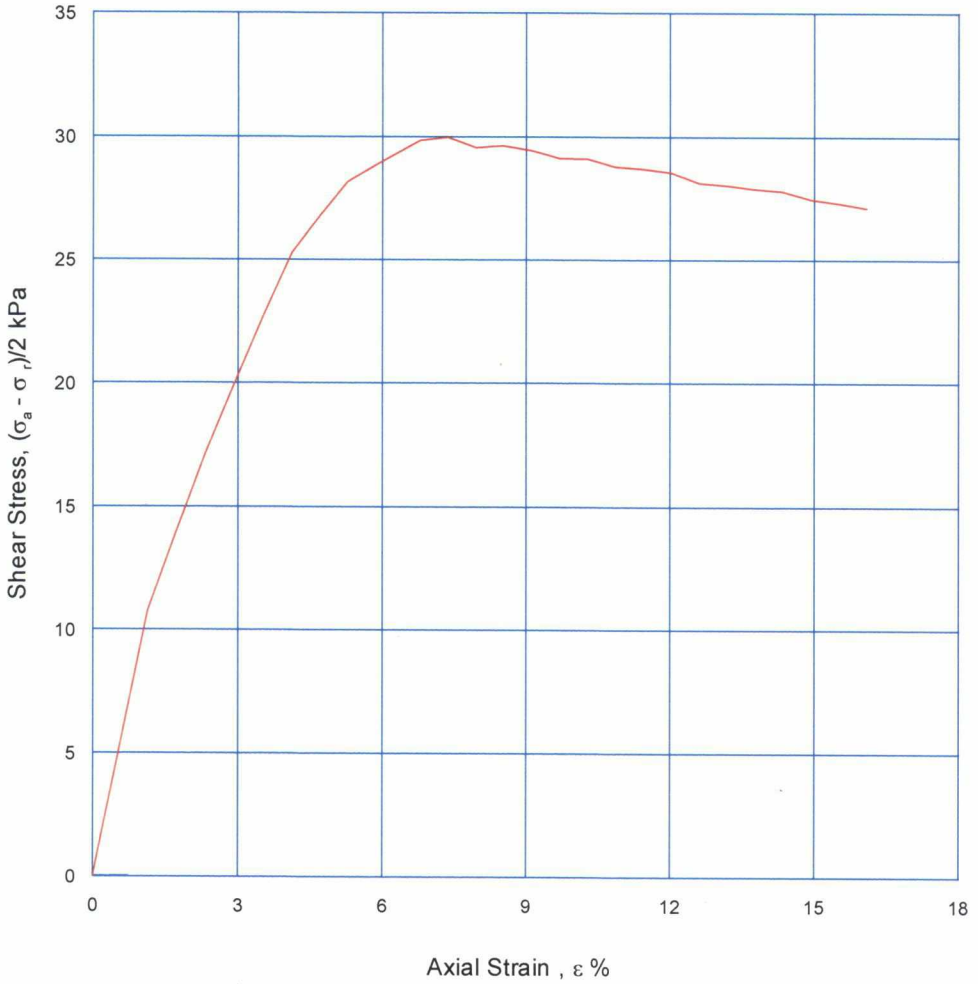
Symbol	Sample No	Depth m	ϕ mm	Ht mm	W %	γ' kN/m^3	σ_r kN/m^2	S_u kN/m^2	S_u (upto 10% strain) kN/m^2	ε_f %	ε_{50} %	Type of Failure
	UU - 30	46.12	72	140	18.7	11.3	1130	483.2	482.3	10.57	3.55	A, B
	UU-36	70.20	72	140	20.9	11.0	1776	435.0	435.0	8.00	2.49	A

ONGC
INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY

PROJECT NO:IEOT/GEO/07/2018-19
PREPARED BY: COMACOE
CHECKED BY: IEOT
LOCATION: GK-28-10 (PLATFORM), GULF OF KUTCH

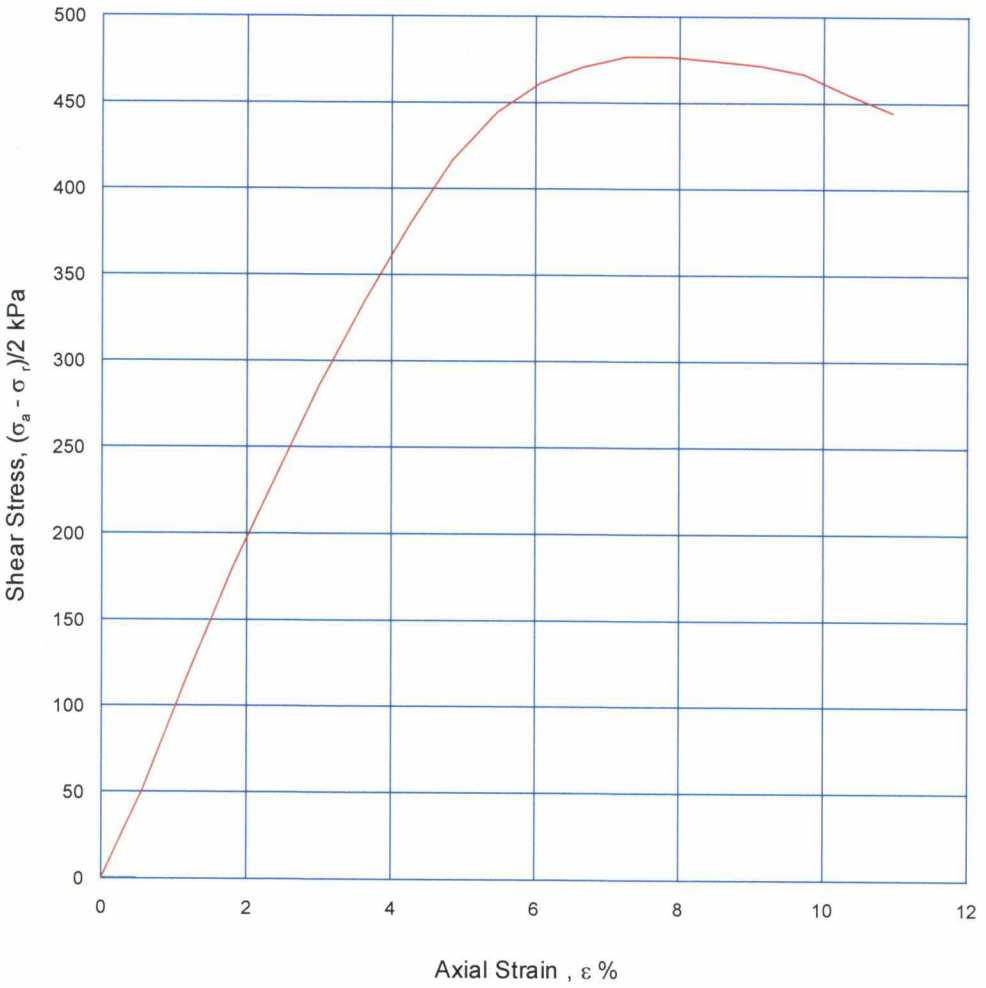
(Alankar Srivastava)

PLATE: UU-2
(ONBOARD)



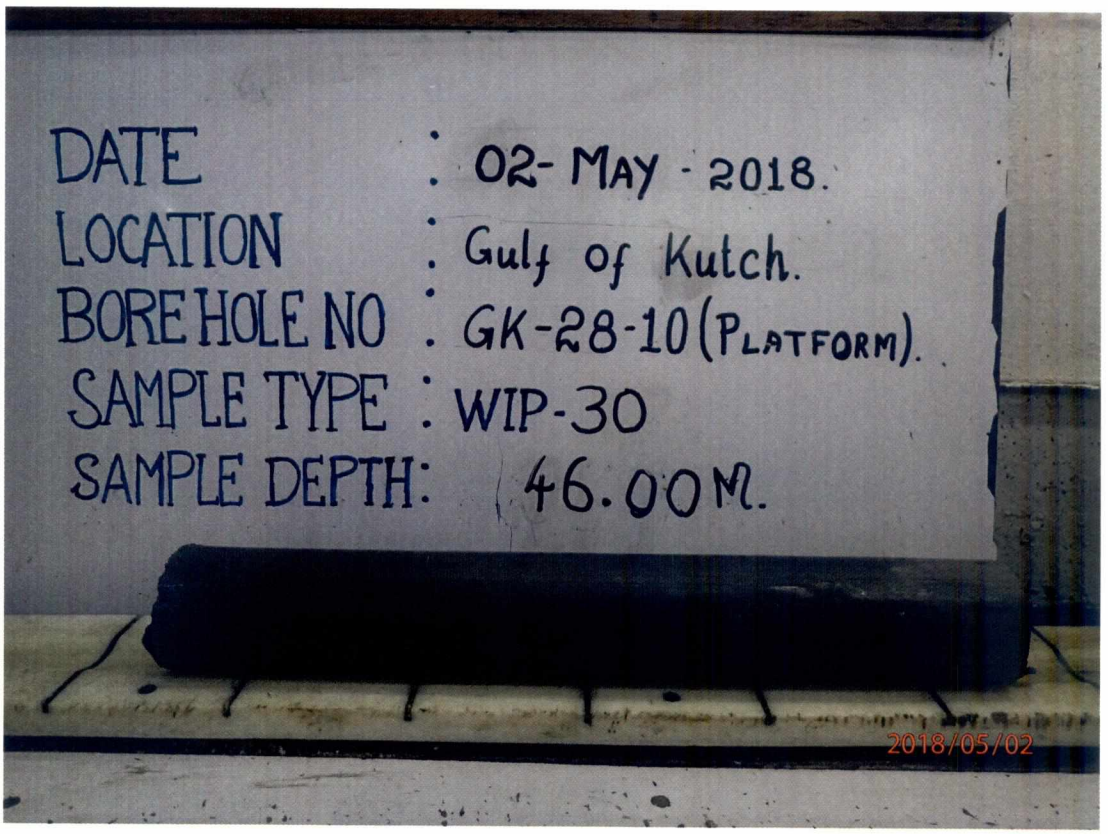
σ_r :800 kPa
S_u :29.98 kPa
ε_f :7.37 %
ε_{50} :1.92 %
Type of failure : Bulging Failure

UNCONSOLIDATED UNDRAINED TRIAXIAL TEST		PLATE	LAB UU-01
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (PLATFORM) FIELD : Gulf of Kutch SAMPLE : W - 19 - I DEPTH : 19.64 m	DATE	1/08/2018
		PREPARED BY	
		CHECKED BY	



σ_r :800 kPa
S_u :477.04 kPa
ε_f :7.43 %
ε_{50} :2.46 %
Type of failure : Multiple Shear Failure

UNCONSOLIDATED UNDRAINED TRIAXIAL TEST		PLATE	LAB UU-02
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : GK-28-10 (PLATFORM) FIELD : Gulf of Kutch SAMPLE : W - 30 DEPTH : 46.29 m	DATE	1/08/2018
		PREPARED BY	
		CHECKED BY	



PH1: Photographs of Clay Samples

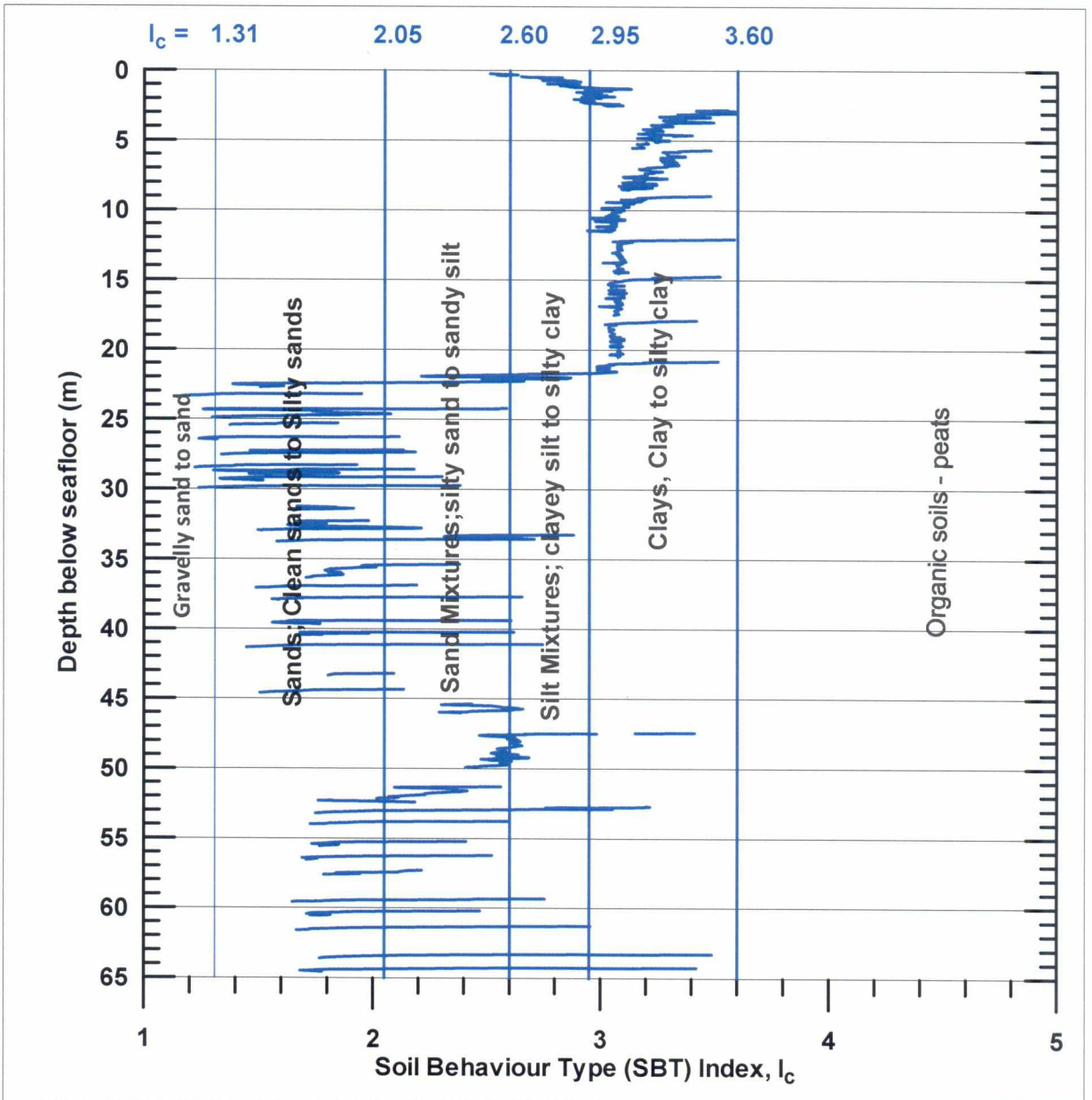
PH2: Photographs of Sand Samples

APPENDIX-C

APPENDIX – C

SUMMARY OF SOIL CONDITIONS

Soil Behaviour Type Index vs. Depth	**C-1-1 & C-1-2**
Water content vs. Depth	**C-2**
Unit weight vs. Depth	**C-3**
Overburden pressure vs. Depth	**C-4**
Unified Soil Classification System	**C-5**
Activity Chart	**C-6**
Liquidity Index vs. Depth	**C-7**
Normalized Undrained Shear Strength vs. Depth	**C-8**
Sensitivity vs. Depth	**C-9**
Overconsolidation Ratio and Relative Density vs. Depth	**C-10-1 & C-10-2**
Clays on Nagaraj Framework	**C-11**

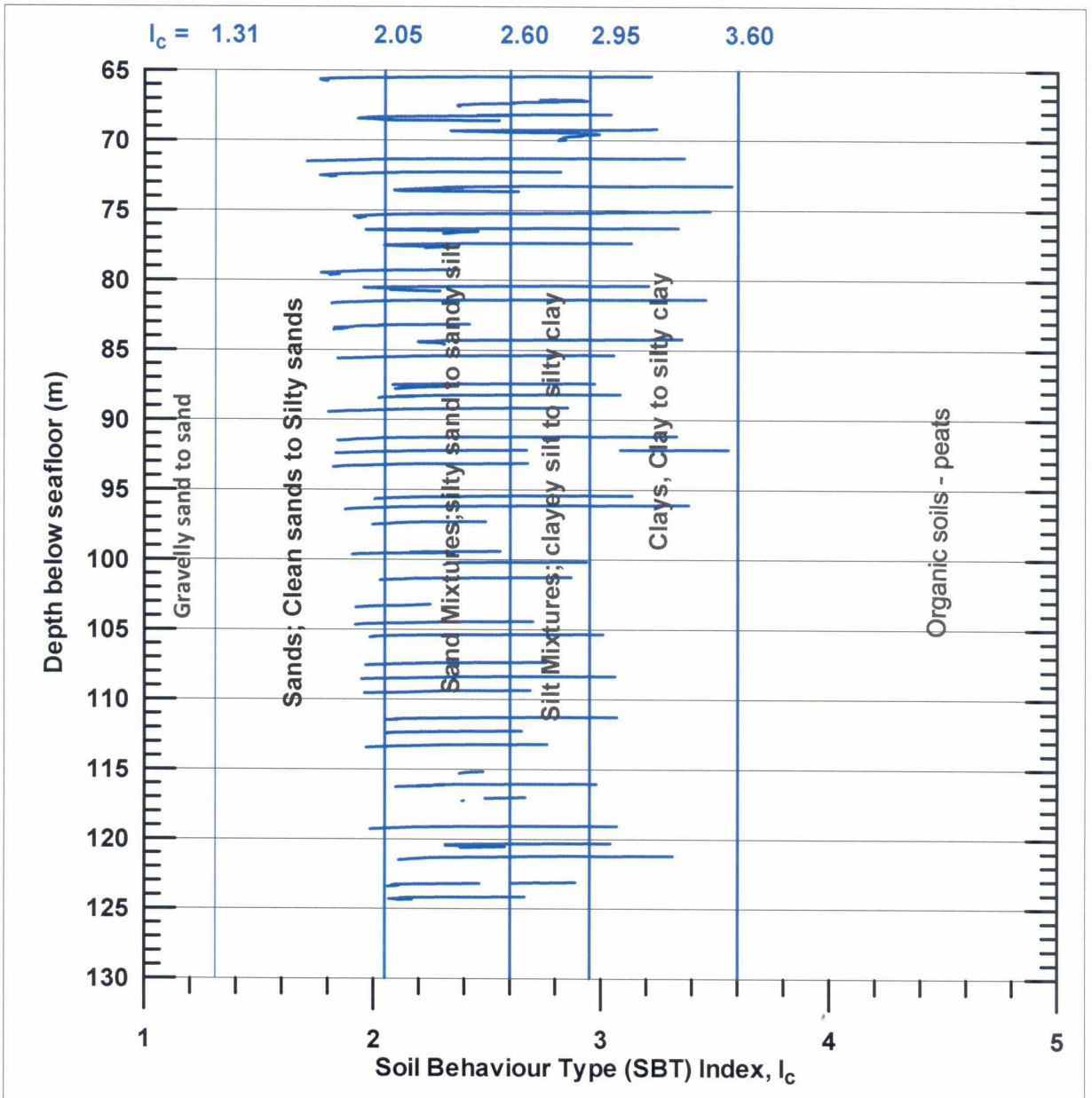


SOIL BEHAVIOUR TYPE(SBT) INDEX vs DEPTH

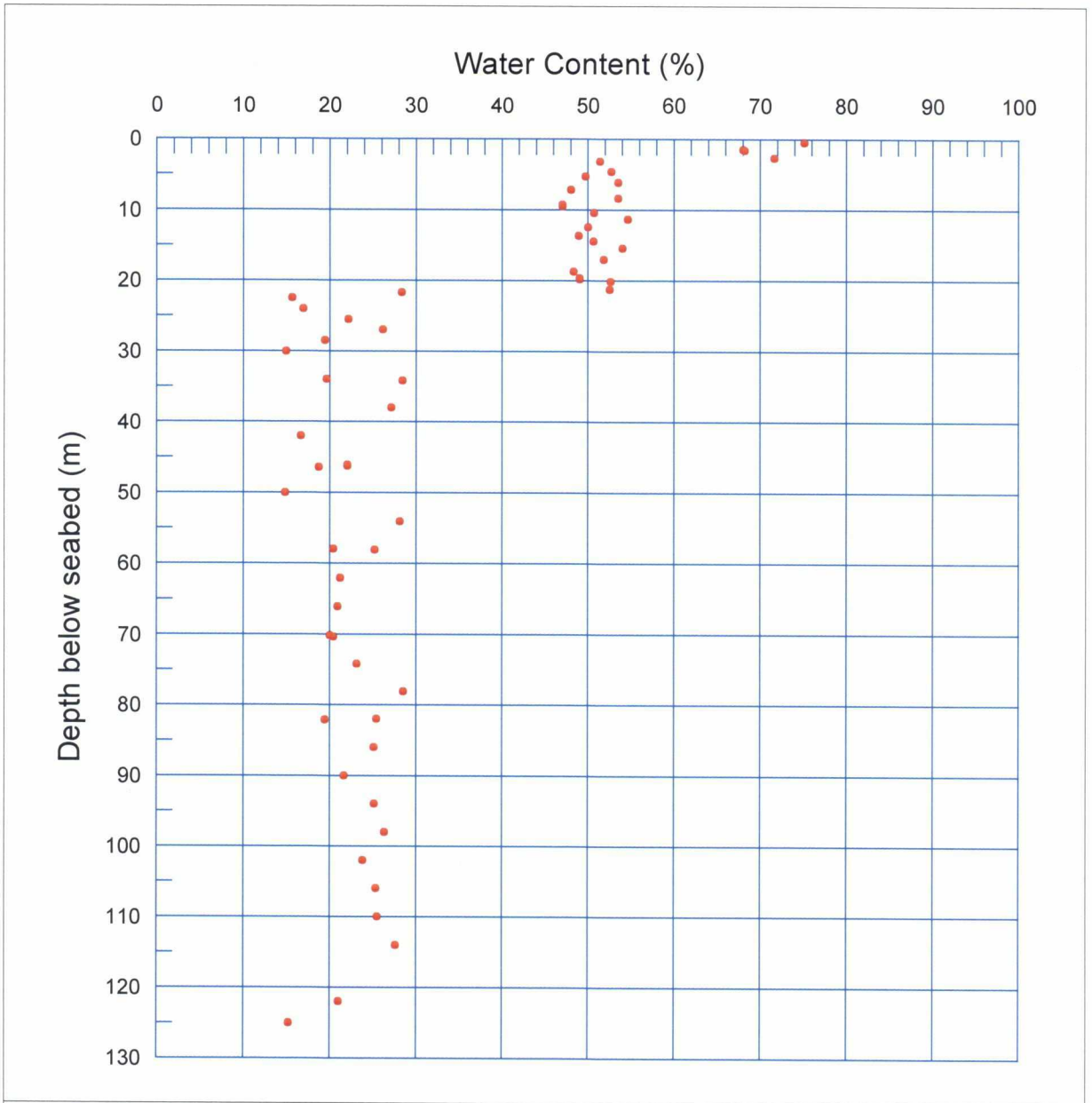
PLATE	C- 1-1		
DATE	16-08-2018		
PREPARED BY	Alankar Srivastava		
CHECKED BY	T. K. Roy		

ONGC
INSTITUTE OF
ENGINEERING AND
OCEAN TECHNOLOGY

PROJECT NO : IEOT/GEO/07/2018-19
REPORT NO : IEOT/GEO/07/2018-19
LOCATION : **GK-28-10 (PLATFORM)**
FIELD : **GULF OF KUTCH**

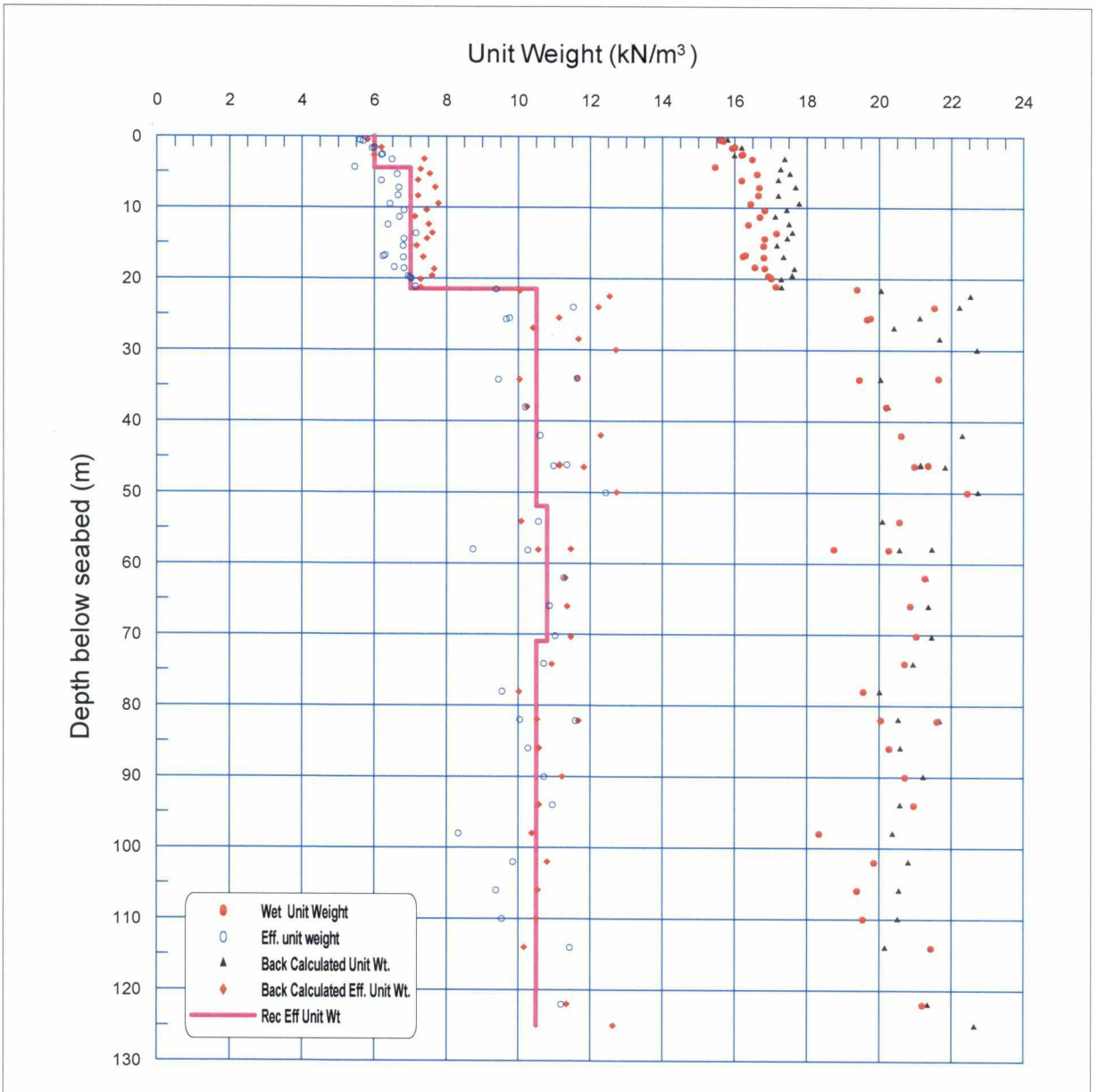


SOIL BEHAVIOUR TYPE(SBT) INDEX vs DEPTH		PLATE	C- 1-2
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : **GK-28-10 (PLATFORM)** FIELD : **GULF OF KUTCH**	DATE	16-08-2018
		PREPARED BY	Alankar Srivastava
		CHECKED BY	T. K. Roy

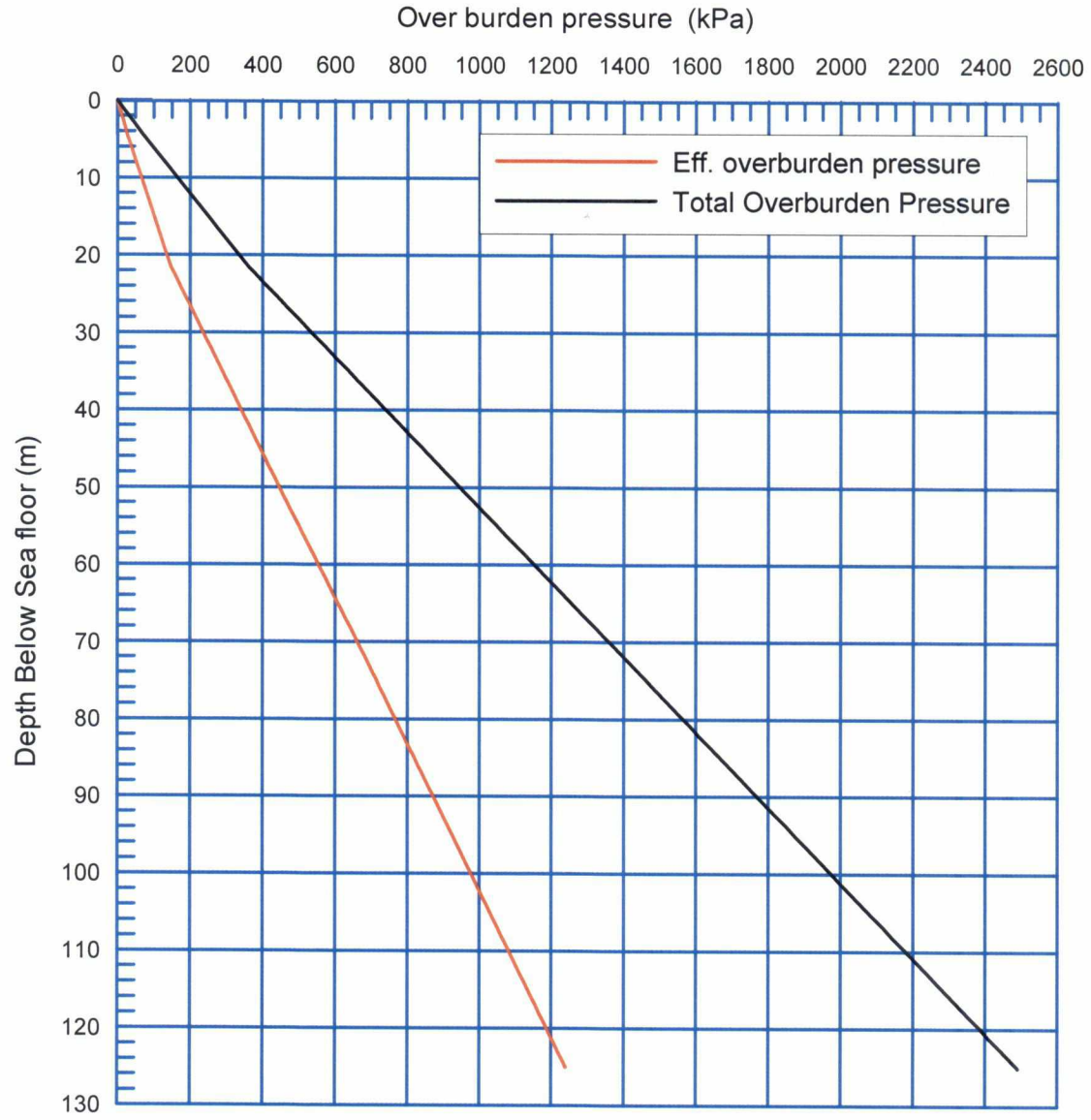


WATER CONTENT vs DEPTH

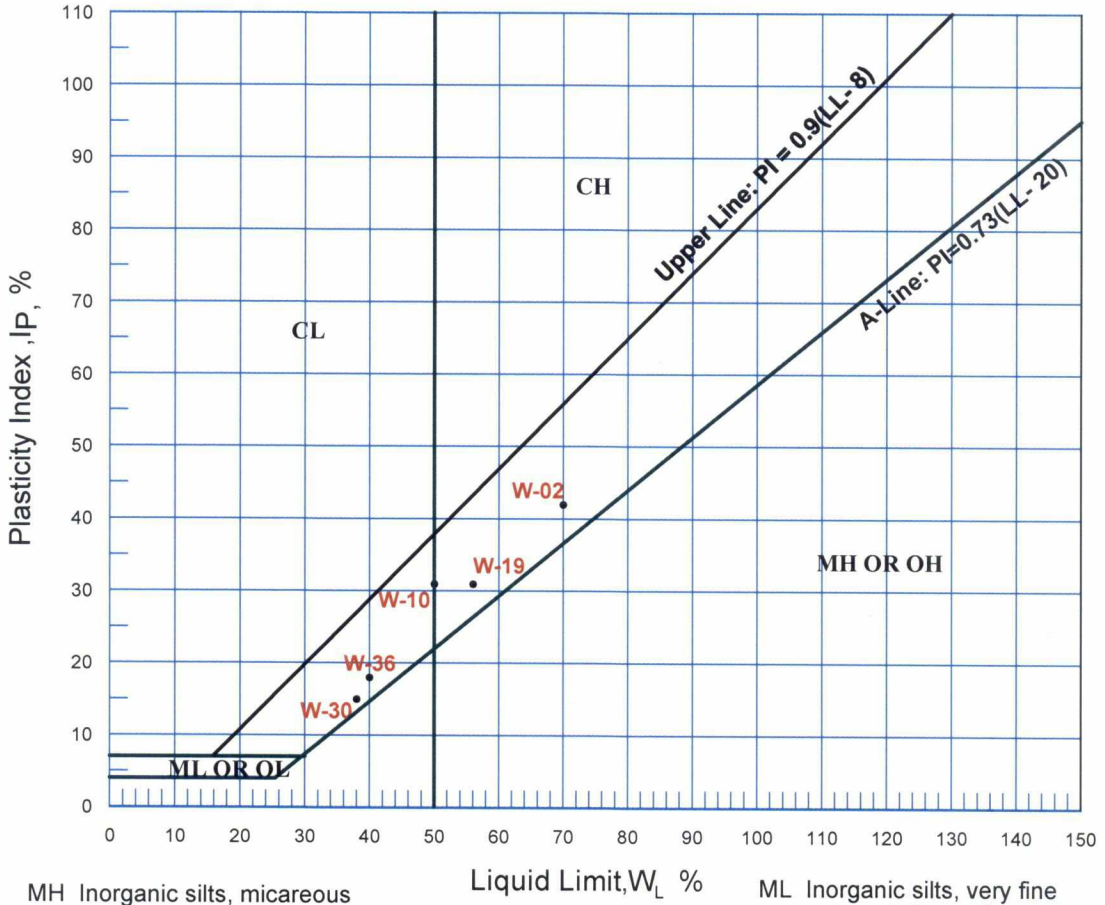
		PLATE	C-2
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : **GK-28-10 (PLATFORM)** FIELD : **GULF OF KUTCH**	DATE	16-08-2018
		PREPARED BY	Alankar Srivastava
		CHECKED BY	T. K. Roy



UNIT WEIGHT vs DEPTH		PLATE	C-3
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : **GK-28-10 (PLATFORM)** FIELD : **GULF OF KUTCH**	DATE	16-08-2018
		PREPARED BY	Alankar Srivastava
		CHECKED BY	T. K. Roy



OVERBURDEN PRESSURE vs DEPTH (Reference seabed)		PLATE	**C-4**
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : **GK-28-10 (PLATFORM)** FIELD : **GULF OF KUTCH**	DATE	16-08-2018
		PREPARED BY	Alankar Srivastava
		CHECKED BY	T. K. Roy



MH Inorganic silts, micareous or diatomaceous fine sands or silts,elastic silts

CH Inorganic clays of high plasticity,fat clays

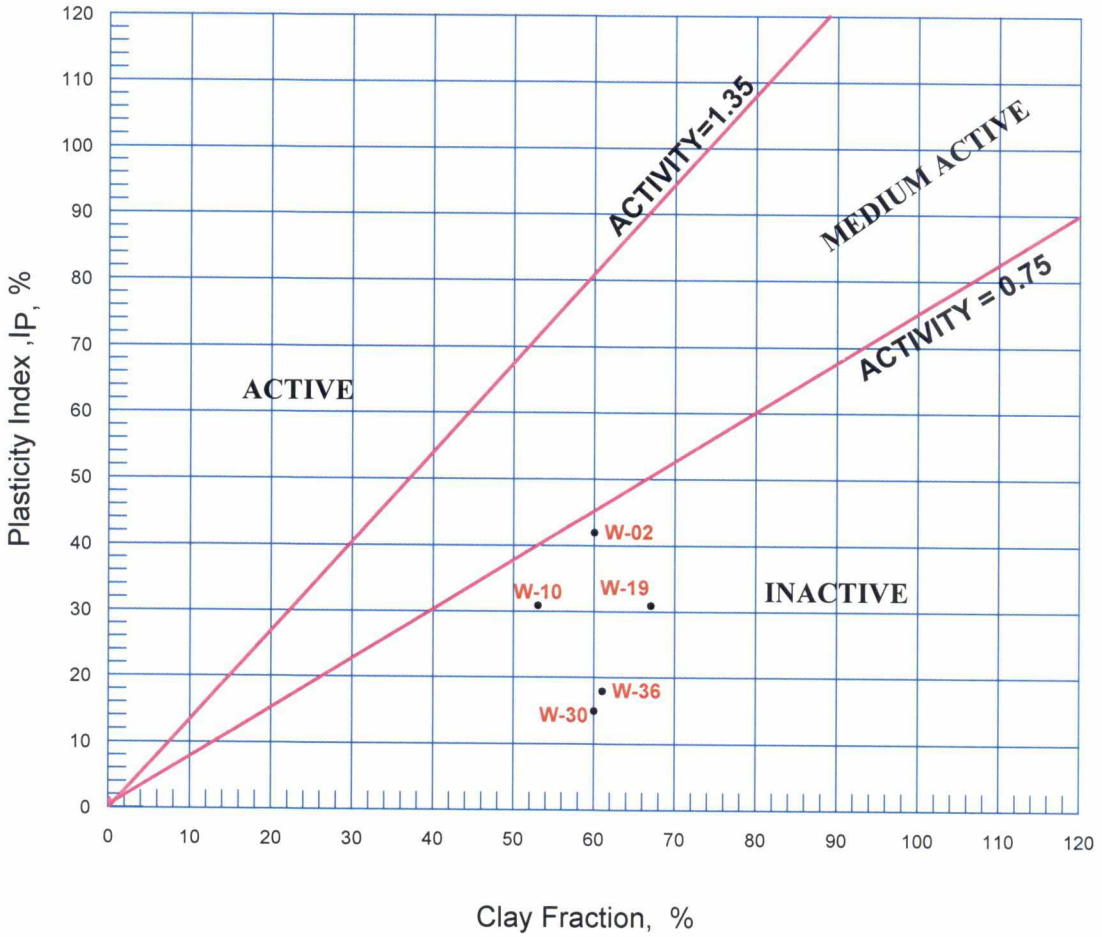
OH Organic clays of medium to high plasticity

ML Inorganic silts, very fine sands,rock flour, silty or clayey fine sands

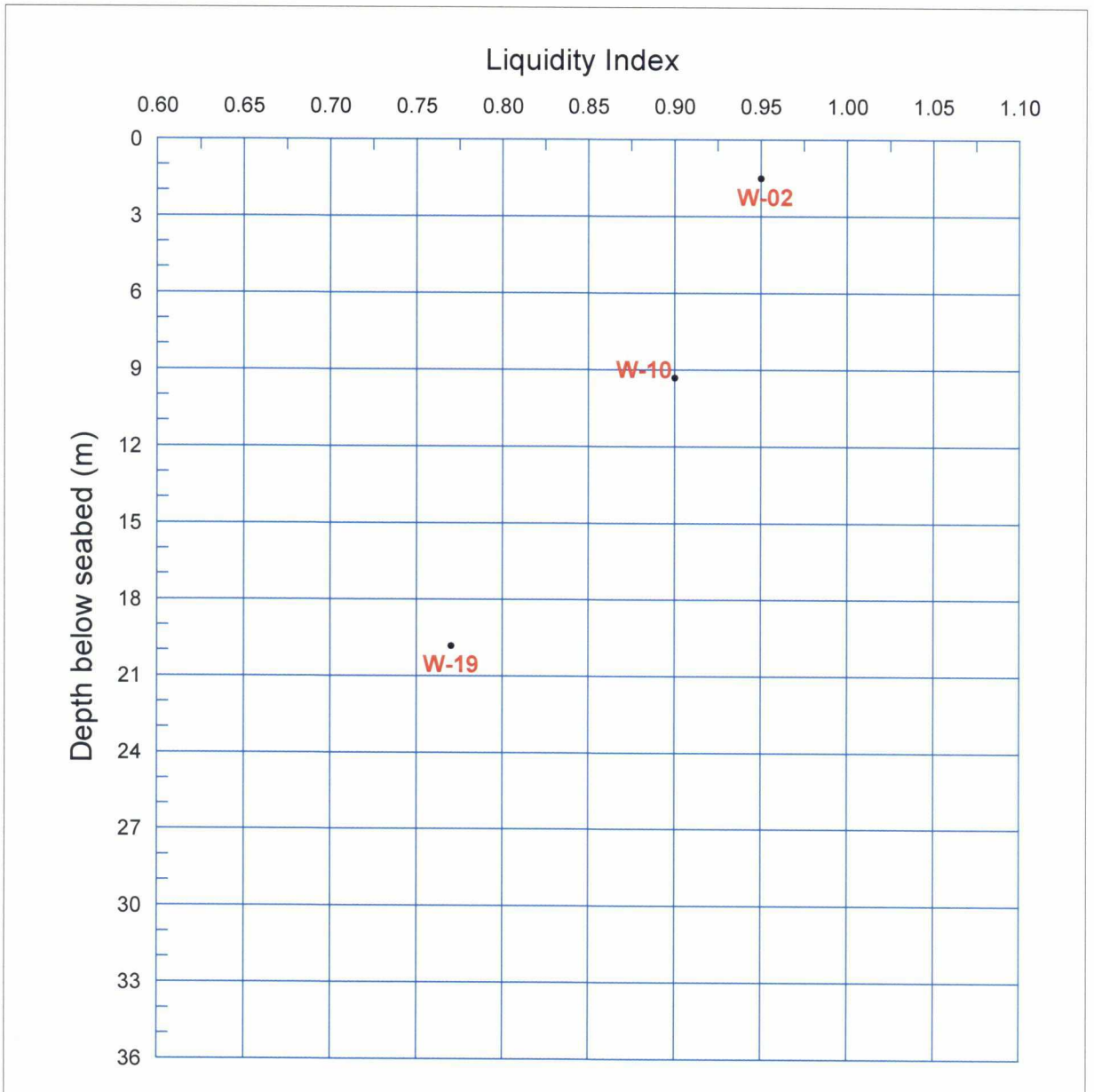
CL Inorganic clays of low to medium plasticity,gravelly clays,sandy clays,silty clays,lean clays

OL Organic silts and organic silty clays of low plasticity

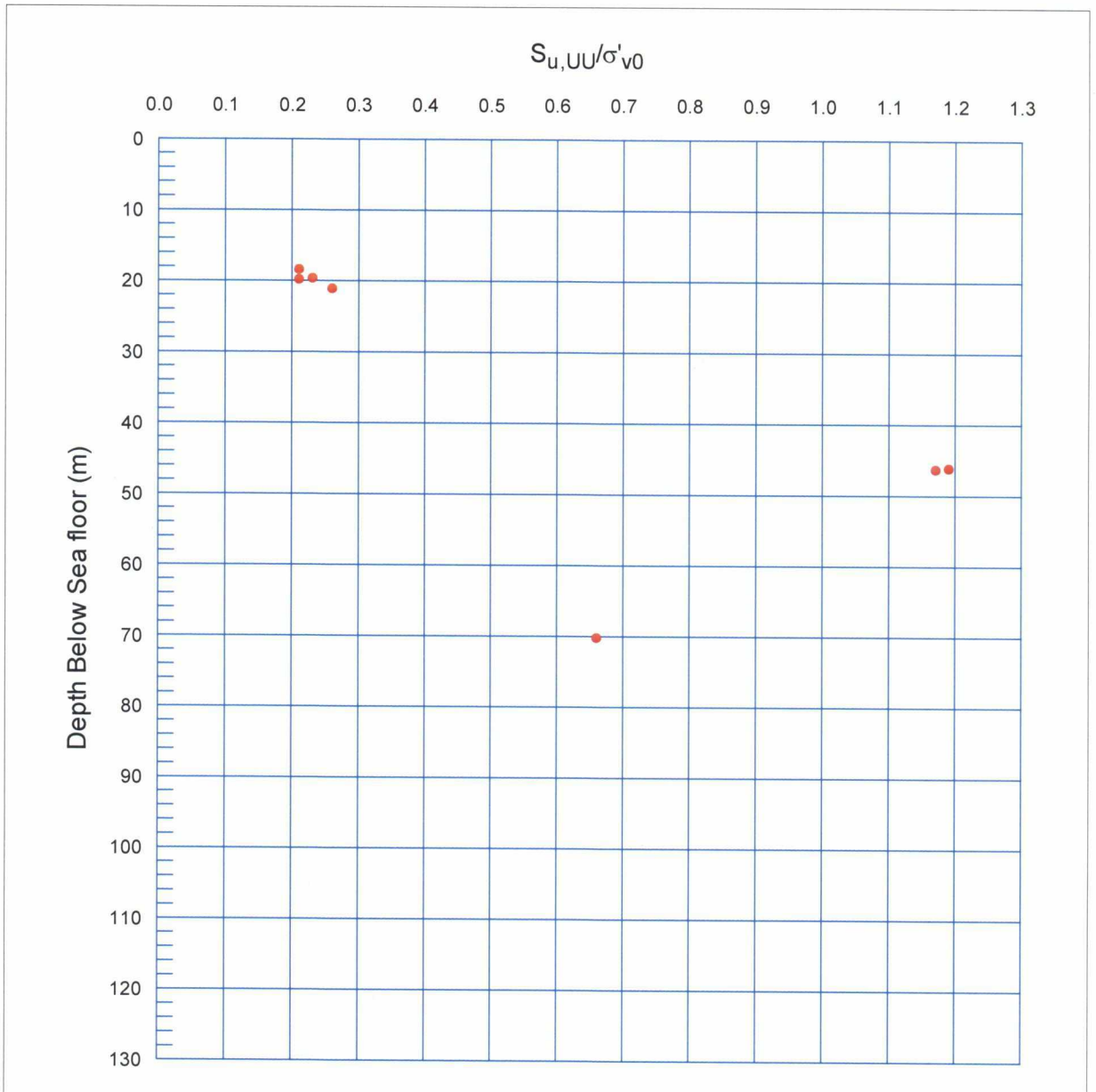
UNIFIED SOIL CLASSIFICATION SYSTEM		Plate	C-5
ONGC, INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : **GK-28-10 (PLATFORM)** FIELD : **GULF OF KUTCH**	Date	16-08-2018
		Prepared by:	Alankar Srivastava
		Checked by:	T. K. Roy



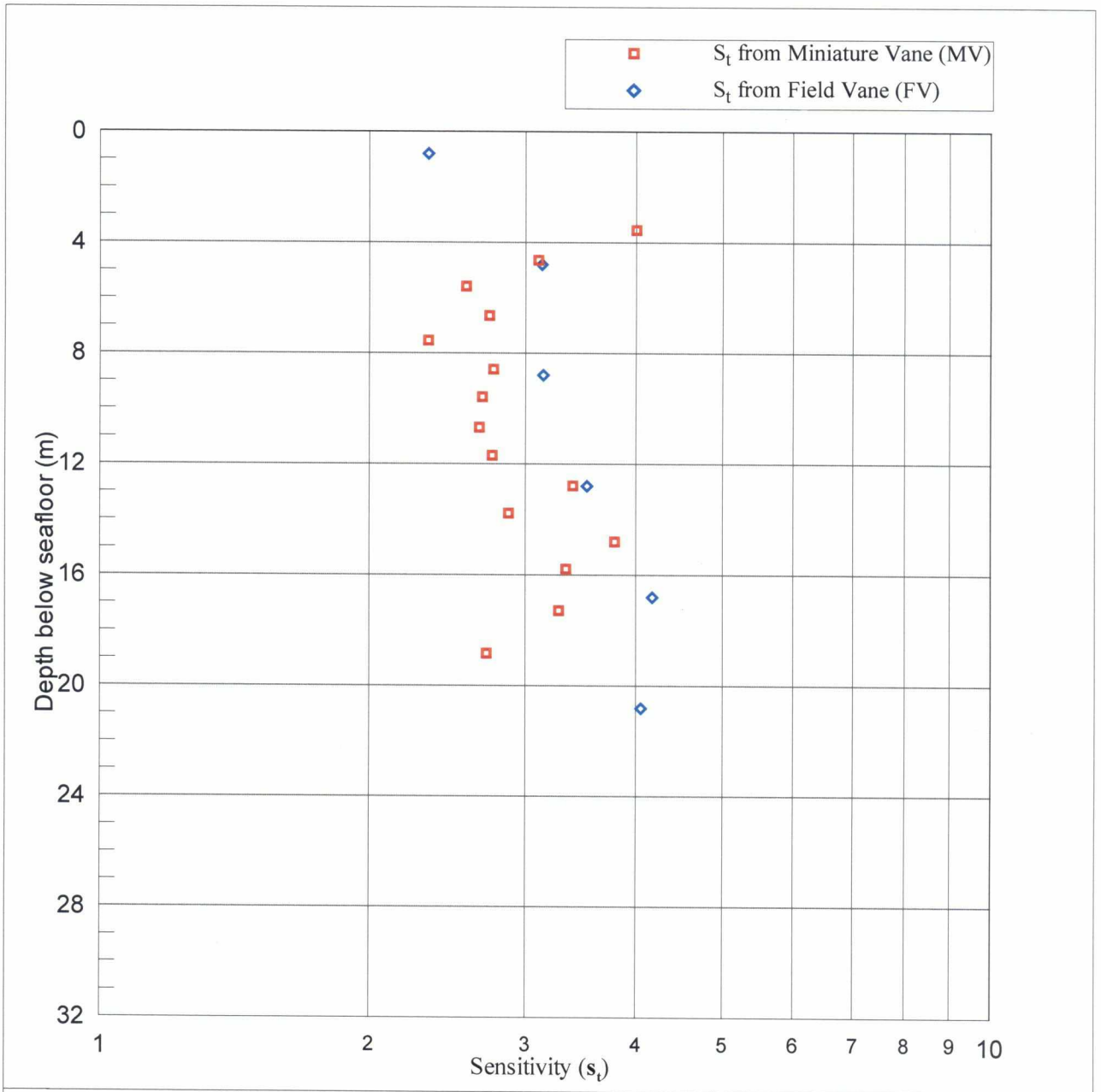
ACTIVITY CHART		PLATE	C-6
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : **GK-28-10 (PLATFORM)** FIELD : **GULF OF KUTCH**	DATE	16-08-2018
		PREPARED BY	Alankar Srivastava
		CHECKED BY	T. K. Roy



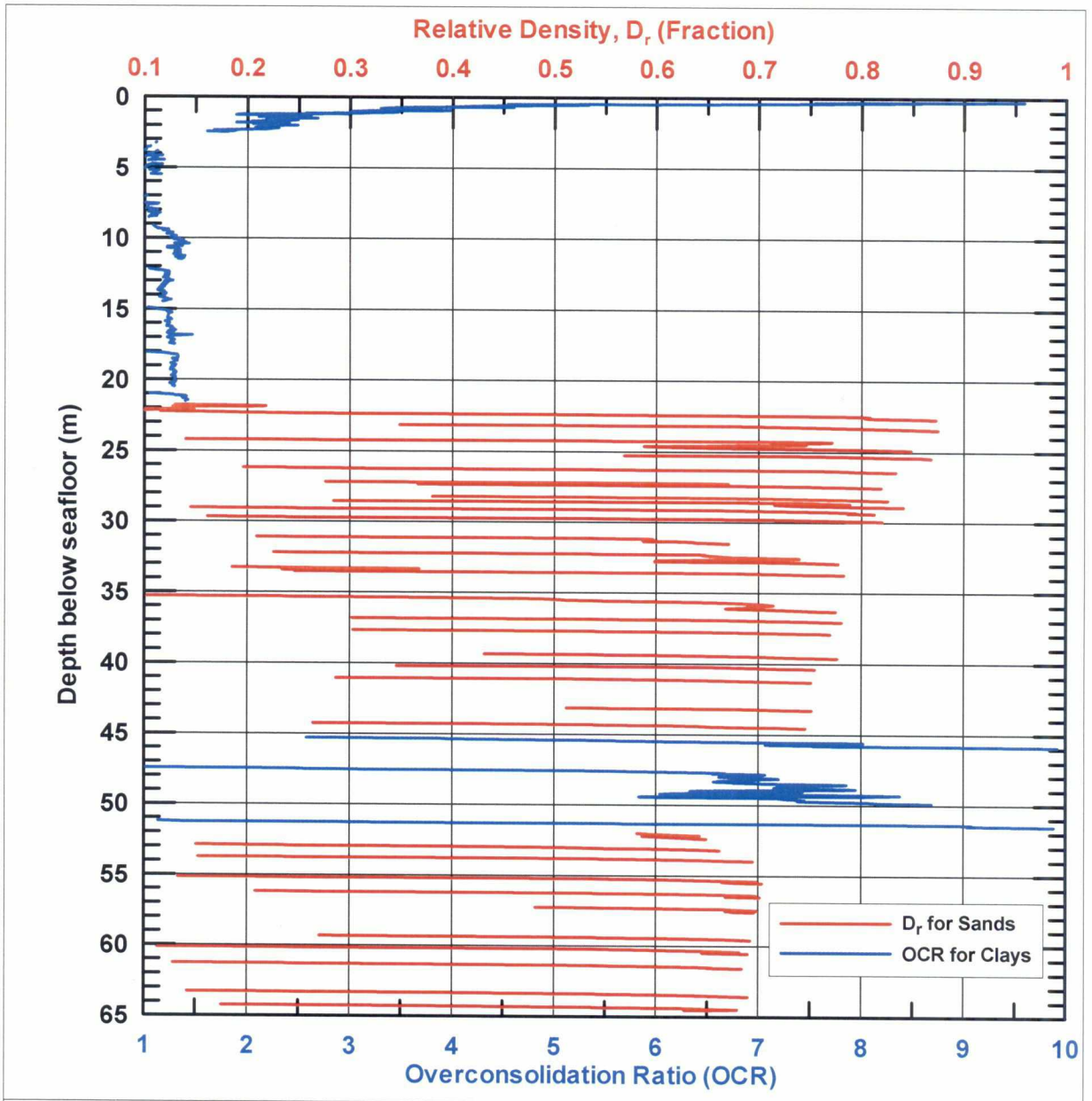
LIQUIDITY INDEX vs DEPTH		PLATE	C-7
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : **GK-28-10 (PLATFORM)** FIELD : **GULF OF KUTCH**	DATE	16-08-2018
		PREPARED BY	Alankar Srivastava
		CHECKED BY	T. K. Roy



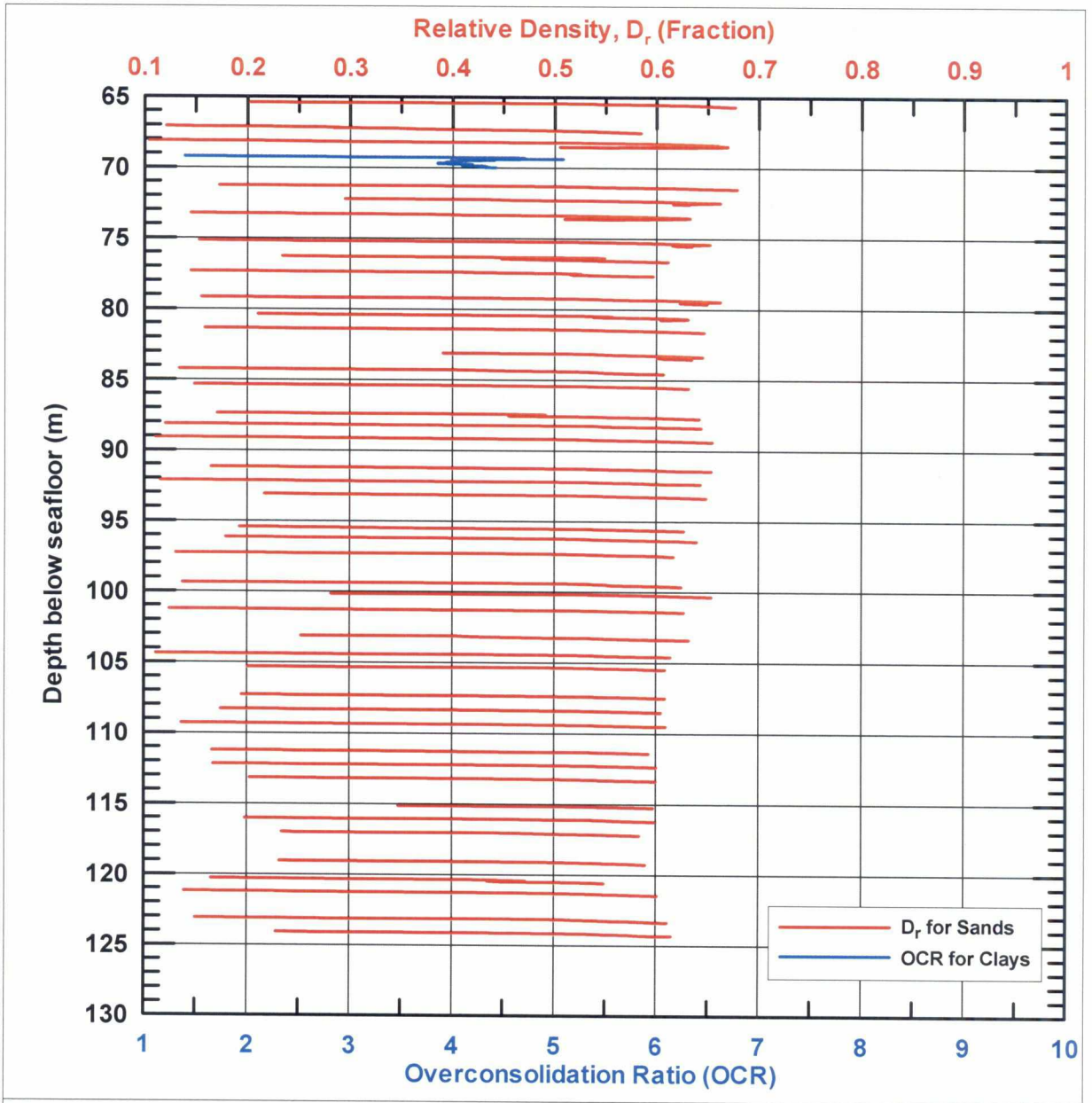
Normalised Shear Strength, S_{uUU}/σ'_{v0} vs Depth		PLATE	**C-8**
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : **GK-28-10 (PLATFORM)** FIELD : **GULF OF KUTCH**	DATE	16-08-2018
		PREPARED BY	Alankar Srivastava
		CHECKED BY	T. K. Roy



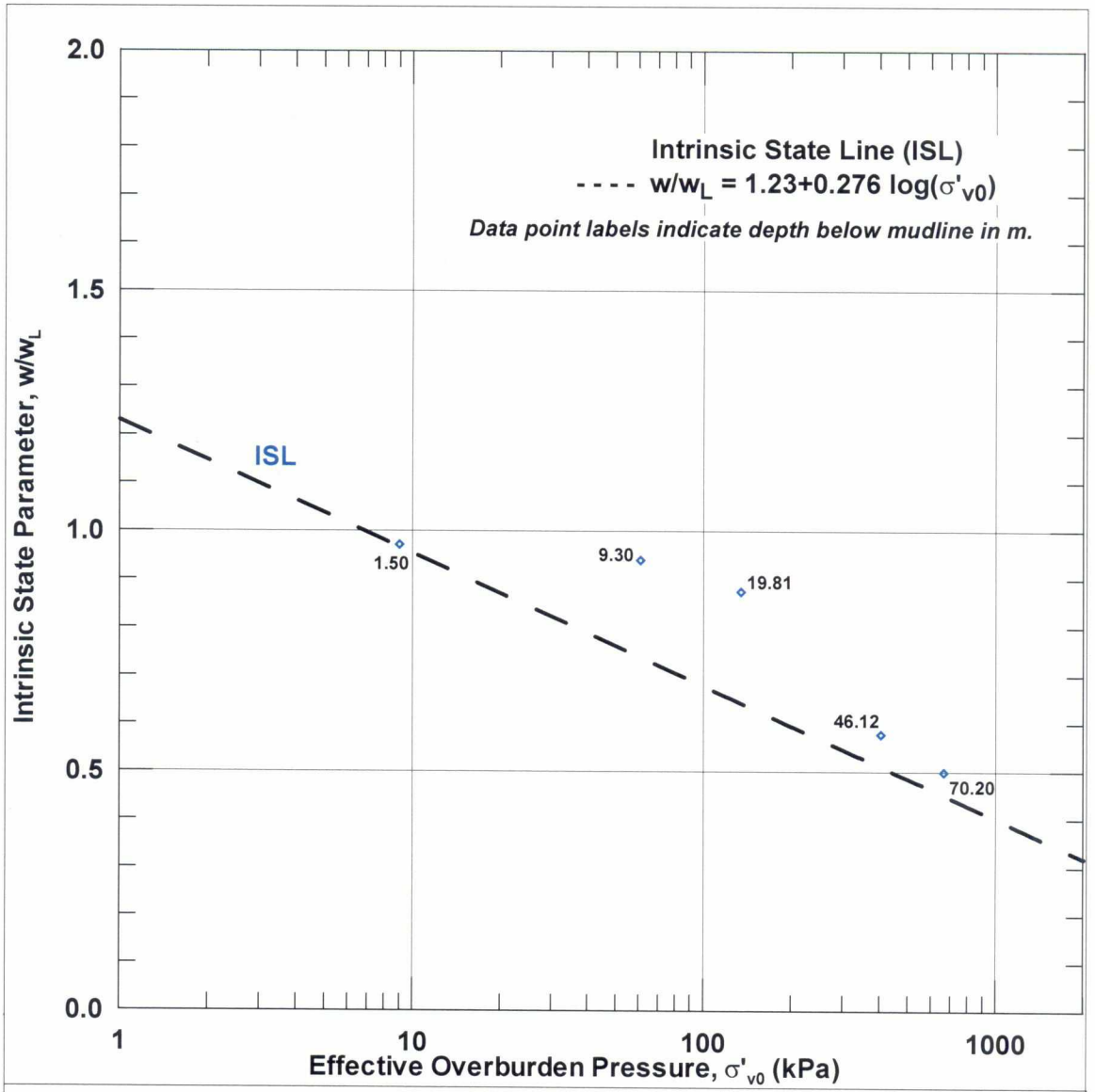
SENSITIVITY VS DEPTH		PLATE	**C-9**
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : **GK-28-10 (PLATFORM)** FIELD : **GULF OF KUTCH**	DATE	16-08-2018
		PREPARED BY	Alankar Srivastava
		CHECKED BY:	T. K. Roy



'OCR for Clays & D_r for Sands' vs DEPTH		PLATE	C- 10-1
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : **GK-28-10 (PLATFORM)** FIELD : **GULF OF KUTCH**	DATE	16-08-2018
		PREPARED BY	Alankar Srivastava
		CHECKED BY	T. K. Roy



'OCR for Clays & D_r for Sands' vs DEPTH		PLATE	C- 10-2
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : **GK-28-10 (PLATFORM)** FIELD : **GULF OF KUTCH**	DATE	16-08-2018
		PREPARED BY	Alankar Srivastava
		CHECKED BY	T. K. Roy



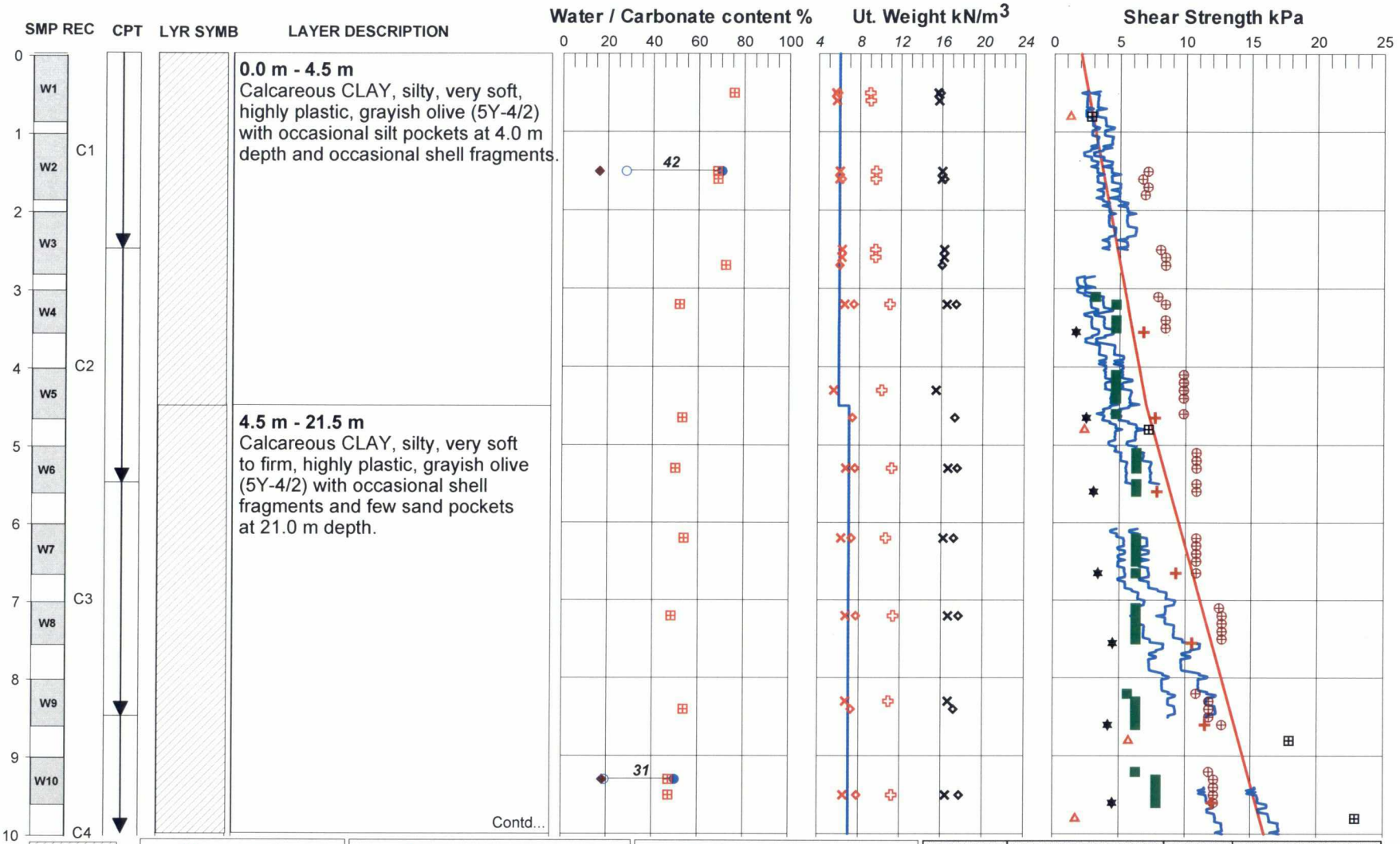
Intrinsic State Parameter vs Effective Overburden Pressure		PLATE	C-11
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : **GK-28-10 (PLATFORM)** FIELD : **GULF OF KUTCH**	DATE	16-08-2018
		PREPARED BY	Alankar Srivastava
		CHECKED BY	T. K. Roy

APPENDIX-D

APPENDIX - D

BORING LOGS, DRILLING LOG AND IN-SITU TEST RESULTS

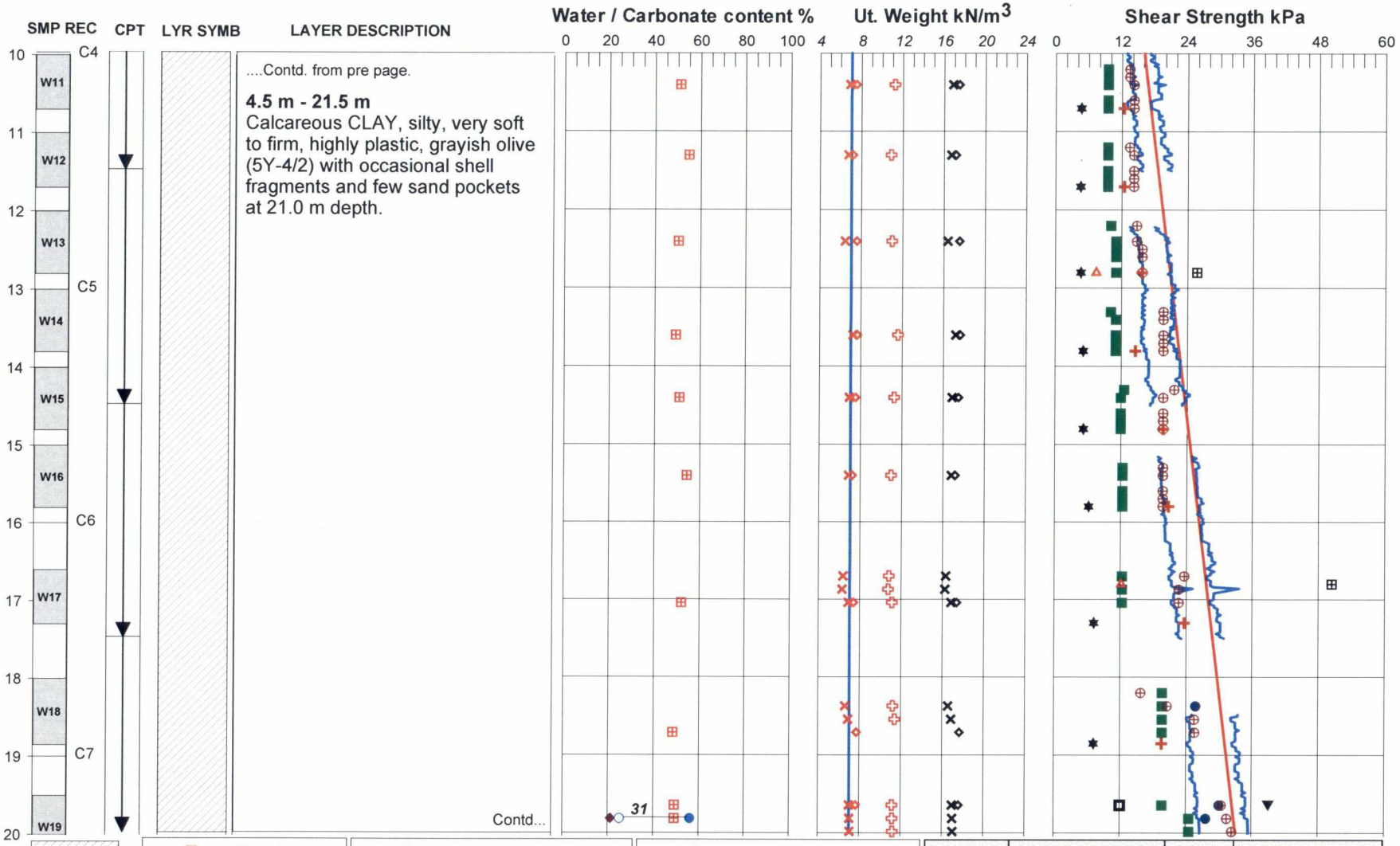
Soil Data Boring	**D 1 - D 13**
Drilling Log	**D 14**
CPTU & Field Vane Shear Tests Plots	



Location	GK-28-10 (PLATFORM)	Field	GULF OF KUTCH
Co-ordinates	WGS-84		ZONE - 42
Latitude	22° 36' 49.614" N	Longitude	068° 26' 08.305" E
Northing	2500876.37 m	Easting	441999.64 m
Project No	IEOT/GEO/07/2018-19	Report No	IEOT/GEO/07/2018-19

Soil Profile	0 - 10 m	Plate	D - 1	Prepared By	Checked By	DATE
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY				Alankar Srivastava	T. K. Roy	17.08.2018

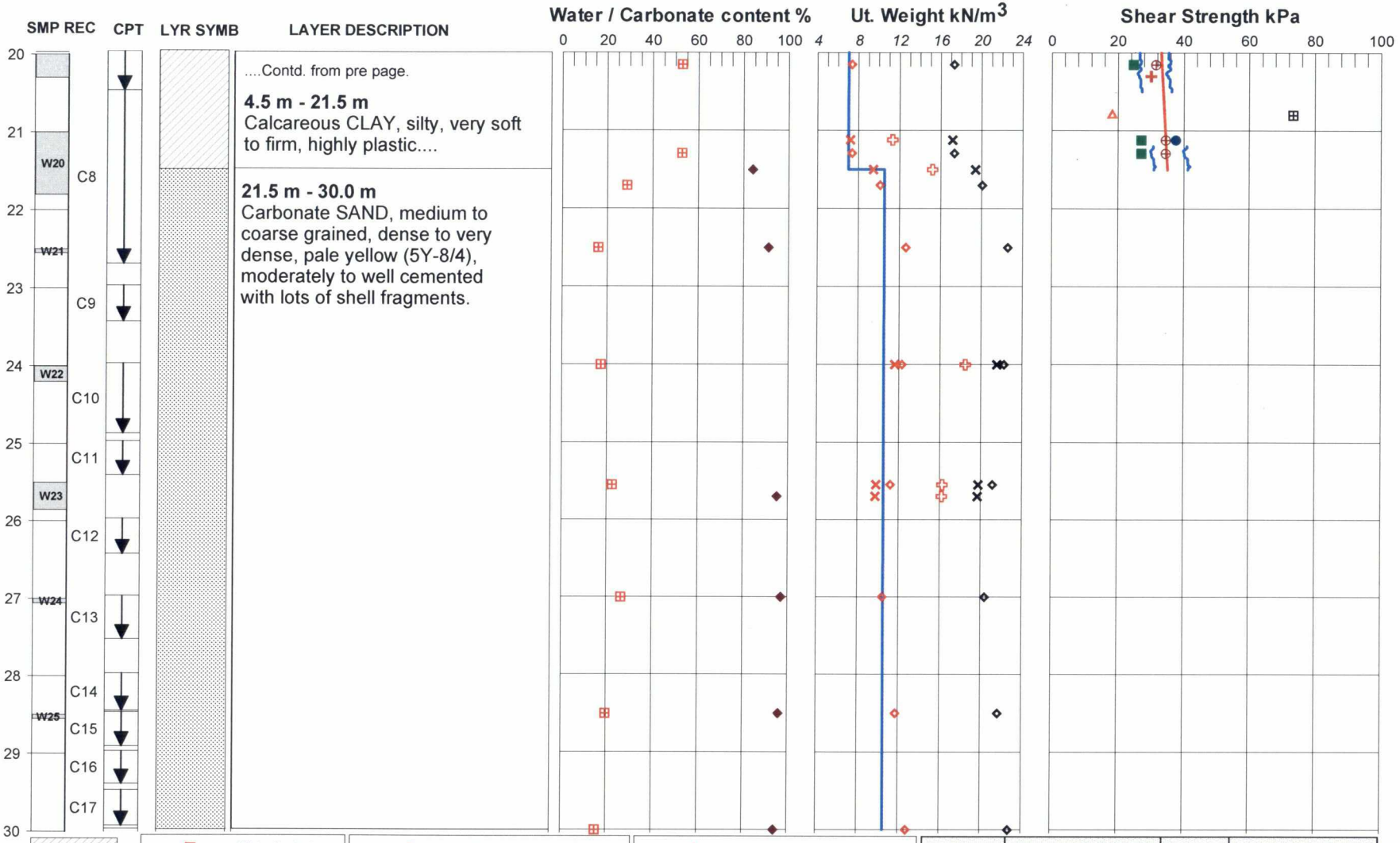
GEOFT 060/01
REV.NO. 00



Location	GK-28-10 (PLATFORM)	Field	GULF OF KUTCH
Co-ordinates	WGS-84		ZONE - 42
Latitude	22° 36' 49.614" N	Longitude	068° 26' 08.305" E
Northing	2500876.37 m	Easting	441999.64 m
Project No	IEOT/GEO/07/2018-19	Report No	IEOT/GEO/07/2018-19

Soil Profile	10 - 20 m	Plate	D - 2	Prepared By	Checked By	DATE
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY				Alankar Srivastava	T. K. Roy	17.08.2018

GEOFT 060/01
REV.NO. 00



Location	GK-28-10 (PLATFORM)	Field	GULF OF KUTCH
Co-ordinates	WGS-84		ZONE - 42
Latitude	22° 36' 49.614" N	Longitude	068° 26' 08.305" E
Northing	2500876.37 m	Easting	441999.64 m
Project No	IEOT/GEO/07/2018-19	Report No	IEOT/GEO/07/2018-19

Soil Profile	20 - 30 m	Plate	D - 3

ONGC
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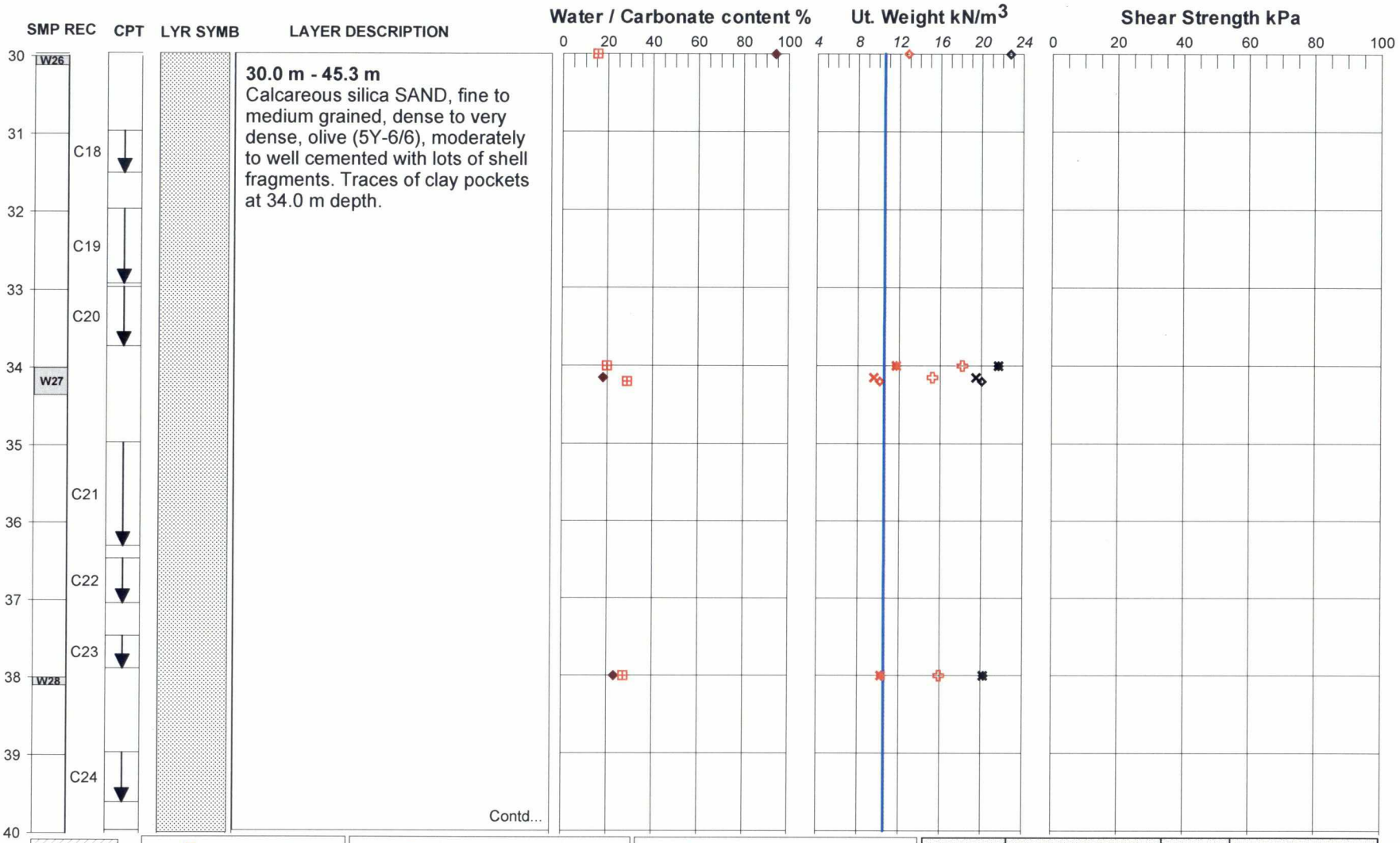
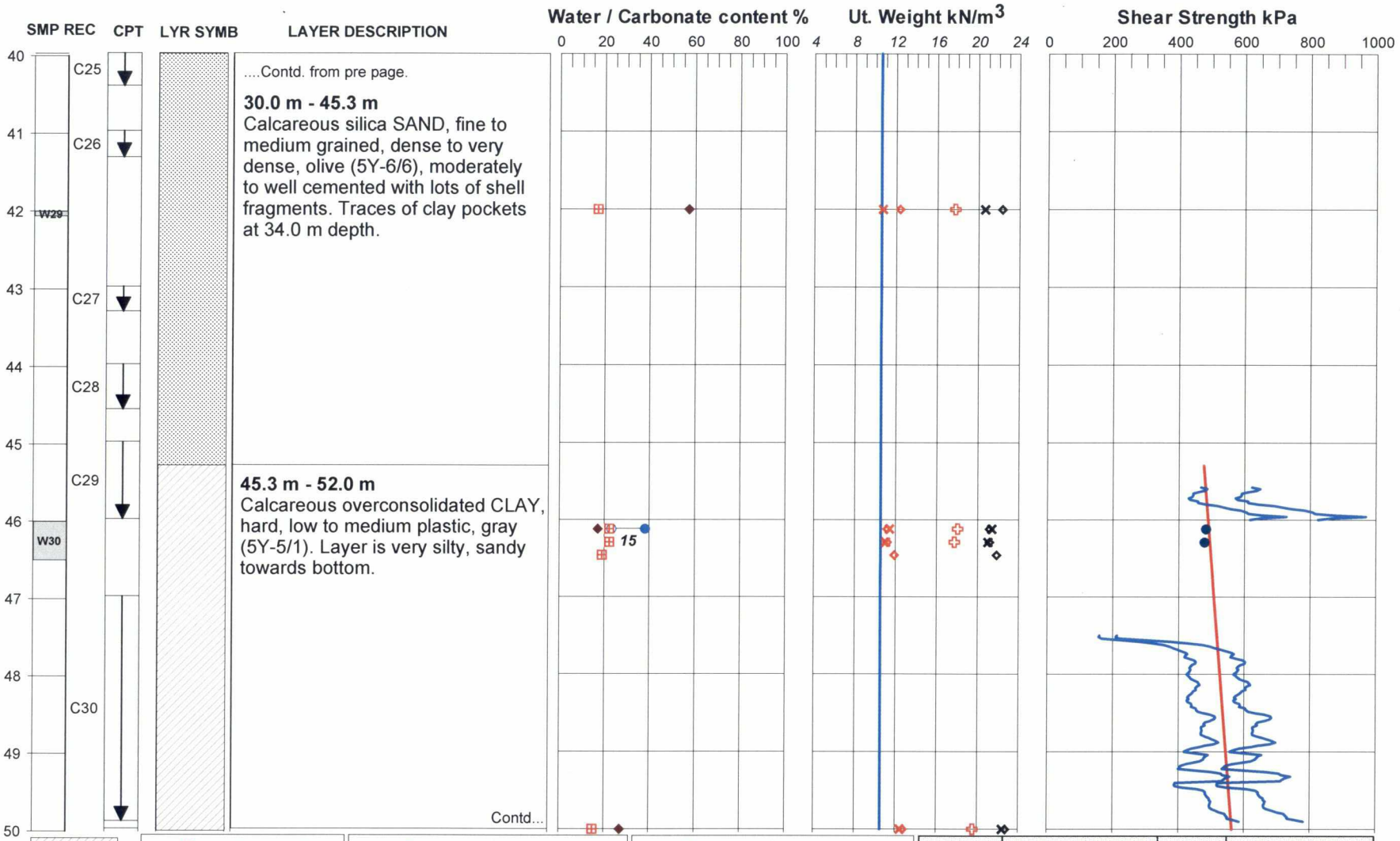
Prepared By	Checked By	DATE
Alankar Srivastava	T. K. Roy	17.08.2018



Location	GK-28-10 (PLATFORM)	Field	GULF OF KUTCH
Co-ordinates	WGS-84		ZONE - 42
Latitude	22° 36' 49.614" N	Longitude	068° 26' 08.305" E
Northing	2500876.37 m	Easting	441999.64 m
Project No	IEOT/GEO/07/2018-19	Report No	IEOT/GEO/07/2018-19

Soil Profile	30 - 40 m	Plate	D - 4	Prepared By	Checked By	DATE
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY				Alankar Srivastava	T. K. Roy	17.08.2018

GEOFT 060/01
REV.NO. 00



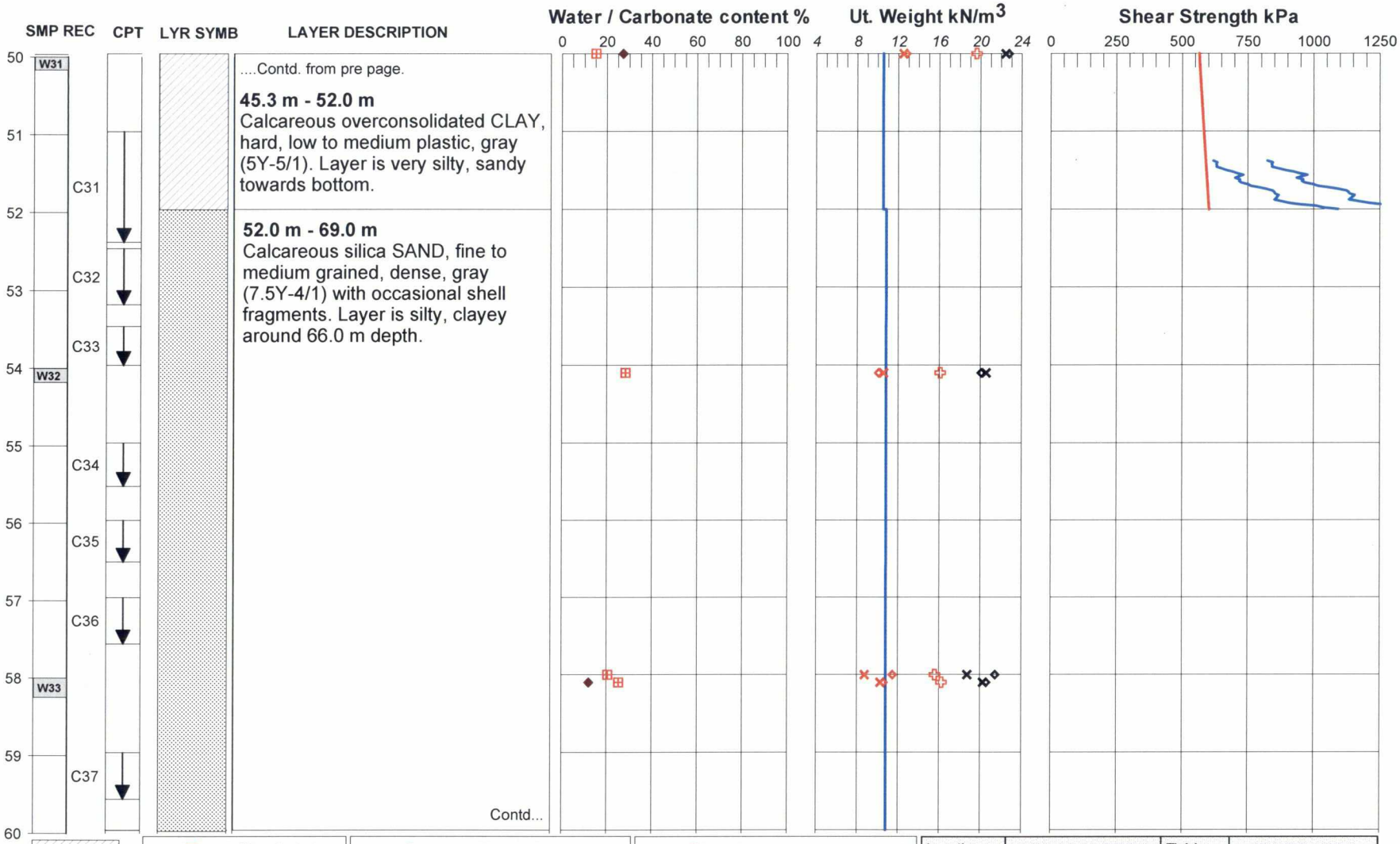
Location	GK-28-10 (PLATFORM)	Field	GULF OF KUTCH
Co-ordinates	WGS-84		ZONE - 42
Latitude	22° 36' 49.614" N	Longitude	068° 26' 08.305" E
Northing	2500876.37 m	Easting	441999.64 m
Project No	IEOT/GEO/07/2018-19	Report No	IEOT/GEO/07/2018-19

Soil Profile	40 - 50 m	Plate	D - 5

GEOFT 060/01
REV.NO. 00

ONGC
INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY

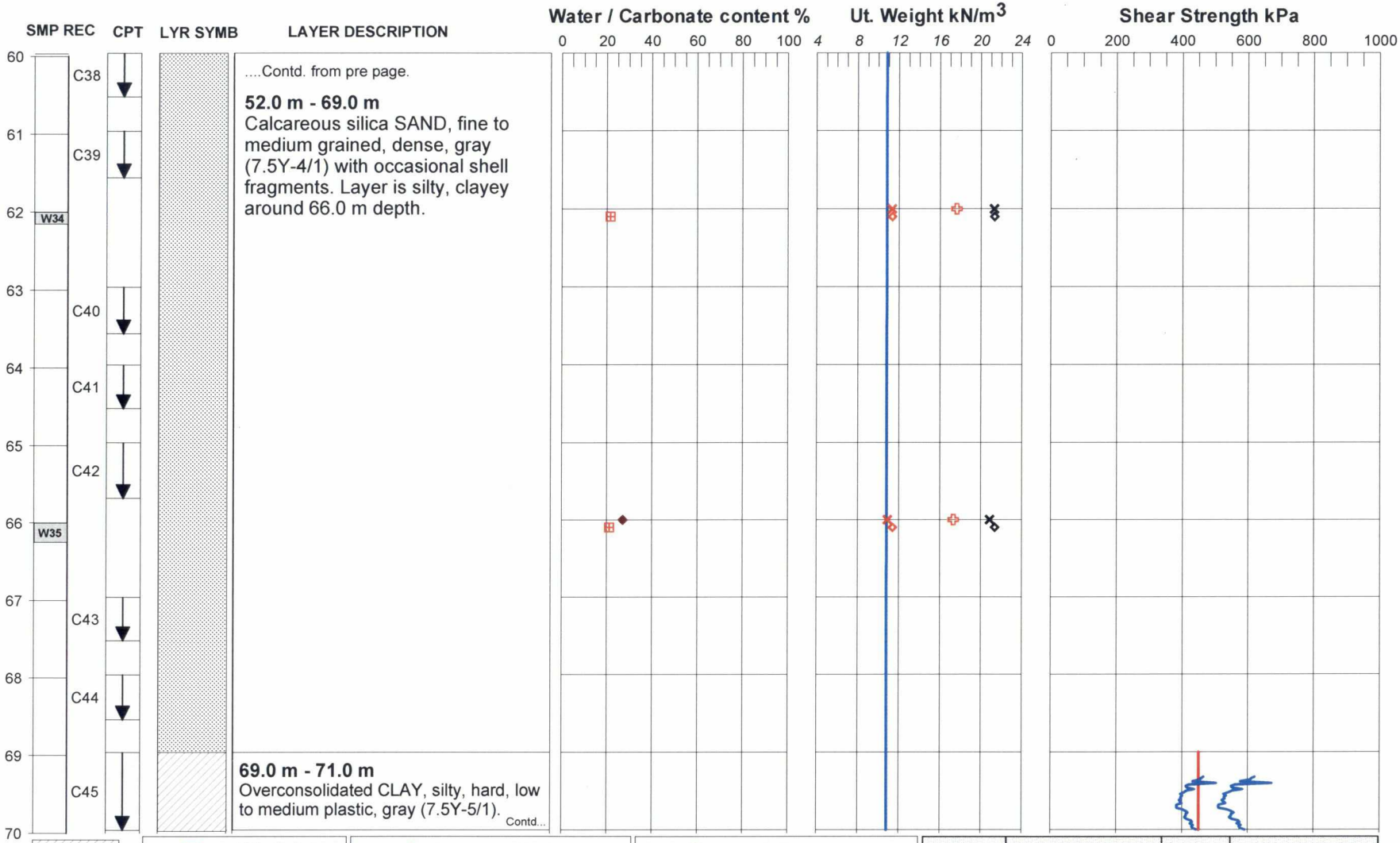
Prepared By	Checked By	DATE
Alankar Srivastava	T. K. Roy	17.08.2018



Location	GK-28-10 (PLATFORM)	Field	GULF OF KUTCH
Co-ordinates	WGS-84		ZONE - 42
Latitude	22° 36' 49.614" N	Longitude	068° 26' 08.305" E
Northing	2500876.37 m	Easting	441999.64 m
Project No	IEOT/GEO/07/2018-19	Report No	IEOT/GEO/07/2018-19

Soil Profile	50 - 60 m	Plate	D - 6	Prepared By	Checked By	DATE
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY				Alankar Srivastava	T. K. Roy	17.08.2018

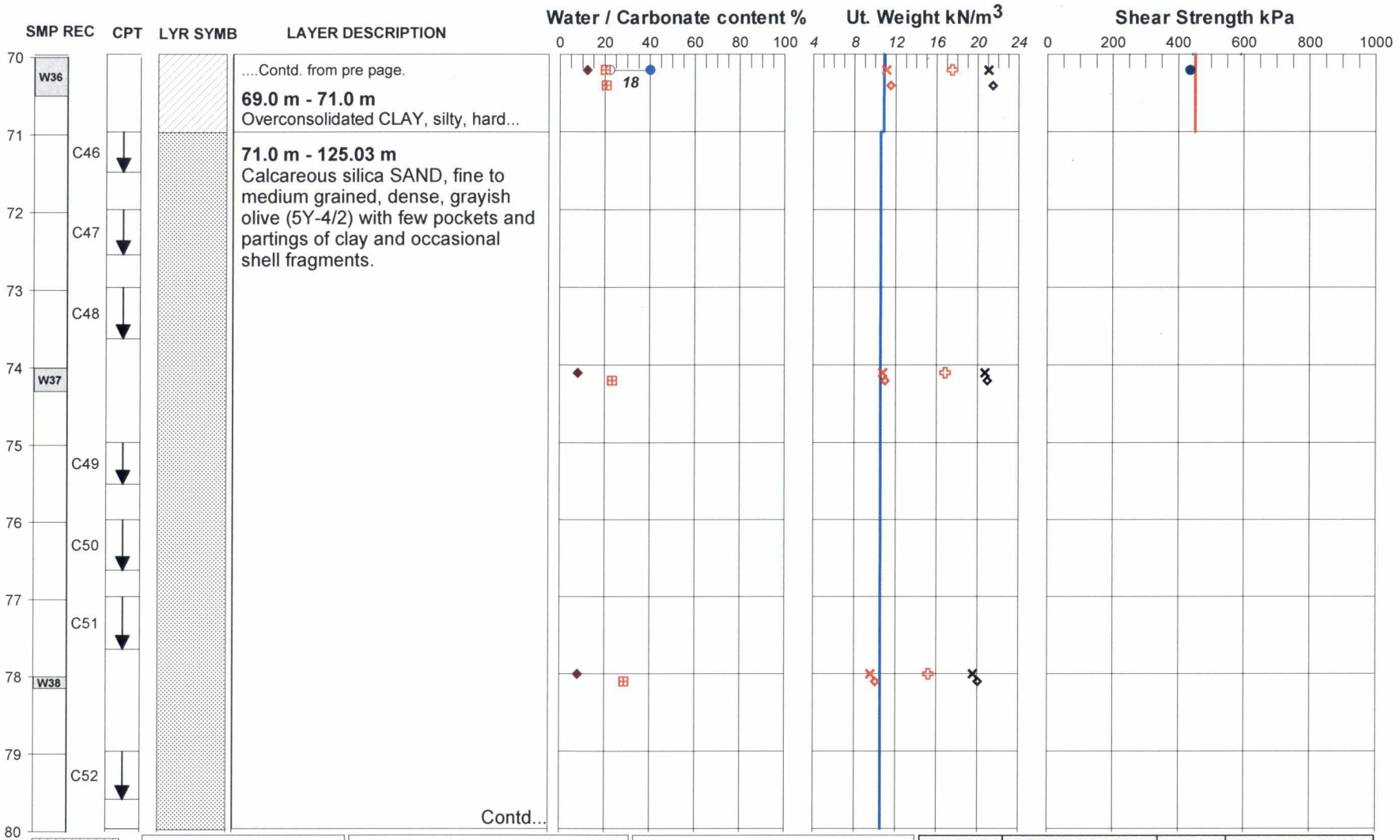
GEOFT 060/01
REV.NO. 00



Location	GK-28-10 (PLATFORM)	Field	GULF OF KUTCH
Co-ordinates	WGS-84		ZONE - 42
Latitude	22° 36' 49.614" N	Longitude	068° 26' 08.305" E
Northing	2500876.37 m	Easting	441999.64 m
Project No	IEOT/GEO/07/2018-19	Report No	IEOT/GEO/07/2018-19

Soil Profile	60 - 70 m	Plate	D - 7	Prepared By	Checked By	DATE
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY				Alankar Srivastava	T. K. Roy	17.08.2018

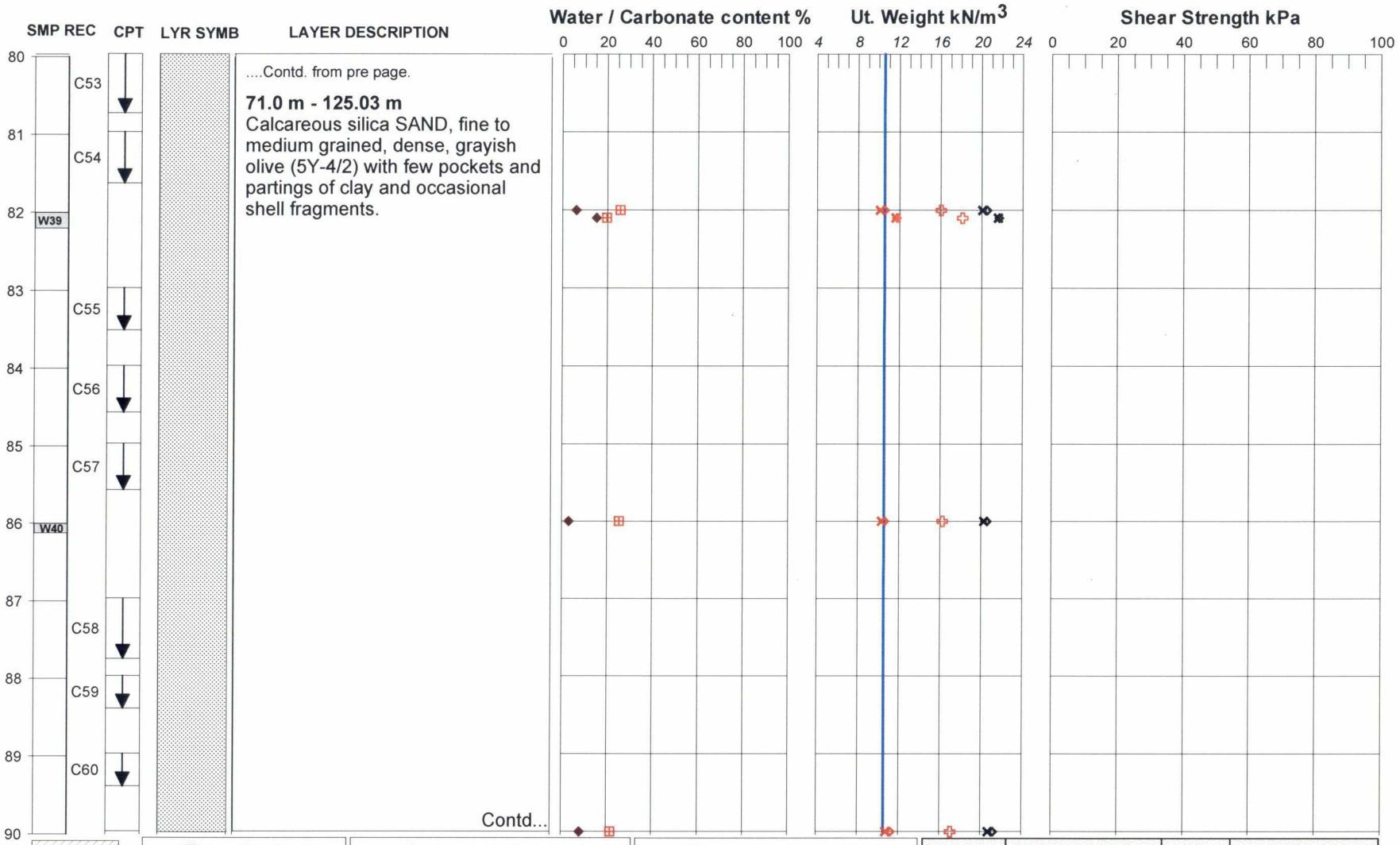
GEOFT 060/01
REV.NO. 00



Location	GK-28-10 (PLATFORM)	Field	GULF OF KUTCH
Co-ordinates	WGS-84		ZONE - 42
Latitude	22° 36' 49.614" N	Longitude	068° 26' 08.305" E
Northing	2500876.37 m	Easting	441999.64 m
Project No	IEOT/GEO/07/2018-19	Report No	IEOT/GEO/07/2018-19

Soil Profile	70 - 80 m	Plate	D - 8	Prepared By	Checked By	DATE
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY				Alankar Srivastava	T. K. Roy	17.08.2018

GEOFT 060/01
REV.NO. 00



Location	GK-28-10 (PLATFORM)	Field	GULF OF KUTCH
Co-ordinates	WGS-84		ZONE - 42
Latitude	22° 36' 49.614" N	Longitude	068° 26' 08.305" E
Northing	2500876.37 m	Easting	441999.64 m
Project No	IEOT/GEO/07/2018-19	Report No	IEOT/GEO/07/2018-19

Soil Profile	80 - 90 m	Plate	D - 9

GEOFT 060/01
REV.NO. 00

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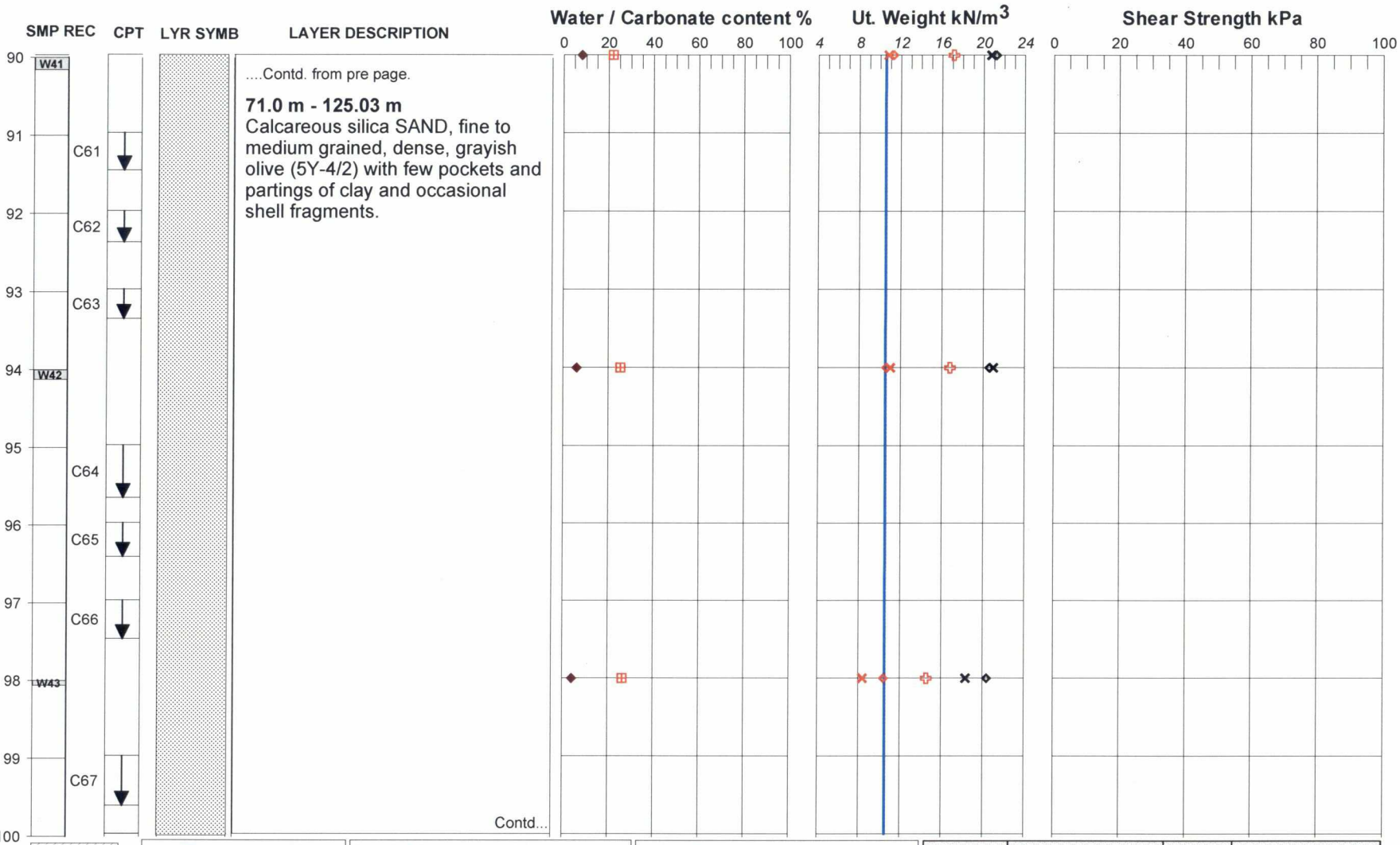
INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY

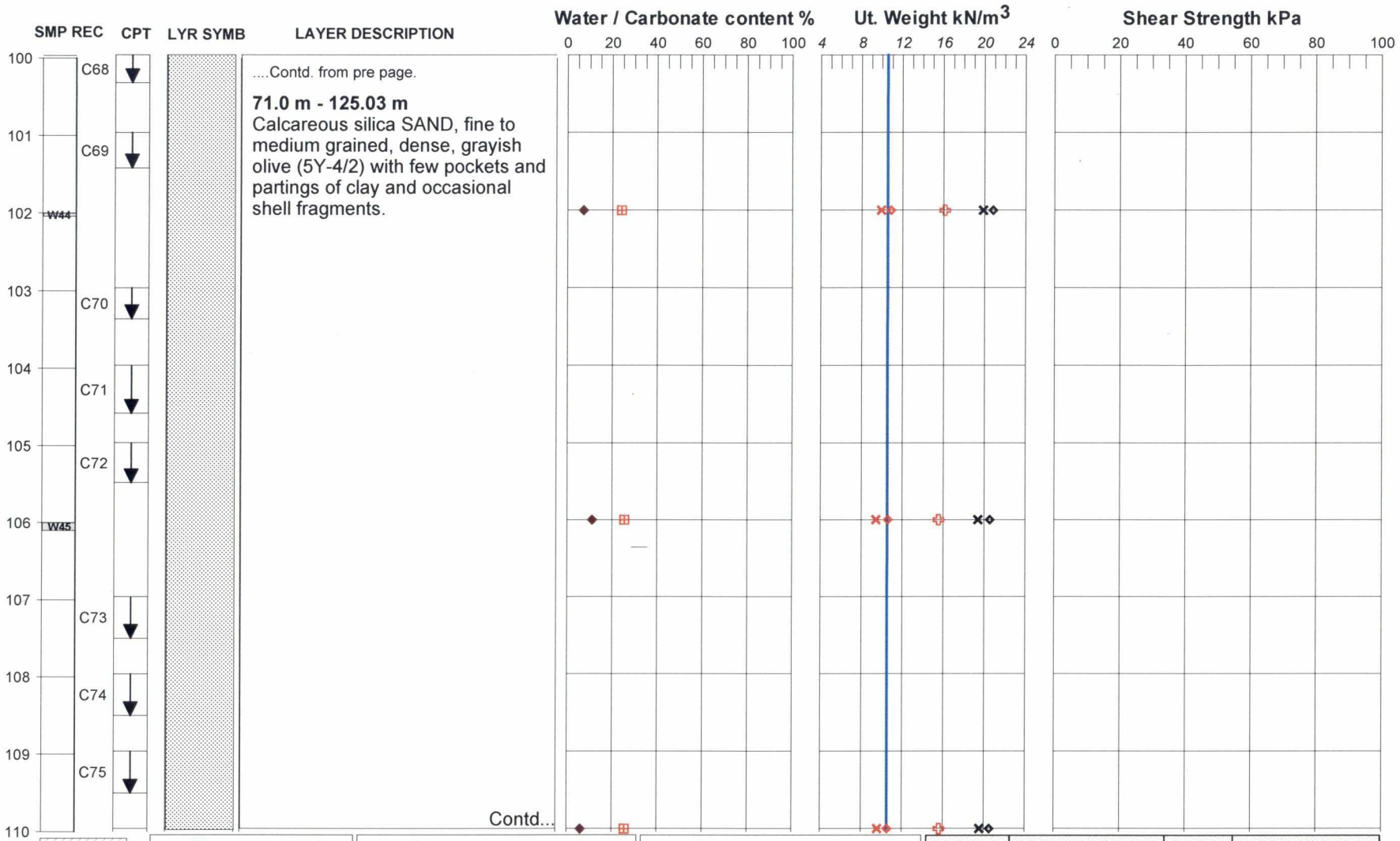
Prepared By	Checked By	DATE
Alankar Srivastava	T. K. Roy	17.08.2018



Location	GK-28-10 (PLATFORM)	Field	GULF OF KUTCH
Co-ordinates	WGS-84		ZONE - 42
Latitude	22° 36' 49.614" N	Longitude	068° 26' 08.305" E
Northing	2500876.37 m	Easting	441999.64 m
Project No	IEOT/GEO/07/2018-19	Report No	IEOT/GEO/07/2018-19

Soil Profile	90 - 100 m	Plate	D - 10	Prepared By	Checked By	DATE
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY				Alankar Srivastava	T. K. Roy	17.08.2018

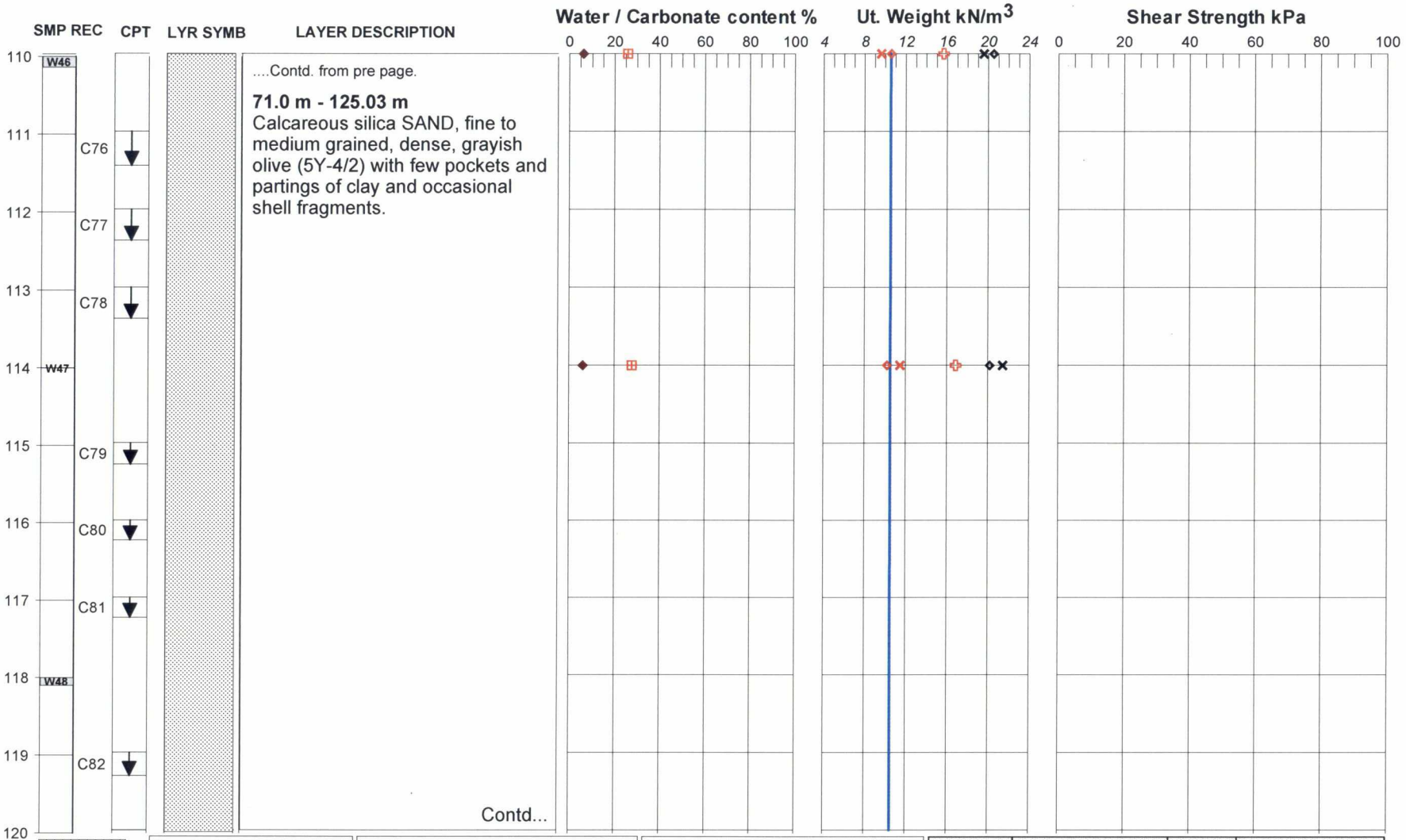
GEOFT 060/01
REV.NO. 00



Location	GK-28-10 (PLATFORM)	Field	GULF OF KUTCH
Co-ordinates	WGS-84		ZONE - 42
Latitude	22° 36' 49.614" N	Longitude	068° 26' 08.305" E
Northing	2500876.37 m	Easting	441999.64 m
Project No	IEOT/GEO/07/2018-19	Report No	IEOT/GEO/07/2018-19

Soil Profile	100 - 110 m	Plate	D - 11	Prepared By	Checked By	DATE
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY				Alankar Srivastava	T. K. Roy	17.08.2018

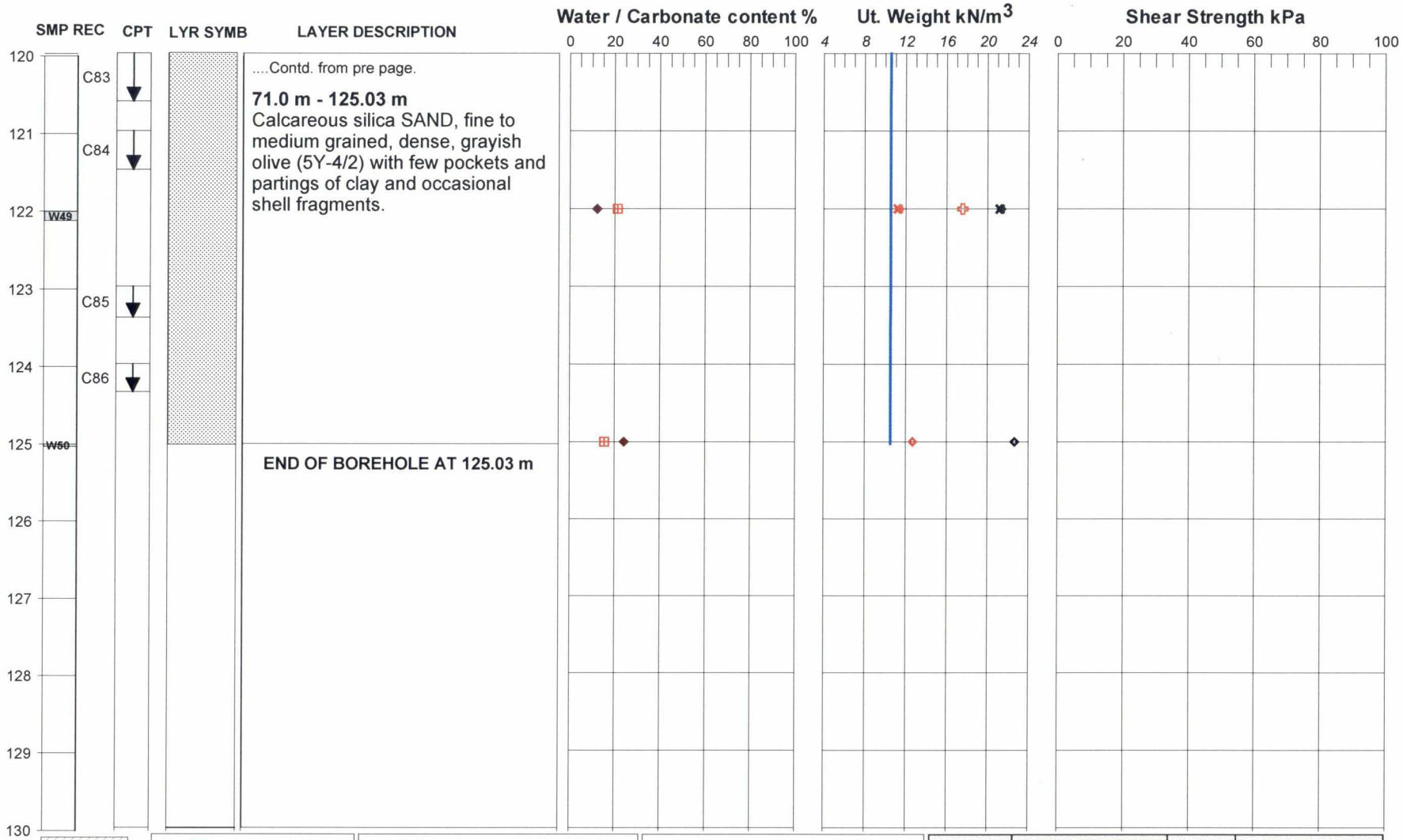
GEOFT 060/01
REV.NO. 00



Location	GK-28-10 (PLATFORM)	Field	GULF OF KUTCH
Co-ordinates	WGS-84		ZONE - 42
Latitude	22° 36' 49.614" N	Longitude	068° 26' 08.305" E
Northing	2500876.37 m	Easting	441999.64 m
Project No	IEOT/GEO/07/2018-19	Report No	IEOT/GEO/07/2018-19

Soil Profile	110 - 120 m	Plate	D - 12	Prepared By	Checked By	DATE
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY				Alankar Srivastava	T. K. Roy	17.08.2018

GEOFT 060/01
REV.NO. 00



CLAY
SILT
SAND

Water Content
Liquid Limit
Plastic Limit
Carbonate Content

Dry Unit Weight
Wet Unit Weight
Eff. Unit Weight
Calculated Ut. Wt.
Calc. Eff. Ut. Wt.
Reco. Eff.Ut.Wt

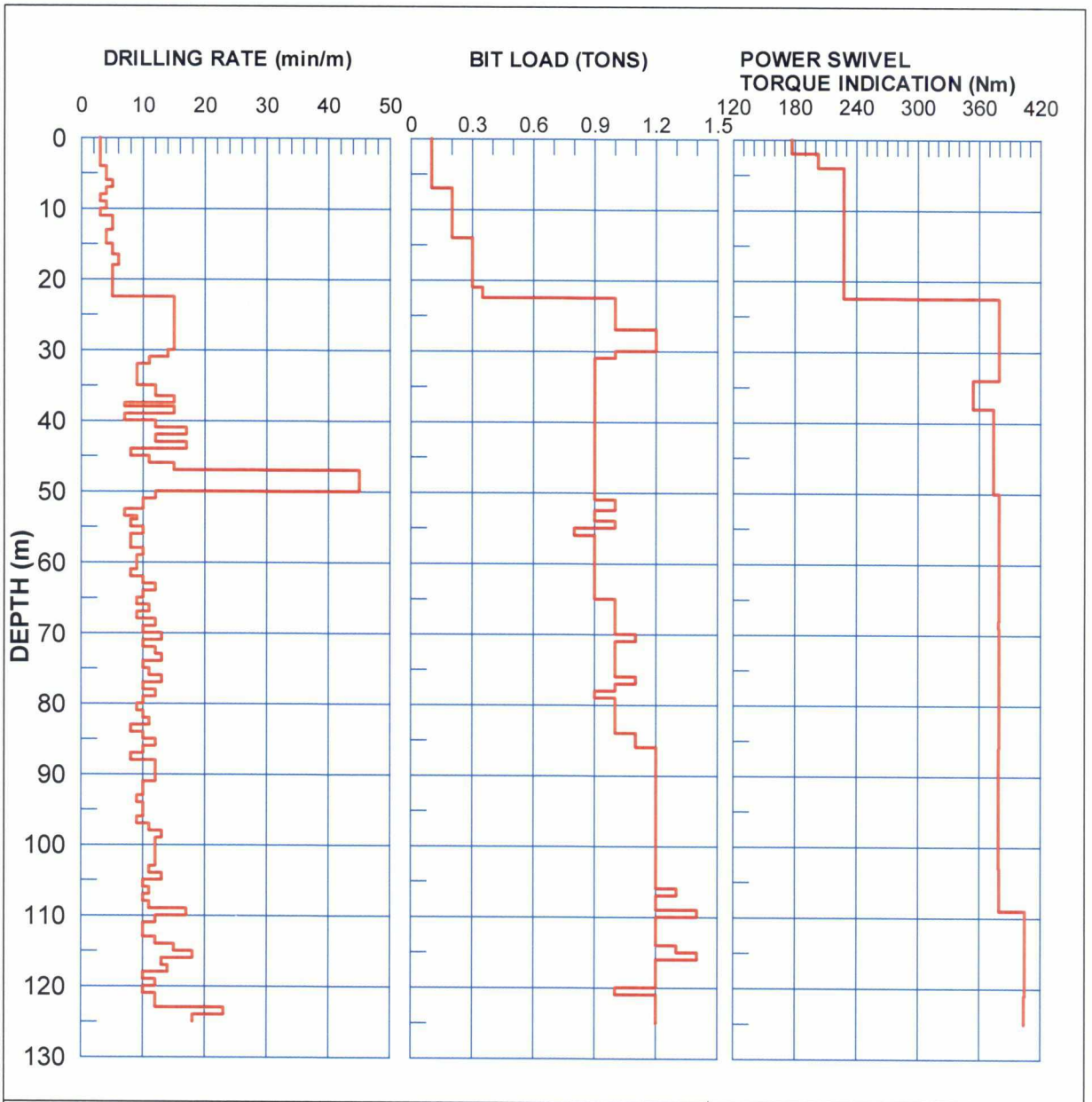
Motor Vane
Motor Vane Remoulded
Fall Cone
Fall Cone Remoulded
Field Vane
Field Vane Remoulded

Torvane
Pocket Penetrometer
UU Triaxial
Range of s_u from PCPT
Recommended s_u

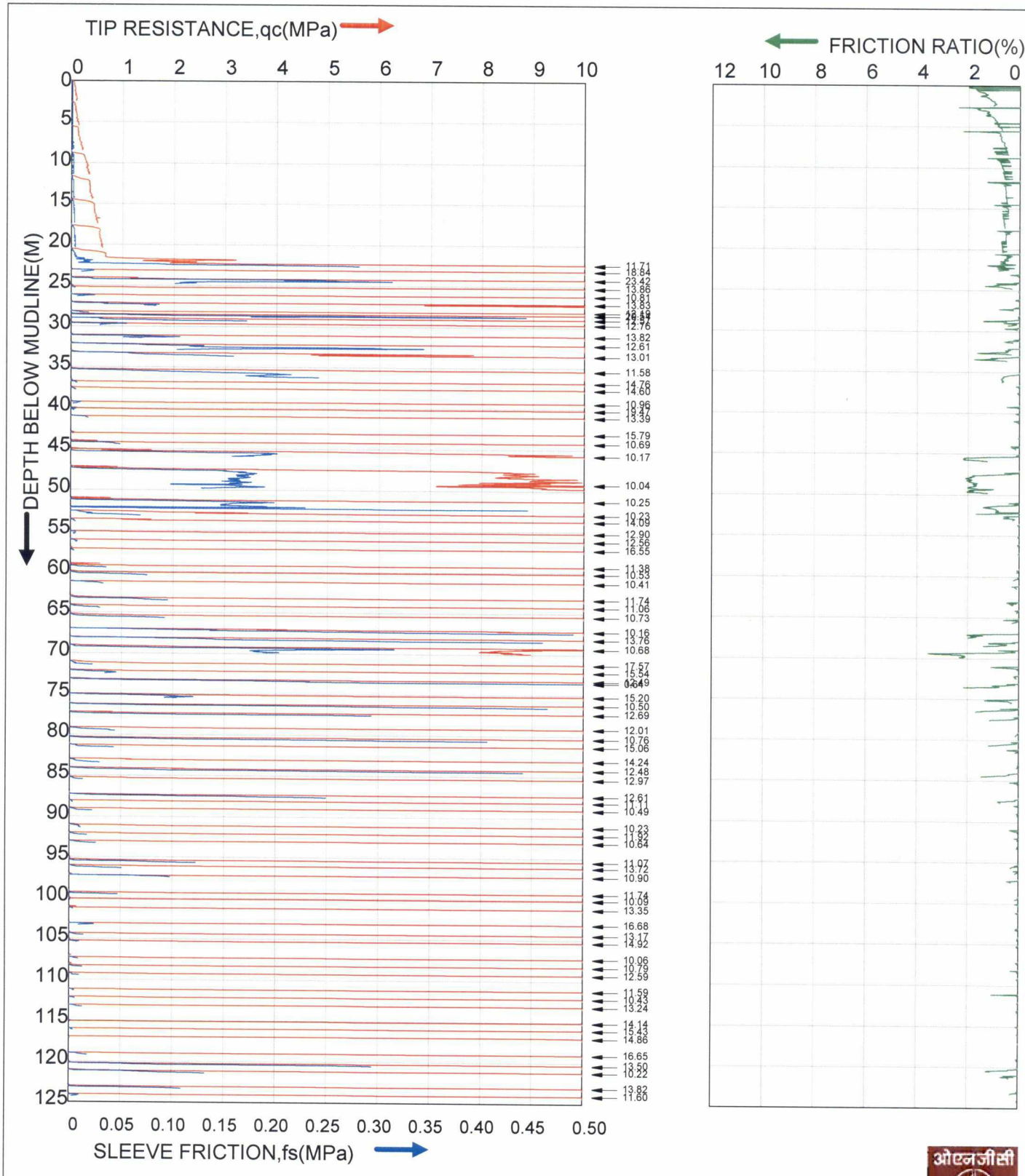
Location	GK-28-10 (PLATFORM)	*Field*	GULF OF KUTCH
Co-ordinates	WGS-84		ZONE - 42
Latitude	22° 36' 49.614" N	*Longitude*	068° 26' 08.305" E
Northing	2500876.37 m	*Easting*	441999.64 m
Project No	IEOT/GEO/07/2018-19	*Report No*	IEOT/GEO/07/2018-19

Soil Profile	120 - 130 m	Plate	D - 13	Prepared By	Checked By	DATE
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY				Alankar Srivastava	T. K. Roy	17.08.2018

GEOFT 060/01
REV.NO. 00



SUMMARY DRILLING LOGS		PLATE	D-14
ONGC INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY	PROJECT NO : IEOT/GEO/07/2018-19 REPORT NO : IEOT/GEO/07/2018-19 LOCATION : **GK-28-10 (PLATFORM)** FIELD : **GULF OF KUTCH**	DATE	17-08-2018
		PREPARED BY	Alankar Srivastava
		CHECKED BY:	T. K. Roy



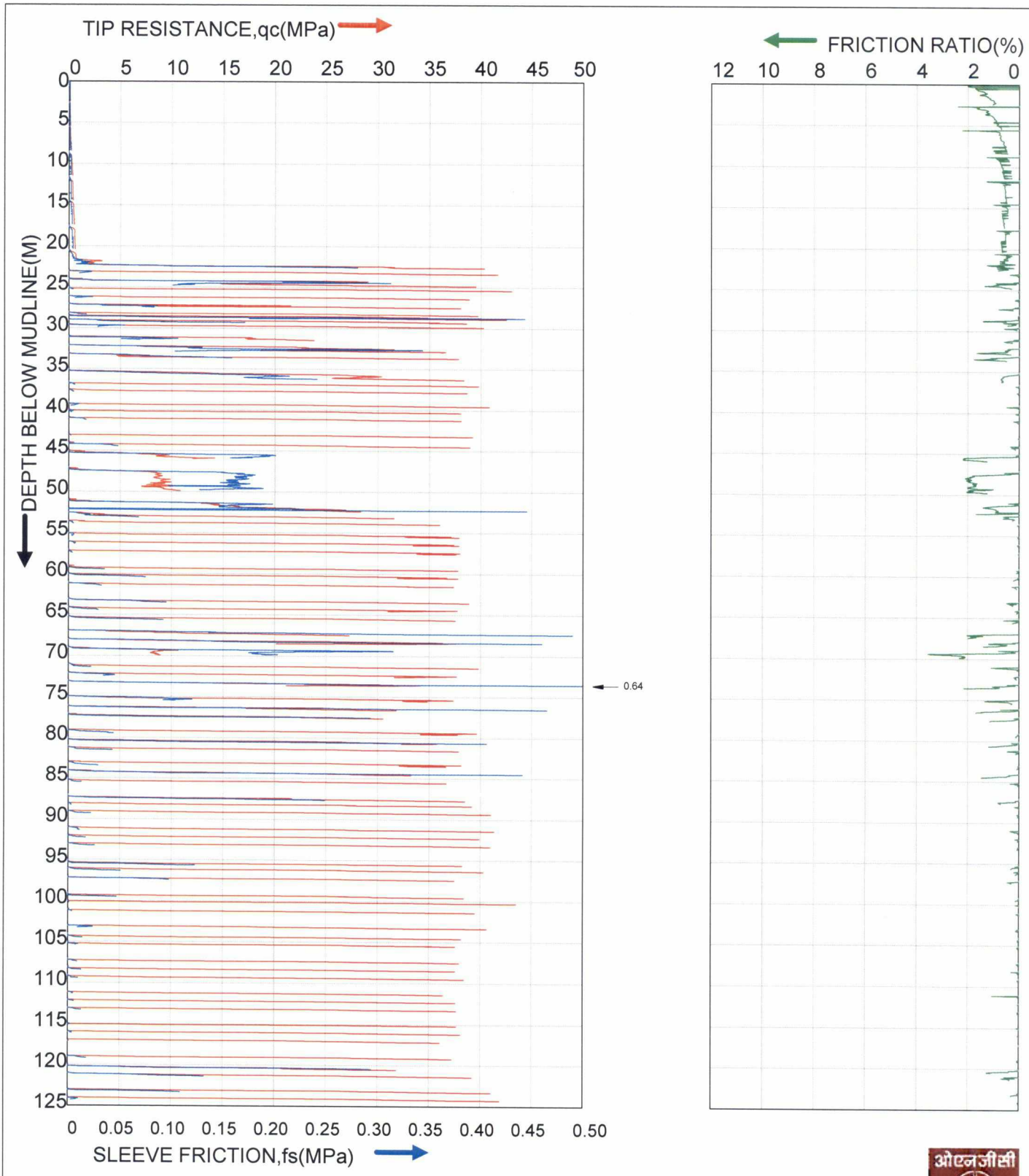
COMACOE

BOREHOLE NO: GK-28-10(PLATFORM)

ओएनजीसी ONGC

LOCATION	:	GULF OF KUTCH	TEST DATE	:	5-5-2018
PROJECT NO	:	CM 15 03 05	CONE TYPE	:	CFP100-10
ZERO REFERENCE	:	SEABED	CONE NUMBER	:	120947
DESCRIPTION	:	PLATFORM LOCATION	OPERATOR	:	DIVAKAR / LALIT

PIEZO-CONE PENETRATION TEST



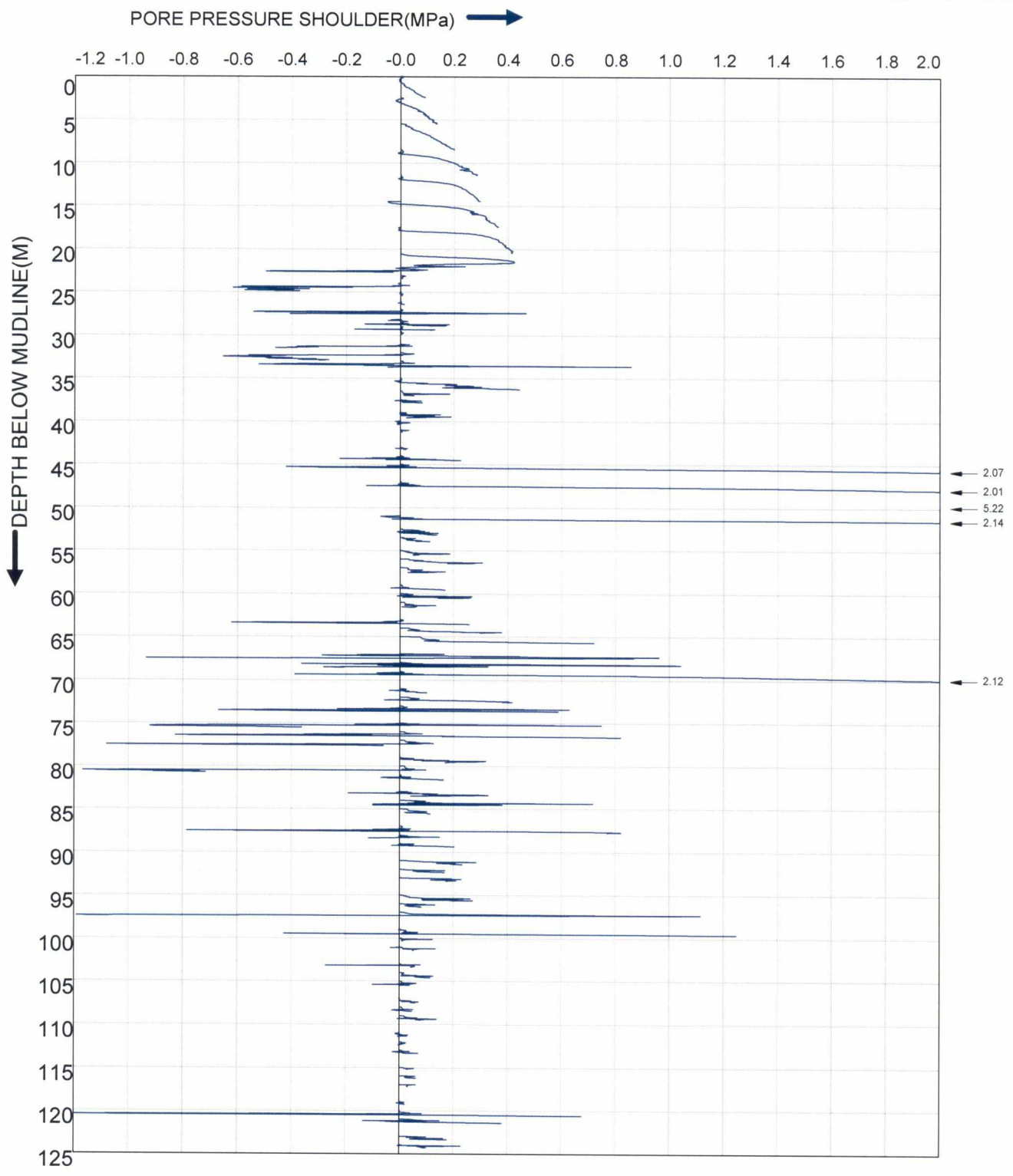
COMACOE

BOREHOLE NO: GK-28-10(PLATFORM)

ओएनजीसी ONGC

LOCATION	:	GULF OF KUTCH	TEST DATE	:	5-5-2018
PROJECT NO	:	CM 15 03 05	CONE TYPE	:	CFP100-10
ZERO REFERENCE	:	SEABED	CONE NUMBER	:	120947
DESCRIPTION	:	PLATFORM LOCATION	OPERATOR	:	DIVAKAR / LALIT

PIEZO-CONE PENETRATION TEST



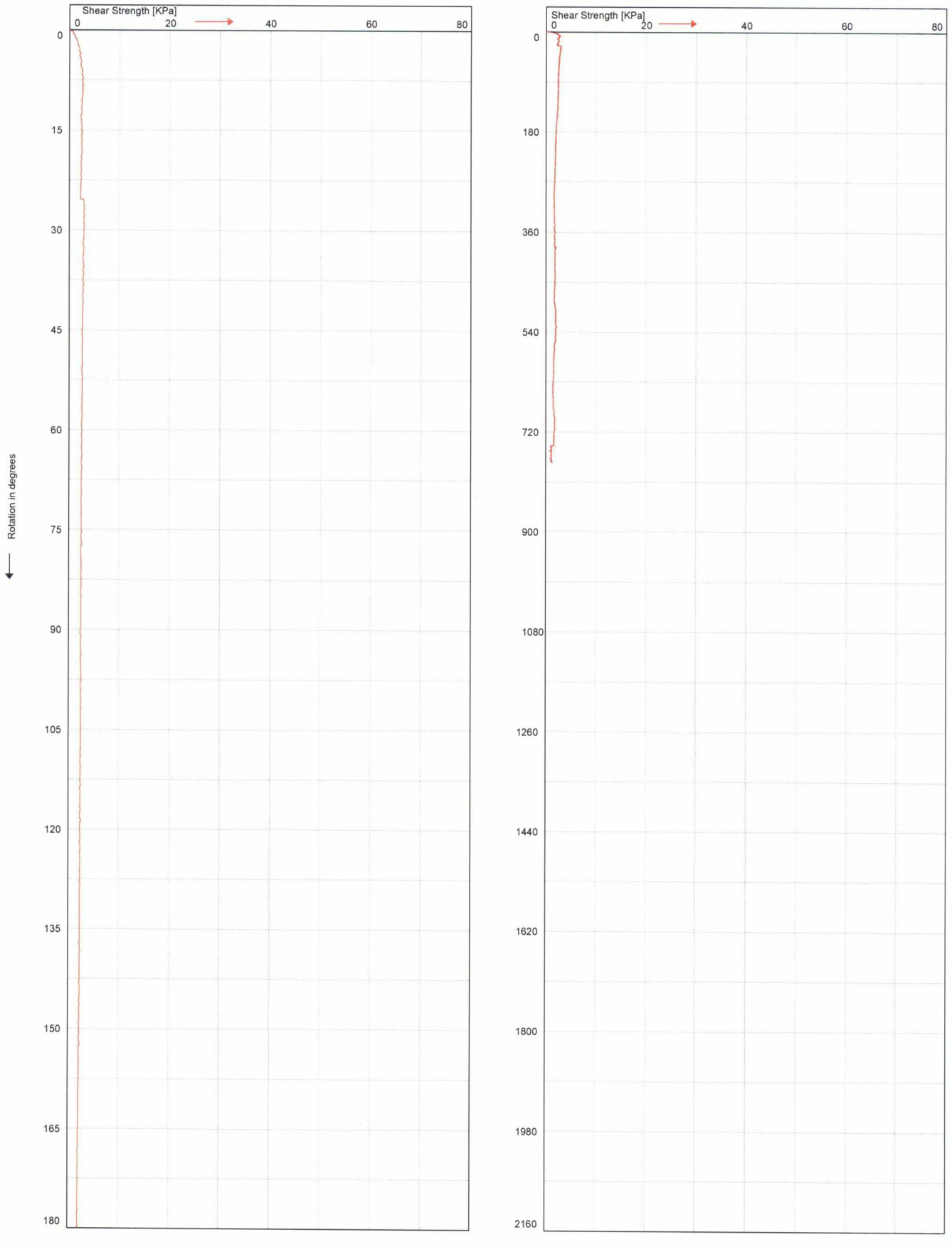
COMACOE

BOREHOLE NO: GK-28-10(PLATFORM)

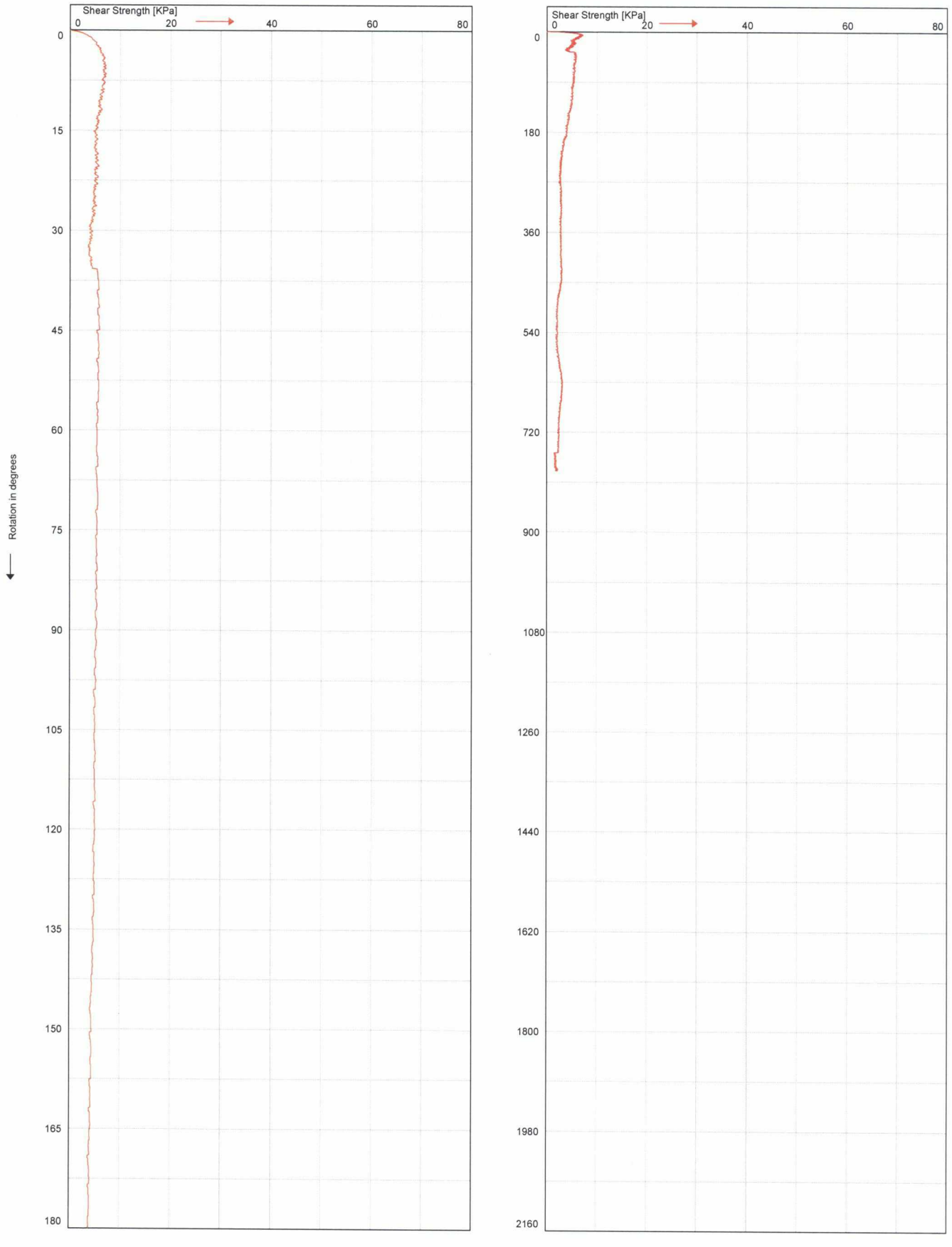
ओएनजीसी ONGC

LOCATION	:	GULF OF KUTCH	TEST DATE	:	5-5-2018
PROJECT NO	:	CM 15 03 05	CONE TYPE	:	CFP100-10
ZERO REFERENCE	:	SEABED	CONE NUMBER	:	120947
DESCRIPTION	:	PLATFORM LOCATION	OPERATOR	:	DIVAKAR / LALIT

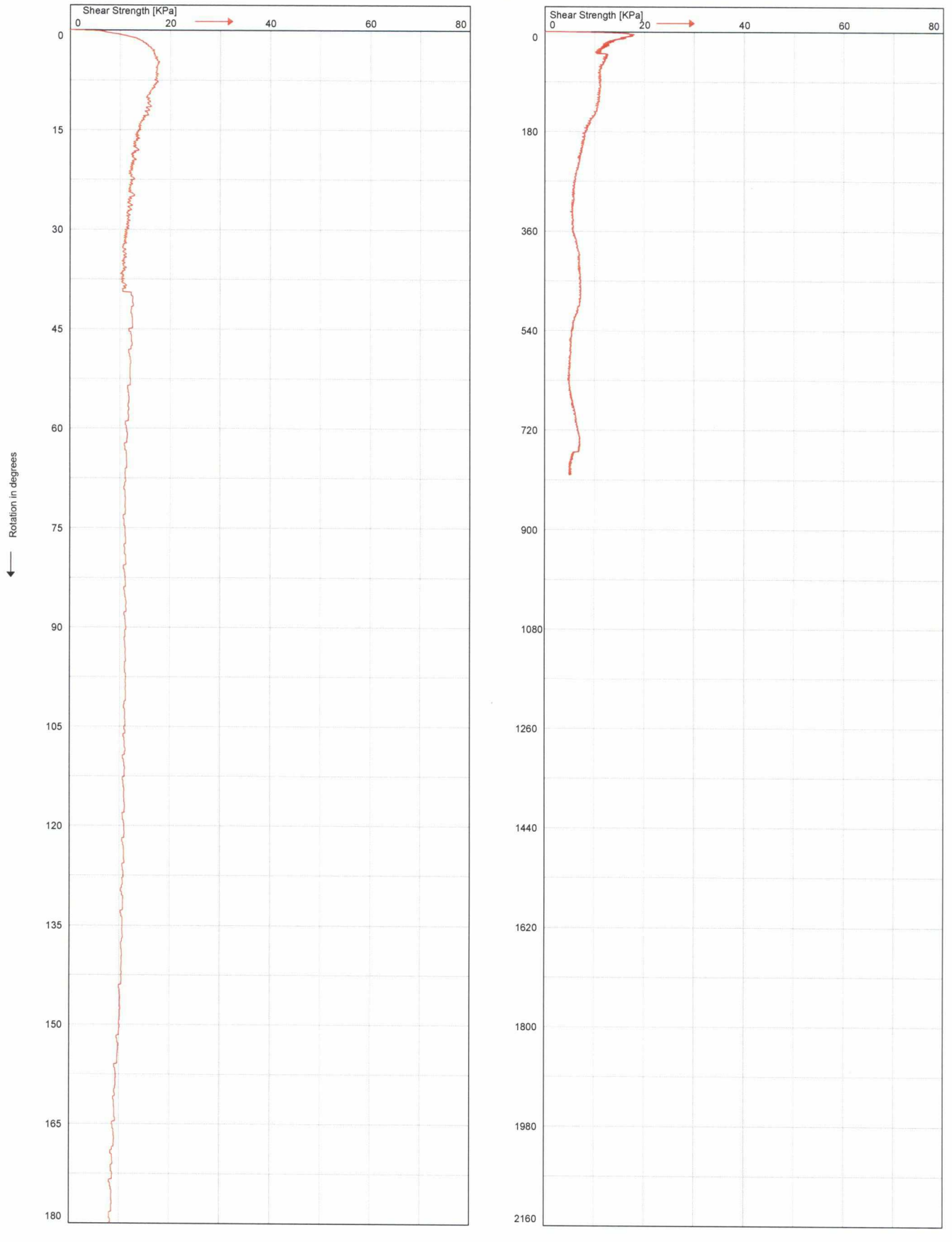
PIEZO-CONE PENETRATION TEST



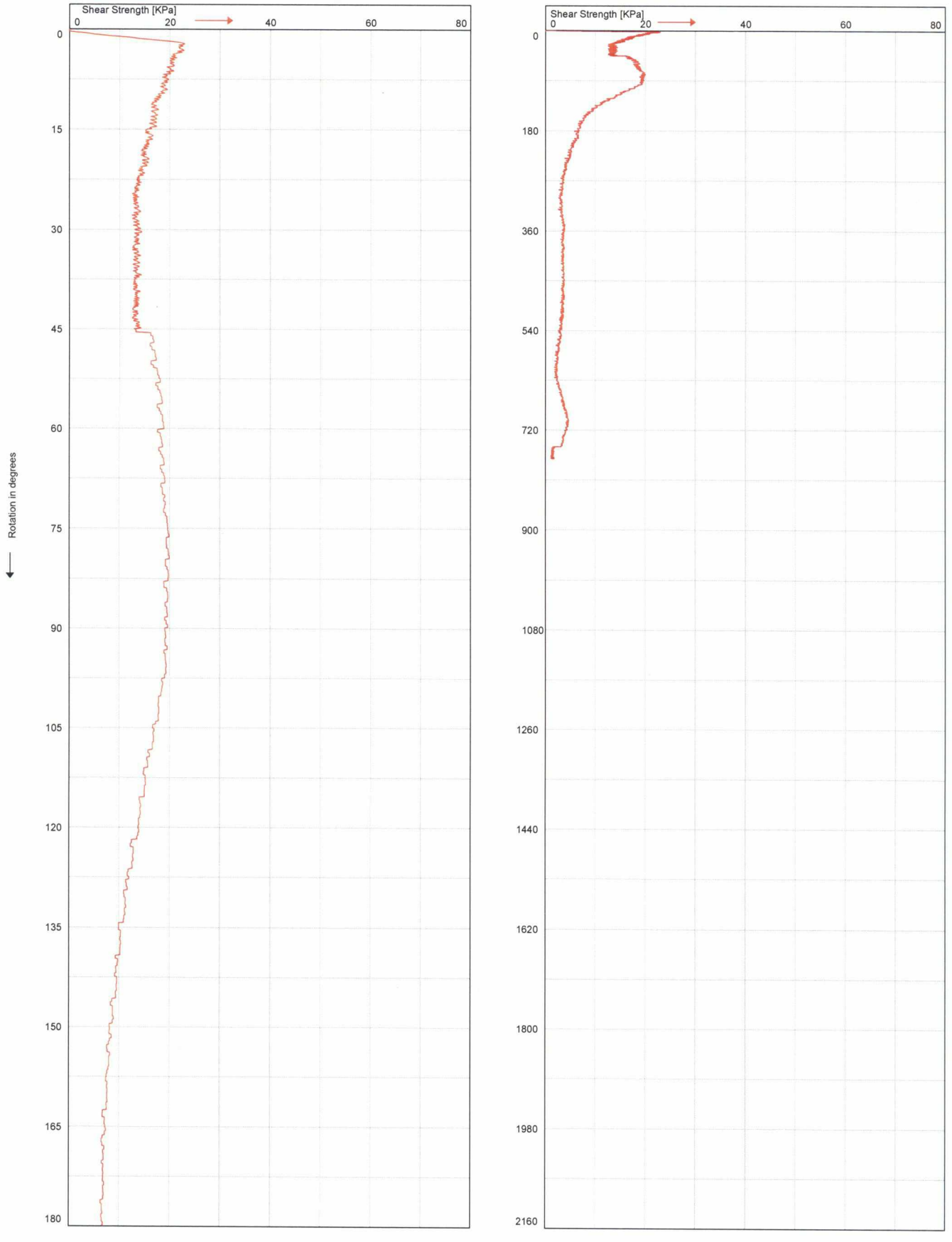
Job	: GK-28-10(PLATFORM)	Testdepth	: 0.8	Vane type	: VANE 65
Test	: 1	Client	: ONGC	Description	: PLATFORM LOCATION
Date	: 5-5-2018	Operator	: DIVAKAR / LALIT		



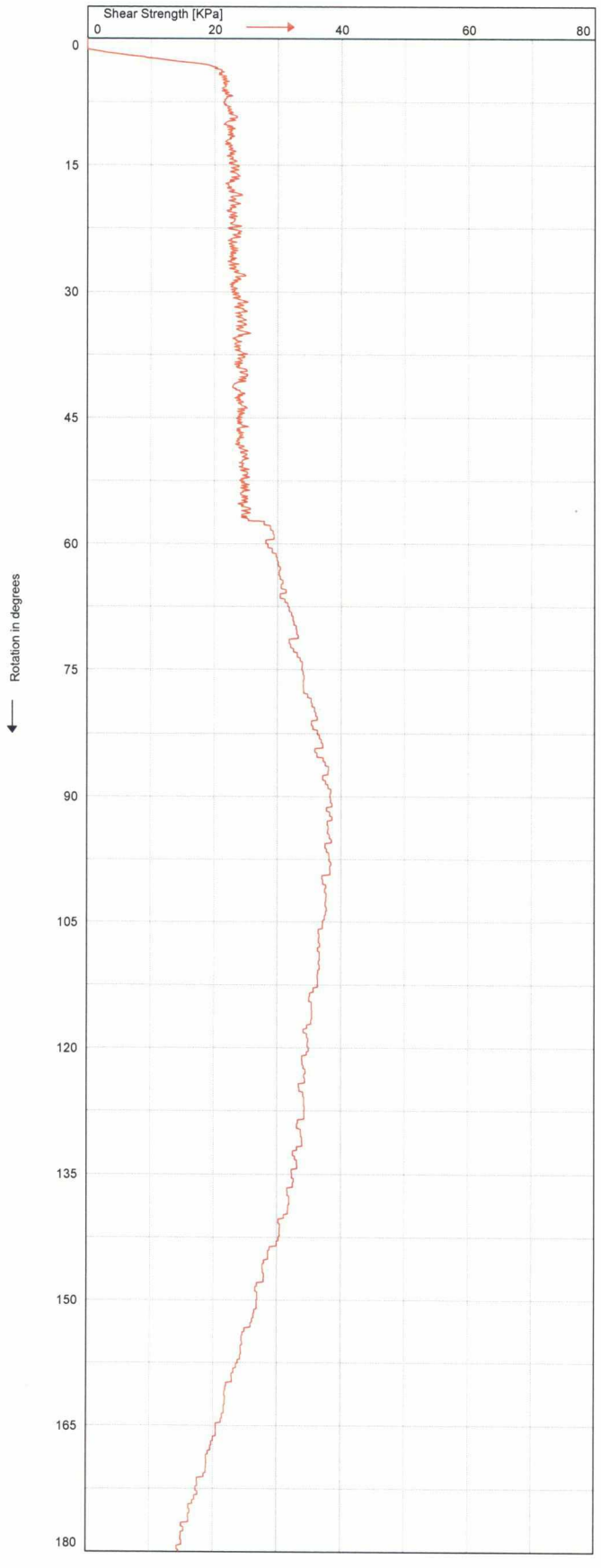
Job	: GK-28-10(PLATFORM)	Testdepth	: 4.8	Vane type	: VANE 65
Test	: 2	Client	: ONGC	Description	: PLATFORM LOCATION
Date	: 5-5-2018	Operator	: DIVAKAR / LALIT		

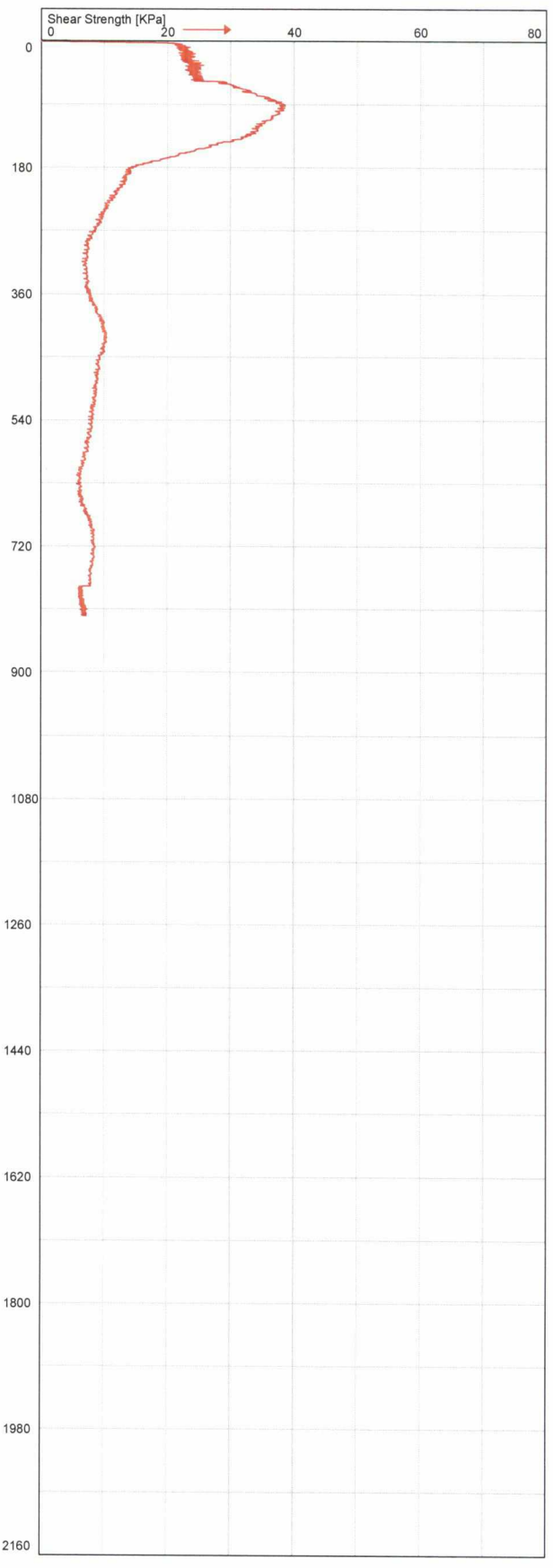


Job	: GK-28-10(PLATFORM)	Testdepth	: 8.8	Vane type	: VANE 65
Test	: 3	Client	: ONGC	Description	: PLATFORM LOCATION
Date	: 5-5-2018	Operator	: DIVAKAR / LALIT		

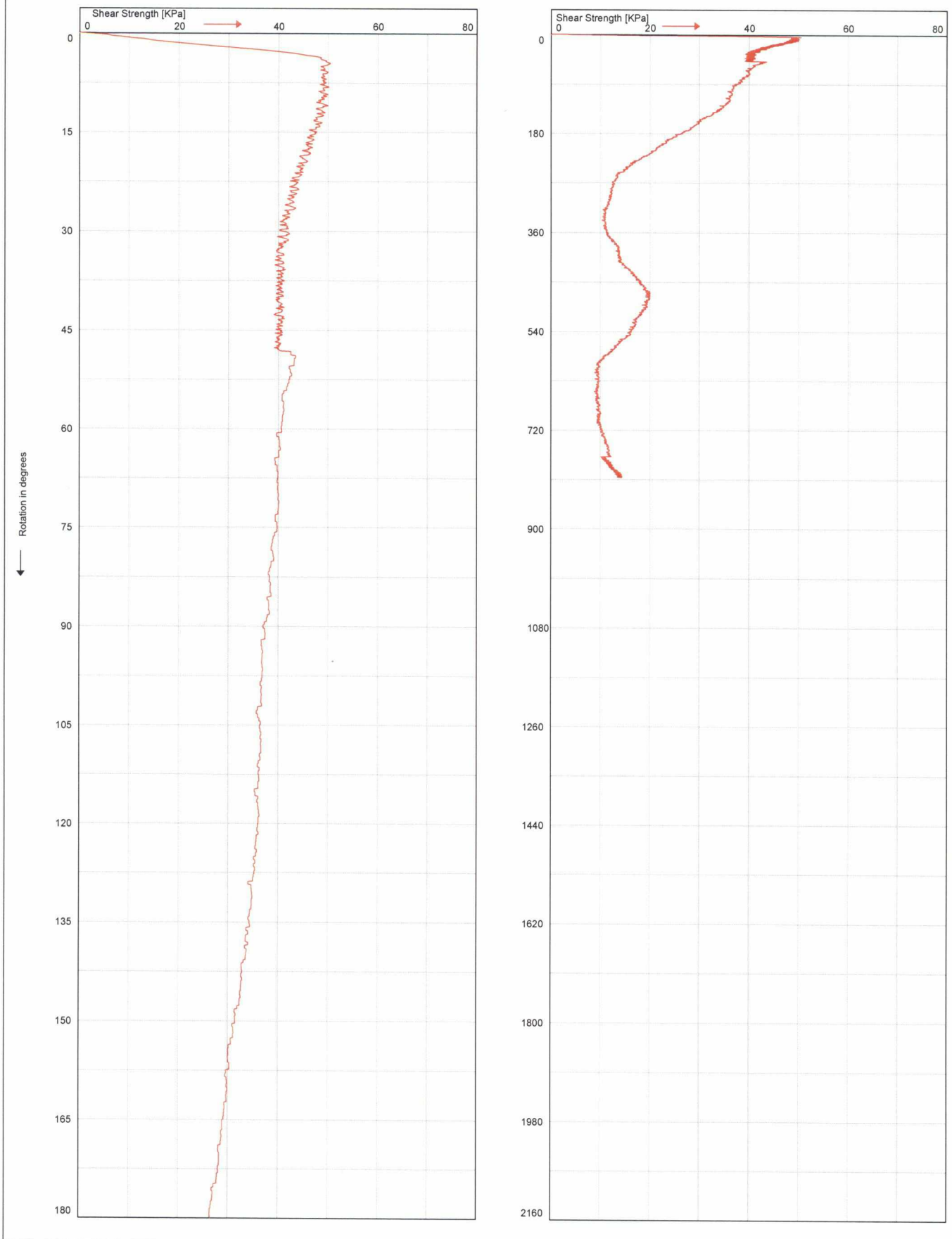


Job	: GK-28-10(PLATFORM)	Testdepth	: 9.8	Vane type	: VANE 50
Test	: 4	Client	: ONGC	Description	: PLATFORM LOCATION
Date	: 6-5-2018	Operator	: DIVAKAR / LALIT		

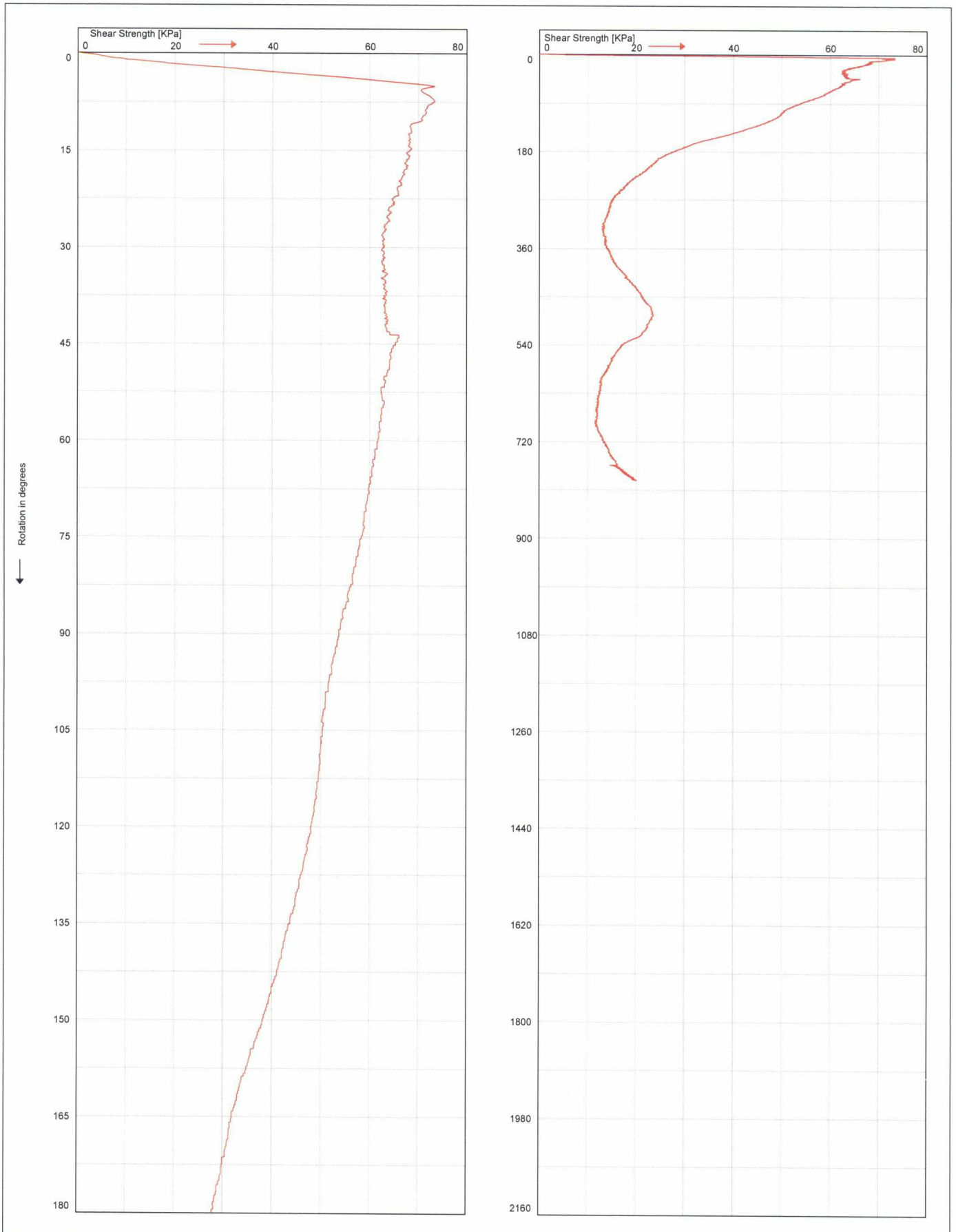




Job	: GK-28-10(PLATFORM)	Testdepth	: 12.8	Vane type	: VANE 50
Test	: 5	Client	: ONGC	Description	: PLATFORM LOCATION
Date	: 6-5-2018	Operator	: DIVAKAR / LALIT		



Job	: GK-28-10(PLATFORM)	Testdepth	: 16.8	Vane type	: VANE 50
Test	: 6	Client	: ONGC	Description	: PLATFORM LOCATION
Date	: 6-5-2018	Operator	: DIVAKAR / LALIT		



Job	: GK-28-10(PLATFORM)	Testdepth	: 20.8	Vane type	: VANE 50
Test	: 7	Client	: ONGC	Description	: PLATFORM LOCATION
Date	: 6-5-2018	Operator	: DIVAKAR / LALIT		

APPENDIX-E

APPENDIX-E

GENERAL INFORMATION

Conversion Factors from SI to other units	**E 1**
Table of Descriptive terms for Soils	**E 2**
Classification Table for carbonate sediments	**E 3**
Clays on Nagaraj Framework	**E 4**
Soil Behaviour Type Index I_c	**E 5.1**
Overconsolidation Ratio (OCR)	**E 5.2**
Relative Density	**E 5.3**

CONVERSION FACTORS FROM S. I. UNITS TO OTHER UNITS

Quantity	From S. I. Units	To Metric Units	Multiply by	From S. I. Units	To Imperial Units	Multiply by
Length	mm	mm	1	mm	in	0.03937
	m	m	1	m	ft	3.281
Area	mm^2	mm^2	1	mm^2	in^2	0.00155
	m^2	m^2	1	m^2	ft^2	10.76
Volume	ml	cm^3	1	ml	in^3	0.06102
	m^3	m^3	1	m^3	ft^3	35.31
Velocity	mm/sec	mm/sec	1	mm/sec	in/sec	0.03937
	m/sec	m/sec	1	m/sec	ft/sec	3.281
Rate of flow	ml/sec	cm^3/sec	1	ml/sec	in^3/sec	0.06102
	m^3/sec	m^3/sec	1	m^3/sec	ft^3/sec	35.31
Mass	g	g	1	g	lb	0.002205
	t	t	1	t	kips	2.205
Density	kg/m^3	kg/m^3	1	kg/m^3	lb/ft^3	0.06243
	t/m^3	t/m^3	1	t/m^3	lb/ft^3	62.43
Force	N	kgf	0.102	N	lbf	0.2248
	kN	kgf	102.0	kN	lbf	224.8
	MN	tf	102.0	MN	kips	224.8
Moment	N-m	kgf-m	0.102	N-m	lbs-ft	0.7376
Pressure,	N/m^2 (Pa)	gf/cm^2	0.0102	N/m^2 (Pa)	lbf/ft^2 (psf)	0.02089
Stress	kN/m^2 (kPa)	kgf/cm^2	0.0102	kN/m^2 (kPa)	lbf/in^2 (psi)	0.145
	MN/m^2 (MPa)	tf/m^2	102.0	MN/m^2 (MPa)	tf/ft^2 (tsf)	10.44
Unit Weight	kN/m^3	gf/cm^3	0.102	kN/m^3	lbf/ft^3 (pcf)	6.366

GEOTECHNICAL GROUP	
IEOT, ONGC, PANVEL	
PLATE	E1

TABLE OF DESCRIPTIVE TERMS FOR SOILS

COHESIVE SOILS

CONSISTENCY	UNDRAINED SHEAR STRENGTH, Su (KPa)
VERY SOFT	0-12.5
SOFT	12.5-25
FIRM	25-50
STIFF	50-100
VERY STIFF	100-200
HARD	200-400
VERY HARD	>400

COHESIONLESS SOILS

DESCRIPTION	RELATIVE DENSITY(%)
VERY LOOSE	0-15
LOOSE	15-35
MEDIUM DENSE	35-65
DENSE	65-85
VERY DENSE	85-100

TERMS FOR ROCK STRENGTH

DESCRIPTION	COMPRESSIVE STRENGTH qu (MPa)
VERY WEAK	<1.25
WEAK	1.25 - 5
MODERATELY WEAK	5 - 12.5
MODERATELY STRONG	12.5 - 50
STRONG	50 - 100
VERY STRONG	100 - 200
EXTREMELY STRONG	>200

GEOTECHNICAL GROUP	
IEOT, ONGC, PANVEL	
PLATE	E2

CLASSIFICATION TABLE FOR CARBONATE SEDIMENTS | PLATE | E- 3

OIL AND NATURAL GAS CORPORATION LIMITED
INSTITUTE OF ENGINEERING AND OCEAN TECHNOLOGY

E 4 – NAGARAJ FRAMEWORK

The basis of Nagaraj framework is a fact first observed by Skempton (1970). Skempton examined the variation of void ratio of normally consolidated natural clays with increasing overburden and observed that the relative positions of different Sedimentation Compression Curves (SCC) i.e. in-situ water content plotted against logarithm of vertical effective stress, depended on the Liquid limit (w_L) of the clays. By representing water content in terms of Liquidity Index (LI), it was shown that the SCCs converge to a narrow band.

Another important concept in the framework is the concept of "Intrinsic Properties" of clays as against properties of naturally sedimented clays. Properties of reconstituted clays (clays remoulded at water content of 1.0 to 1.5 times the Liquid Limit, w_L and consolidated one dimensionally), called Intrinsic Properties, are considered independent of the influence of processes of natural sedimentation and geological history. It has been proposed that these Intrinsic Properties can be taken as reference to assess a clay's natural state. Compression Curves of such reconstituted clays are termed "Intrinsic Compression Curves (ICC)".

Starting with micro level analysis Nagaraj concluded that Liquid Limit is a reference state for clays since at Liquid Limit all clays have:

- Similar pore size distribution and micro-structure
- Shear strength of the same order equal to 1.7 to 2.2 kPa
- Permeability of the same order
- Consolidation / suction of the same order equal to 5 to 6 kPa

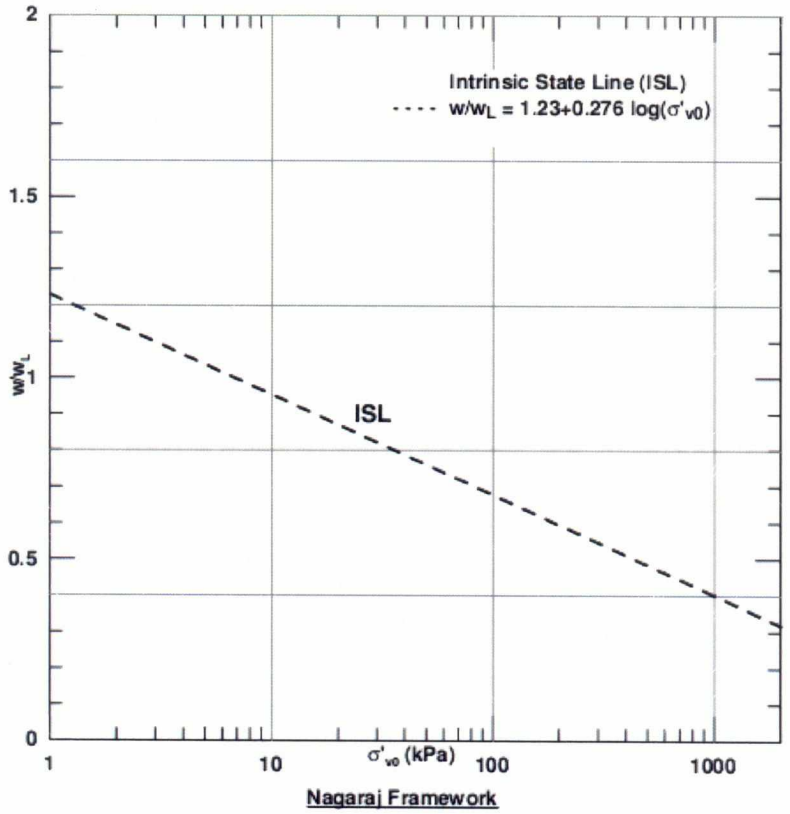
and,

- it can be used as a parameter for generalizing mechanical behaviour of saturated clays.

By normalizing water content, which is a measure of structure (in terms of fabric of a clay) with Liquid Limit, it was shown that ICCs of clays of widely ranging plasticity characteristics form almost a unique curve / line. This line was termed Intrinsic State Line (ISL) and Parameter e/ e_L (or w/w_L), Intrinsic / Generalized State Parameter. The ISL was defined by equation:

$$\frac{w}{w_L} \text{ or } \frac{e}{e_L} = 1.23 - 0.276.\log_{10}(\sigma'_{v0})$$

Where "w" is natural water content, w_L is Liquid limit, "e" and "e_L" the corresponding void ratios, and, σ'_{v0} is the effective vertical overburden pressure.



It was postulated that in the $e/e_L - \log_{10}(\sigma'_{v0})$ space, if the in situ state of a clay

- plots around the Intrinsic State Line then that soil is considered to be a "normally consolidated saturated uncemented soil"
- plots below the Intrinsic State Line then that soil is considered to be an "Overconsolidated (with or without cementation) saturated stiff clay"
- plots above the Intrinsic State Line then that soil is considered to be a "naturally cemented saturated soil"

References

Nagaraj, T.S., and Miura, N. (2001). "Soft Clay Behaviour – Analysis and Assessment", *A. A. Balkema / Rotterdam/ Brookfield*.

Skempton, A.W. (1970). "The consolidation of clays by gravitational compaction", *Quart. J. Geol. Soc.* London, 125, pp. 373-411.

E 5 - CPT INTERPRETATION

E 5.1 - SOIL BEHAVIOUR TYPE (SBT) INDEX, I_c

Using the concept of Soil Classification Index proposed by Jefferies and Davies (1993) Robertson and Wride (1998) proposed a Soil Behaviour Type Index, I_c as a parameter defining boundaries between various soil type zones as proposed in the Q_t-F_r chart of Robertson (1990). I_c is defined as:

$$I_c = [(3.47 - \log_{10} Q_t)^2 + (\log_{10} F_r + 1.22)^2]^{0.5}$$

Where:

- Normalized Cone Resistance, $Q_t = (q_t - \sigma_{v0})/\sigma'_{v0}$
- Normalized friction ratio, $F_R = f_s/(q_t - \sigma_{v0})$
- q_t is cone resistance corrected for pore pressure effect, f_s is the measured sleeve friction, σ_{v0} total vertical overburden pressure and σ'_{v0} the effective vertical overburden pressure.

Soil type based on I_c by Robertson and Wride (1998)

Soil Type	Zone no.	I_c Range
Gravelly sand to sand	7	$I_c < 1.31$
Sands – clean sands to silty sands	6	$1.31 < I_c < 2.05$
Sand mixtures – silty sand to sandy silt	5	$2.05 < I_c < 2.6$
Silt mixtures - Clayey silt to silty clay	4	$2.6 < I_c < 2.95$
Clays – clay to silty clay	3	$2.95 < I_c < 3.60$
Organic soil - peats	2	$I_c > 3.60$

I_c approximates the boundaries between various soil types defined by zones 2 to 7 in Robertson (1990) chart which uses normalized cone resistance and normalized friction ratio. However, this classification has an advantage that by plotting I_c against depth, the stratigraphy can easily be visualized at a given location. The SBT Index proposes classification of soils as per their "behaviour" and does not strictly corresponds to soil classification based on grain size distribution.

E 5.2 - OVERCONSOLIDATION RATIO (OCR)

Since 1980s, a number of relationships between OCR for clays and various forms of normalised pore pressure and normalised cone resistance values have been proposed. Most commonly used relationship is of the form:

$$OCR = k.\frac{(q_t - \sigma_{vo})}{\sigma'_{vo}}$$

where k is in the range of 0.2 to 0.5 and other terms are defined in E 6.1.

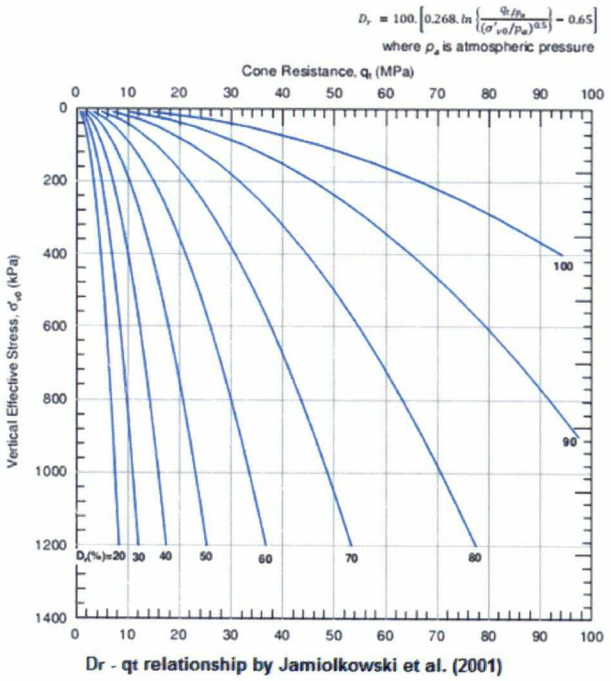
Higher values of k are recommended in aged heavily overconsolidated clays (Powell et al, 1988). Mayne (2007) suggests an average value of 0.33 for k for first order determination of OCR and states that this relationship under predicts OCR values for fissured clays. The OCR values presented in this report are derived using a k value of 0.33.

E 5.3 - RELATIVE DENSITY, (D_r)

The relative densities of sand deposits can be estimated using the CPT parameters. The relative densities presented in this report have been estimated using the relation proposed by Jamiolkowski et al. (2001):

$$D_r = 100.\left[0.268.\ln\left\{\frac{q_{t/p_a}}{(\sigma'_{v0}/p_a)^{0.5}}\right\} - 0.65\right]$$

Where q_t is the corrected cone resistance, p_a is the atmospheric pressure and σ'_{v0} is the effective vertical overburden pressure. The relationship is shown graphically in the following figure:



Dr - qt relationship by Jamiolkowski et al. (2001)

References

Jamiolkowski, M.; Lo Presti, D.C.F. and Manassero, M. (2001). "Evaluation of Relative Density and Shear Strength of Sands from Cone Penetration Test and Flat Dilatometer Test", Soil Behavior and Soft Ground Construction (GSP 119), ASCE, Reston, Virginia, 2001, 201-238.

Jefferies, M.G. and Davies, M.P. (1993). "Use of PCPT to estimate equivalent SPT N60". Geotechnical Testing Journal 16 (4), pp. 458-468.

Mayne, P.W. (2007). "Cone Penetration Testing State-of-Practice", Final Report, NCHRP Project 20-05, Topic 37-14, Transportation Research Board Synthesis Study.

Powell, J.J.M., Quarterman, R.S.T. and T. Lunne. (1988). "Interpretation and Use of the Piezocone Test in UK Clays", Penetration Testing in the UK, Thomas Telford, London, 151-156.

Robertson, P.K. (1990). "Soil classification using the cone penetration test". Canadian Geotechnical Journal, Vol. 27, No. 1, pp. 151 - 158.

Robertson, P.K. and Wride, C.E. (1998). "Evaluating cyclic liquefaction potential using the cone penetration test". Canadian Geotechnical Journal 35 (3), 442–459.

Re: : Priority of soil testing and analysis for Platform locations.
P Prdeep kumar
to:
RK GHANEKAR
18-05-2018 14:45
Cc:
M SAMPATHKUMAR, Sumit Adhana, SRI HARSHA GAMIDI
Hide Details
From: P Prdeep kumar/ONGCL
To: RK GHANEKAR/ONGCL@ONGCL
Cc: M SAMPATHKUMAR/ONGCL@ONGCL, Sumit Adhana/ONGCL@ONGCL, SRI HARSHA GAMIDI/ONGCL@ONGCL
Sir,

With reference to trailing mail, pointwise reply is as under:

1. The priority given on 04.04.2018 was tentative and the revised priority was forwarded vide letter dated 16.05.2018.

2. Soil investigation at GK-28-10, GK-42-1, GK-28-7A platform locations were completed and soil sample boxes are being delivered to Nhava. The same will be delivered to IEOT after receipt at Nhava. Presently, vessel is carrying soil coring at WO-24-3-P (SE-1) platform location. Exact date of delivery soil sample boxes to IEOT will be intimated in advance.

Revised tentative priority of Soil testing is as follows :
1. WO-24-3-P (SE-1)
2. B-192-5A (new location added)
3. GK-28-10
4. GK-42-1
5. GK-28-7A

Thank you for completing soil testing of B-192-10-P & D30-2-P platform location.

Regards

P Pradeep Kumar
DGM (Elec)
Marine Survey Division,
OES,11 High
Mumbai
9969228246.

-----RK GHANEKAR/ONGCL wrote: -----

> To: P Prdeep kumar/ONGCL@ONGCL
> From: RK GHANEKAR/ONGCL
> Date: 05/18/2018 10:14AM
> Cc: SUMESH KUMAR/ONGCL@ONGCL, M SAMPATHKUMAR/ONGCL@ONGCL
> Subject: Re: : Priority of soil testing and analysis for Platform locations.
>
> Dear Mr Pradeep Kumar
>
> We find the following abiout the subject of your letter:
>
> 1. The priorities given in Marine Survey mail dated 04.04.2018 and your present letter do not match.
> 2. Except B-192-10-P and D-30-2-P locations we have not received samples of any other locations.